CFA Institute

Investment Foundations Course 1: Industry Overview and Structure

Introduction to This Reading Material

Thank you for purchasing/downloading the reading packet for DNU-02-23 M-IF1: Industry Overview and Structure.

Begin your online course by reading the syllabus; it contains the information you need to successfully complete the course. As you begin, you will notice that each module includes a brief introduction, learning objectives for the module, and reading assignments. The reading material for each module is included in this packet, as well as in the online course.

In addition to the readings, the modules have computer-graded assignments and self-check exercises associated with them. These assignments and other interactive activities, such as videos, are available only through the online course.

This packet is designed to give you the best experience for reading the detailed lesson content and taking notes. To complete the course, you will need access to the online assignments and exams.

Best wishes for your success in this course!

Module 1: Industry Overview and Structure

Have you ever wondered why the financial services industry exists and how it operates? At its best, the industry efficiently matches those who need money with those who have savings to invest while minimizing the costs to each and facilitating financial support for the most productive businesses and projects.

In this module, you will be introduced to the structure of the financial services industry at large, the investment industry in particular, and the types and functions of industry participants.

By the end of this module, you will be able to:

- Describe the financial services industry
- Explain how economies benefit from the existence of the investment industry
- Explain how investors benefit from the existence of the investment industry
- Describe the types of institutions, services, and functions of participants in the investment industry

Let’s get started!

Lesson 1: What Is the Financial Services Industry?

Welcome to Lesson 1 of Module 1!

In this lesson, you will explore the meaning and function of “savers” and “spenders” in the financial services industry, as well as different forms of assets.

By the end of this lesson, you will be able to describe the financial services industry.

Let’s get started!

The Financial Services Industry

The financial system links savers, who have money, to spenders, who need money. Within the financial system, the **financial services industry** offers products and services to savers and spenders and channels funds between them.

“Savers” include individuals, households, companies, and governments that have money to invest. “Spenders” also include individuals, companies, and governments. For example, individuals borrow to pay for houses, education, and other expenses. Companies borrow to invest in land, buildings, and machinery. Governments borrow when their tax receipts are below their spending plans.

Terminology

Typically, the term “saver” characterizes those who have accumulated savings, which are often invested.

When savers make investments, they are typically called investors and they become providers of capital.

If their investment is a loan—that is, money that is expected to be repaid with interest—investors are often referred to as lenders.

Similarly, the term “spender” characterizes those who need money. When spenders have received the money they need and start using it, they become users of **financial capital**. If they are recipients of a loan, they are typically called borrowers.

Note: There are times when the terms savers, investors, lenders, and providers of capital; spenders, borrowers, and users of capital; or money, cash, funds, capital, and financial capital are used interchangeably.

Real and Financial Assets

Savings can be invested in a wide range of **assets**. Assets are items that have value and include **real assets** and **financial assets**.

Real assets are physical assets, such as land, buildings, machinery, cattle, and gold. A company's production may depend on them, and they are sometimes referred to as **physical capital**.

In contrast, financial assets are claims on real assets or possibly other financial assets, and they frequently come in the form of a certificate or legal contract. For example, a share of stock represents ownership in a company. This share gives its owner, who is called a shareholder, a claim to some portion of the company's assets and earnings. An investor's total holdings of financial assets is usually called a **portfolio** or investment portfolio.

Financial assets that can be traded are called **securities**. The two largest categories of securities are debt and equity.

Learn more about debt and equity securities in Exhibit 1.

Exhibit 1: Debt and Equity Securities

Debt Securities	Equity Securities
Debt securities are loans that lenders make to borrowers. Lenders expect the borrowers to repay these loans and to make interest payments until the loans are repaid. You may already be familiar with this concept in personal finance because many people use debt to pay for big expenses, such as cars and homes. Because interest payments on many loans are fixed, debt securities are also called fixed-income securities. They are also known as bonds, and investors in bonds are referred to as bondholders. More information about debt securities is provided in Course 3, Investment Instruments.	Equity securities are also called stocks or shares. Shareholders (also known as stockholders) have ownership in a company. The company has no obligation to repay the money that shareholders paid for their shares or to make regular payments to them, which are called dividends. But investors who buy shares expect to earn a return by selling their shares at a higher price than they bought them, and possibly by receiving dividends. Equity securities are discussed further in Course 3, Investment Instruments.

Markets are places where buyers meet sellers to trade. Buyers and sellers trade securities in securities markets or **financial markets**. How securities are issued, bought, and sold is explained in Course 2, Types and Functioning of Markets.

Overview of the Financial Services Industry

Exhibit 2: The Financial Services Industry

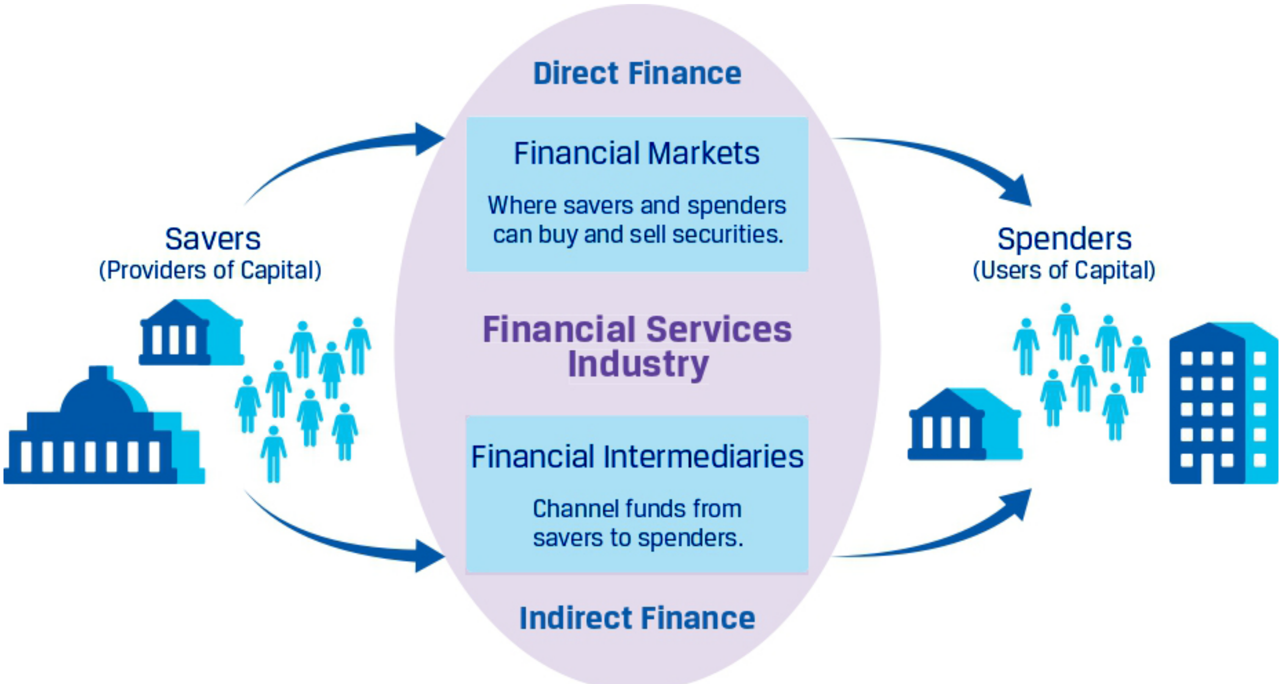


Exhibit 2 shows how the financial services industry channels funds between those who have money to invest (the savers that become providers of capital) and those who need money (the spenders that become users of capital).

When providers and users of capital interact, the providers usually have a direct claim on the users. For example, if the provider owns a share of Nestlé, that gives them a right to certain assets and earnings generated by Nestlé, the user. This is **direct finance**.

Providers and users of capital often rely on **financial intermediaries** to find each other and to channel funds between each other. This process is **indirect finance** because financial intermediaries sit between savers and spenders; the former do not have direct claims on the latter. Financial intermediaries may also create new products and securities that depend on other assets.

Financial intermediaries play an important role in the financial services industry. Many savers do not have the time or the expertise to identify and select individuals, companies, and governments to lend to or invest in.

Once savers have loaned money, they have to monitor borrowers' behavior and financial health to ensure that they will get their money back, a task that is time-consuming and costly. Matching savers and borrowers and monitoring borrowers' behavior and financial health are functions financial intermediaries can perform better and more cheaply than most investors can do on their own.

Lesson 1 Knowledge Check

The following knowledge check is designed to reinforce your understanding of the concepts taught in this lesson. It is ungraded, but you are strongly encouraged to complete it.

Question 1

GlobalBank credits Krause's account for USD100,000 and makes a USD50,000 loan to a new customer. This means that GlobalBank is a:

a. provider of capital
b. user of capital
c. both

Answers for Lesson 1 Knowledge Check

Question 1

a. **Incorrect.** When GlobalBank credits Krause's account, it is a user of capital as well as being a provider of capital when it makes a loan to a new customer.

b. **Incorrect.** When GlobalBank makes a loan to a new customer, it is a provider of capital as well as being a user of capital when it credits Krause's account.

c. **Correct.** GlobalBank's crediting Krause's account and loaning to another customer classifies it as both a provider and a user of capital. In crediting Krause's account, GlobalBank acknowledges that it borrowed money from Krause and is therefore a user of capital. In lending USD50,000, GlobalBank is acting as a provider of capital.

Up Next

You have completed Lesson 1! We hope that this lesson has helped you to expand your knowledge of the financial services industry, its participants, and how it works.

You should now be able to describe the financial services industry.

In the next lesson, you will explore how economies benefit from the existence of the investment industry.

Lesson 2: How Economies Benefit from the Existence of the Investment Industry

Welcome to Lesson 2 of Module 1!

In this lesson, you will explore the way the investment industry operates within economic systems, how resources are allocated in economic systems, and how the investment industry fosters economic growth.

By the end of this lesson, you will be able to explain how economies benefit from the existence of the investment industry.

Let's get started!

How Economies Benefit from the Existence of the Investment Industry

The **investment industry** is a subset of the financial services industry. It contains the participants who help savers invest their money and the participants who help spenders raise capital in financial markets.

Economic Systems

The investment industry and its participants do not exist in isolation; they operate within economic systems that vary from country to country. Economic systems can take many forms, from pure **capitalism** with free markets to planned economies with centralized authorities. A goal of most economic systems is the efficient allocation of scarce resources to the most productive uses.

Resources, such as labor, real assets, and financial capital, are necessary to produce goods and services. Desire for goods and services is not limited, but resources are limited.

Consider this concept of scarcity by picturing individuals that have a limited budget; in other words, their financial capital is scarce. Should they spend their money buying food, paying their mortgage, purchasing a new car, or going on holiday?

Similarly, should a company focus its resources on an existing product or on a new one that might produce a higher profit? And should governments spend money on healthcare, education, defense, or infrastructure?

Because resources are scarce, decisions must be made regarding the allocation of these resources. Participants in economic systems must address three questions:

1. Which goods and services should be produced?
2. How should the goods and services be produced?
3. Who should receive the goods and services that are produced?

Economic planners prefer that scarce resources are applied to produce goods and services in a way that satisfies the needs of consumers.

Capitalism promotes private ownership and markets. Private owners produce goods and services, and markets allocate them based on scarcity.

In a purely free market capitalist economy, there is no central authority, such as a government, directing economic activity. Instead, individuals and companies make their own decisions about what goods and services to provide, and they keep the profits from their activities. When well-functioning markets support the efficient allocation of scarce resources, economies can grow and society benefits.

It is important to note that pure free market capitalism exists only in theory. In the real world, governments play a role in all economic systems.

In some capitalistic economies, such as Western economies, the government's role in business may be relatively minimal.

In economies largely dependent on the extraction of natural resources, such as in some Middle Eastern, African, and South American countries, the government may maintain significant control over key national industries.

In transition economies, which are moving from planned economies to market economies, the government may play a significant role in business.

How the Investment Industry Fosters Economic Growth

What role does the investment industry play in supporting the creation of goods and services and, ultimately, enhancing the lives of consumers? We have considered the investment industry's role in channeling funds between savers who have money to invest and spenders who need money to finance businesses and projects.

The role of the investment industry in the economy includes elements such as:

- **Resource allocation:** The investment industry contributes to the efficient allocation of resources in the economy. Without the investment industry, savers would have to spend significant resources finding individuals, companies, and governments offering suitable investment opportunities. Resources would also be

spent on the search for capital rather than on considering how best to use it, which would reduce efficiency. Capital that is put to more productive use fosters economic growth and boosts the willingness of savers to supply funds to those who need them.
- **Data collection and analysis:** The investment industry helps investors collect and analyze data about economies and information about individuals, companies, and governments. It also assists investors in determining the value of real and financial assets. The types of inputs and tools used by investment industry participants are described in Course 4, Investment Inputs and Tools.
- **Investment services:** Investment industry participants package investment opportunities so that they satisfy the needs of investors. In particular, the investment industry offers a wide range of services and products that make it easier for savers to invest.
- **Liquidity:** The investment industry provides liquidity, which is the state of a market's ability to transfer its products between buyers and sellers with ease. Highly liquid markets make it easy to buy or sell an asset without affecting the price at which other people will participate in a similar transaction. Some assets, such as real estate, are inherently illiquid: selling your house will likely take some time even if it is priced fairly. Other assets, such as shares that trade actively, are more liquid.

Consider an investor who holds 100 shares. She can probably sell her shares quickly without affecting the share price that others have to use for their transactions with the same share. But if she owned 100,000 shares, selling them quickly could lower the share price if the market is not liquid enough to be indifferent to that volume of shares.

Liquidity is a very important aspect of well-functioning financial markets. Highly liquid markets allow investors to complete a transaction quickly and to reverse it quickly if they change their minds. They give investors confidence about getting a fair price at that particular moment.

Lesson 2 Knowledge Check

The following knowledge check is designed to reinforce your understanding of the concepts taught in this lesson. It is ungraded, but you are strongly encouraged to complete it.

Question 1

The investment industry benefits the economy by providing:

Select all that apply.

a. Liquidity
b. Relevant financial information
c. Capital gains

Answers for Lesson 2 Knowledge Check

Question 1

a. & b. **Correct.** Market liquidity exists when investors can buy and sell easily without impacting the trade price. Liquidity improves when transaction costs are low. The investment industry provides access to the financial data needed to evaluate investment opportunities. When financial data is difficult to obtain, liquidity declines, sellers raise their asking prices, and buyers lower their bids, reflecting the risk that they may not be aware of relevant information.

c. **Incorrect.** Capital gains are a potential outcome of investing (for example, when an asset price appreciates). But capital gains are not a structural feature of markets.

Up Next

You have completed Lesson 2. We hope that this lesson has helped you to expand your knowledge of the way the investment industry operates within economic systems, how resources are allocated in economic systems, and how the investment industry fosters economic growth.

You should now be able to explain how economies benefit from the existence of the investment industry.

In the next lesson, you will explore how investors benefit from the existence of the investment industry.

Lesson 3: How Investors Benefit from the Existence of the Investment Industry

Welcome to Lesson 3 of Module 1!

In this lesson, you will explore the characteristics of a well-functioning investment industry, the benefits experienced by investors, and how finance professionals can help investors.

By the end of this lesson, you will be able to explain how investors benefit from the existence of the investment industry.

Let’s get started!

Characteristics of Well-Functioning Markets

A well-functioning investment industry depends on investor confidence. Such confidence arises from a feeling of trust that all parties are adhering to a code of ethics and standards for professional conduct, and that trust must be earned and verified consistently.

Some of the things that work against that trust and lead to inefficient markets include the following:

- Unfair transactions
- Shortfalls in integrity
- Unreliable advice
- Inefficient handling of trades

If transactions require increased analysis or due diligence, delays and costs are introduced. A robust investment industry supports efficiency and benefits investors.

One aspect of a well-functioning investment industry is a broad range of investment opportunities that meet investors’ needs. Investors can invest in debt and equity securities, and they can invest in derivatives and alternative investments.

Investors may also choose to save by using investment vehicles that exist solely to hold investments on behalf of their shareholders, partners, or unit holders. We can think of unit holders as similar to shareholders, but they hold a unit in an investment vehicle that could take many forms beyond a simple share or a bond, and such vehicles frequently invest in shares and bonds.

A general name for such an arrangement is a pooled investment vehicle because groups of individual investors put their money together to invest in the same customized financial arrangement that is professionally managed.

Participants may buy and sell real and financial assets and then package them to create new assets that suit the needs of investors better than the original assets. Mortgage-backed securities are an example; they represent a claim on the money generated by a large number of mortgages that have been grouped together in a process called securitization.

In addition to being able to choose from many investment opportunities, investors benefit from a broad range of services that support better investment decisions, such as the following:

- Financial planning
- Information and research
- Trading support

How investment industry participants assess and serve the needs of investors is discussed further in Course 5, Serving Client Needs.

Learn more about some other characteristics of the investment industry that benefit investors in Exhibit 1.

Exhibit 1: Competitive and Liquid Markets

Competitive Markets	Liquid Markets and Low Transaction Costs
Investors benefit when financial markets are competitive. Competitive markets lead to fair prices, which ensure that buyers and sellers transact at a reasonable price. Markets in general and financial markets in particular are competitive if a large number of participants compete with one another without any one of them having an undue influence on supply or demand. Competitive markets promote production efficiency and keep the prices of goods and services down, including investment products and services.	Investors benefit when financial markets are liquid and transaction costs are low. As covered earlier, better liquidity lets investors quickly buy or sell an asset without unwanted price effects. If we picture an active market with many buyers and sellers, such as a stock exchange, the costs incurred when a trade happens are called transaction costs. Because transaction costs reduce the return on investments, the lower the transaction costs the better.

How Financial Professionals Help Investors

To make reasonable judgements about what to invest in, savers need relevant and reliable information about the companies and governments to which they may provide capital. The data collecting and processing done by financial professionals benefits investors, and the timeliness of this information is critical because securities prices may change quickly in response to new, relevant information.

Consider a biotech company that just announced that its new wonder drug received regulatory approval; its share price will likely increase as investors anticipate higher profits.

A well-functioning investment industry supports society as it deals with risk, the uncertainty of future outcomes. Although the word “risk” can be taken as negative by individuals who seek safe investments, uncertainty is unavoidable, and all investor choices carry risk.

Investors can rely on investment industry expertise to determine whether a particular risk is appropriate to their financial circumstances and justified by the expected investment return.

It is the financial adviser’s role to explain financial risk in terms that are understandable and to propose solutions for managing it.

Risk management solutions in investment portfolios may range from complex hedging strategies to simply replacing a large investment with a diversified basket of smaller investments. We will explore the concept of hedging in Course 2, Types and Functioning of Markets and Course 3, Investment Instruments.

The industry’s professionals provide advice on mitigating investment risk, and they can also step in to reduce risk by providing insurance. Certain participants will be willing to take on more risk by selling insurance or investments that allow others to reduce their risks.

Lesson 3 Knowledge Check

The following knowledge check is designed to reinforce your understanding of the concepts taught in this lesson. It is ungraded, but you are strongly encouraged to complete it.

Question 1

Investors benefit in many ways from a well-functioning investment industry. Which of the following is *not* a benefit?

a. Access to a diverse range of investment products structured to meet investor needs
b. Availability of investment management and advisory services
c. Low market volatility
d. Access to financial and economic data
e. Fair competitive pricing in liquid markets

Answers for Lesson 3 Knowledge Check

Question 1

a. **Incorrect.** This is a benefit. Access to a diverse range of investment products facilitates the construction of portfolios that reflect investors' goals and constraints.

b. **Incorrect.** This is a benefit. The availability of investment management and advisory services allows investors that lack sufficient time, financial skills, or desire to design and implement efficient investment strategies.

c. **Correct.** Market volatility is not necessarily low in a well-functioning investment industry. Without a benchmark or further definition, "low" has little meaning.

d. **Incorrect.** This is a benefit. Access to financial and economic data is needed to evaluate investment opportunities and construct appropriate portfolios.

e. **Incorrect.** This is a benefit. Fair pricing in liquid markets increases investor confidence and willingness to trade, resulting in a more efficient allocation of capital.

Up Next

You have completed Lesson 3. We hope that this lesson has helped you to understand the characteristics of a well-functioning investment industry, the benefits experienced by investors, and how finance professionals can help investors.

You should now be able to explain how investors benefit from the existence of the investment industry.

In the next lesson, you will explore investment industry participants.

Lesson 4: Investment Industry Participants

Welcome to Lesson 4 of Module 1!

In this lesson, you will explore the roles of various professionals within the investment industry through a hypothetical example. By engaging with this example, you will learn how investment professionals are categorized and the key services they offer to assist investors and companies as well as interact with example personas representing key industry participants.

By the end of this lesson, you will be able to describe the types of institutions, services, and functions of participants in the investment industry.

Let’s get started!

How Companies and Governments Raise Capital

As a way of introducing investment industry participants, let’s consider the hypothetical firm, Primo. This small, privately owned company has experienced steady growth over multiple years and is now looking for investors to provide the capital it needs for more aggressive expansion.

The financial services industry can step in to raise the money Primo needs and provide investors with ways to participate in the company’s growth. We first introduce those who can help the company raise capital, and then we discuss investors and the professionals who help them invest.

Primo wants to issue shares to raise capital. Until now, it has been a private company; it has not raised money by issuing shares publicly. It wants to take the opportunity to become a public company and have its shares listed on a major stock exchange. Stock exchanges are organized and regulated financial markets that allow buyers and sellers to trade securities with each other.

Primo contacts an investment bank for guidance. **Investment banks**, also known as merchant banks, are financial intermediaries that have expertise in assisting companies and governments with raising capital, often by organizing the sale of shares and bonds to the public. In the case of Primo, the equity sale is called an initial public offering, or IPO, because it is the first time the company will sell shares to the public.

Investment banks match investors with the companies and governments seeking capital. The investment banks pay close attention to the types of investments that investors want so that they can design and propose securities that will suit the capital needs of the company or government and also appeal to investors. By offering securities that investors want to purchase, companies and governments can obtain capital at a lower cost.

The investment bank works with Primo to determine the price at which the new shares will be issued, and that involves gauging investor interest in the company's new shares. The investment bank's analysts, often called **sell-side analysts** because they work for the organization selling the securities, collect and analyze information about the company and its competitors, and then prepare a report that is shared with potential investors.

When Primo becomes a public company, it will have to comply both with the stock exchange rules where its shares are listed and with relevant national regulations. It may, for instance, be required to file audited financial statements. Auditors evaluate the company's internal controls and financial reporting to ensure that investors receive relevant and reliable financial information, a key feature of well-functioning financial markets.

How the Investment Industry Helps Savers Invest Their Money

Primo may sell its shares to various investors whose needs vary. A distinction is often made between individual and institutional investors; an individual investor is simply a person who holds investments, whereas institutional investors are typically organizations that invest either for themselves or on behalf of others.

Individual investors can be classified more specifically, but there is no universal standard, and distinctions vary across countries, currencies, and investment firms. Generally, **retail investors** have the least investible assets and **high-net-worth investors** have substantial investible assets.

The advice and services that individual investors receive depend on their wealth and investment knowledge, as well as on the regulatory environment. Retail investors tend to receive standardized services, whereas wealthier investors often receive services tailored to their needs.

Institutional investors come in different forms, such as the following:

- **Pension plans: Pension plans** hold and manage investment assets for the benefit of current and future retired individuals, called beneficiaries.

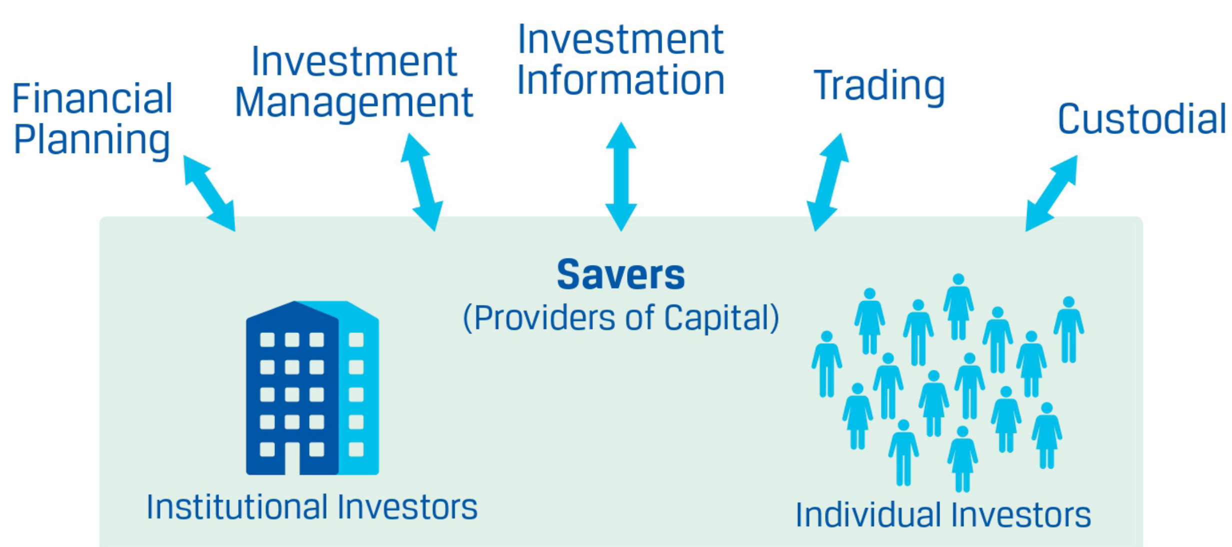
- **Endowment funds: Endowment funds** are long-term funds belonging to not-for-profit institutions, such as universities, colleges, schools, museums, theaters, opera companies, hospitals, and clinics. (In this context, "funds" refers to formal financial entities with dedicated investment criteria, rather than functioning as a synonym for money or capital.)
- **Foundations: Foundations** are grant-making institutions funded by financial gifts and by the investment income that they produce.
- **Sovereign wealth funds: Sovereign wealth funds** are overseen by governments, which may gather more tax than they intend to spend by selling natural resources or by financing the trade of goods and services. Such a surplus is usually invested, and some governments create sovereign wealth funds to invest their surpluses for the benefit of their citizens.

Institutional investors that invest on behalf of others include investment firms and financial institutions, such as banks and insurance companies.

Investor Services

Despite the differences among investors and their range of needs, many of the services they require are common to all of them. Some of these services are shown in Exhibit 1.

Exhibit 1: Services Available to Investors



When investors want to buy or sell shares, they need to find another investor who is willing to buy or sell shares. Brokers and dealers are trading service providers that facilitate this trading. Brokers act as agents—that is, they do not trade directly with investors but help buyers and sellers find and trade with each other.

In contrast, dealers act as principals—that is, they use their own accounts and their own capital to trade with buyers and sellers in what is known as proprietary trading. (Proprietary traders will be expanded on in the Trading Services lesson in Module 3, Introduction to Investment Services). They “make markets” in securities by acting as buyers when investors want to sell and as sellers when investors want to buy.

Dealers often have thousands of clients, so if one client wants to sell shares at a certain price, the dealer can usually identify another client who is willing to buy the shares at a similar price. Brokers and dealers provide liquidity and reduce transaction costs, established above as beneficial to investors and well-functioning markets.

Also providing trading services are clearing houses and settlement agents, which confirm and settle trades after they have been agreed on. Custodians and depositories hold money and securities on behalf of their clients.

Institutional investors may employ analysts to review potential investments. These analysts are called **buy-side analysts** because they work for the organization buying the securities.

To gather data about a company and the markets in which it operates, analysts often rely on investment information providers, such as data vendors, or investment research and report providers.

Individual investors often do not have the time, inclination, or expertise to manage the entire investment process on their own, so many of them seek the help of investment professionals. Financial planners help clients to define their investment goals and understand their future needs. Investment management service providers, such as asset managers, help their clients make investment decisions or make them on their behalf to achieve the clients’ investment goals.

Many investors, particularly retail investors, are willing to invest but lack the resources to hire an investment manager to look after their investments. Instead, these investors may buy investment products offered by investment firms, banks, and insurance companies. An individual planning for their retirement, for example, may need a convenient and inexpensive way to invest money regularly. In order to do this, they could buy shares in a mutual fund, which is a professionally managed vehicle that invests in a range of securities. Mutual funds will be expanded on in Course 2, Types and Functioning of Markets.

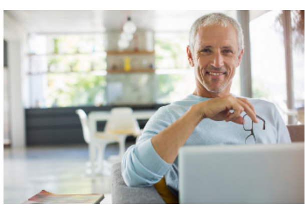
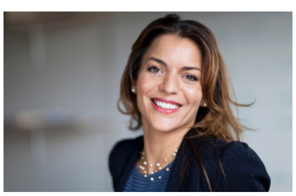
In the next section, we will explore examples of investment professionals and learn about their roles, their areas of expertise, and the products they offer. We will also meet investors to learn about their needs and the products they use.

Meet Some of the Industry Participants

The following hypothetical industry participants are a mix of investors and financial professionals working in the investment industry. You will continue to engage with these types of industry participants throughout the course as you expand your knowledge of various investment-related topics.

Exhibit 2 shows the investors' goals and needs, and the roles that the financial professionals play in assisting other investors.

Exhibit 2: Investors' Goals and Needs

Investor	Goals, Needs, and Roles
	**Zhang Li** is a retail investor. She earns SGD5,000 a month and wants to save for a deposit on an apartment in the suburbs of Singapore. She also wants to save for her son's university education. She goes to her bank for investment advice.
	**Mike Smith** is a high-net-worth investor who lives in California. He recently sold his technology company and has USD10 million to invest. He hires a financial planner to help him define his investment objectives in terms of return and risk.
	**Anna Huber** works for an institutional investor called Euro Pension Fund in Frankfurt, Germany. The fund's purpose is to finance the retirement of its members, and so it invests the money it receives and pays out money to the retired members. Its asset management team, which Huber is a part of, devises the fund's strategy and implements it, managing a EUR50 billion portfolio invested in a wide range of assets.

Peter Robinson is an asset manager for Aus Ltd., based in Sydney. Aus Ltd. invests money in shares, bonds, and alternative investments on behalf of its clients. It works with a data vendor to access global economic data and historical market prices, and it hires two investment research providers to keep it updated with the latest market developments. Aus Ltd. has a broker that carries out trades on its behalf and a custodian that safeguards clients' money and assets.

Amina Al-Subari is a broker at Middle East Corp., based in Dubai. Investors, such as Aus Ltd. and Euro Pension Fund, ask her to find and trade assets in the market. Middle East Corp. can find and trade these assets on an exchange and also deal directly with other investors who want to sell their assets.

James Armistead is with Big Bank Financial Services, a custodian with offices all over the world. When an investor, such as Euro Pension Fund or Aus Ltd., buys securities, the trade is confirmed by a clearing house and settlement agent. The custodian then holds the security on behalf of the investor and makes sure a proper record is kept of the security and its price.

Lesson 4 Knowledge Check

The following five questions are designed to reinforce your understanding of the concepts taught in this lesson. They are ungraded, but you are strongly encouraged to complete them.

For each scenario, select the service provider *best* suited to help the industry participant with their investment needs.

Question 1

Primo Company needs to raise capital by issuing stock.

a. Buy-side analyst
b. Investment bank
c. Financial planner
d. Broker/dealer
e. Custodian

Question 2

Zhang Li, a retail investor, needs to invest assets now to cover future expenses, such as her son’s college tuition and her retirement.

a. Investment bank
b. Broker/dealer
c. Custodian
d. Buy-side analyst
e. Financial planner

Question 3

Peter Robinson, asset manager for Aus Ltd., needs the latest market research to decide which investments to make on behalf of his clients.

a. Custodian
b. Investment bank
c. Broker/dealer
d. Financial planner
e. Buy-side analyst

Question 4

Mike Smith, a high-net-worth investor, needs various advisory and trading-related services.

a. Buy-side analyst
b. Investment bank
c. Financial planner
d. Broker/dealer
e. Custodian

Question 5

Anna Huber, an investor at Euro Pension Fund, needs to have her firm’s securities held with accompanying account statements for their records.

a. Investment bank
b. Broker/dealer
c. Custodian
d. Buy-side analyst
e. Financial planner

Answers for Lesson 4 Knowledge Check

Question 1

b. **Correct.** The Primo Company needs the assistance of an investment bank to issue new stock.

Incorrect. The Primo Company needs the assistance of an investment bank to issue new stock.

Question 2

e. **Correct.** Well done! Zhang Li will need the help of a financial planner to assist with the investment of her assets for her financial future.

Incorrect. Zhang Li will need the help of a financial planner to assist with the investment of her assets for her financial future.

Question 3

e. **Correct.** Peter Robinson will need a buy-side analyst to provide him with the information and advice needed to invest on behalf of his clients.

Incorrect. Peter Robinson will need a buy-side analyst to provide him with the information and advice needed to invest on behalf of his clients.

Question 4

d. **Correct.** Mike Smith needs the help of a broker or dealer for advisory and trading-related services.

Incorrect. Mike Smith needs the help of a broker or dealer for advisory and trading-related services.

Question 5

c. **Correct.** Well done! Anna Huber needs a custodian to safeguard her firm's securities and account statements.

Incorrect. Anna Huber needs a custodian to safeguard her firm's securities and account statements.

Up Next

You have completed Lesson 4. We hope that this lesson has helped you to expand your knowledge of different professionals in the investment industry.

You should now be able to describe the types of institutions, services, and functions of participants in the investment industry.

Module 1 Summary

Without the financial services industry, it would be difficult for money to find its way from savers to individuals, companies, and governments that have businesses and projects to finance but insufficient capital to do so themselves.

At its best, the industry efficiently matches those who need money with those who have savings to invest, minimizing the costs to each and allowing money to support the most productive businesses and projects. The investment industry acts on behalf of savers, helping them to navigate the financial markets. When the investment industry is efficient and trustworthy, economies and individuals benefit.

We began this module by defining the financial services industry and examining the actors within it, primarily savers and spenders, and how money in the form of assets and securities is transferred from one to the other.

- The financial services industry exists to provide a link between savers/lenders/providers of capital who have money to invest and spenders/borrowers/users of capital who need money.
- Financial intermediaries go between savers and spenders.

The investment industry exists within an economic system. These systems differ from country to country, but the goal of most economic systems is the efficient allocation of scarce resources to the most productive uses.

- The investment industry, a subset of the financial services industry, includes all participants that help savers invest their money and spenders raise capital in financial markets.
- Financial markets and the investment industry help allocate capital, a scarce resource, to the most productive uses, which fosters economic growth and benefits society.
- The investment industry provides numerous benefits to the economy, including the efficient allocation of scarce resources, better information about investment opportunities, products and services that are appropriate for providers and users of capital, and liquidity.

Investor confidence is key to fostering a well-functioning investment industry.

- Trust is essential to the functioning of the investment industry as well as to the broader financial services industry. Investor confidence arises from a feeling of trust that all parties are adhering to a code of ethics and standards for professional conduct, and that trust must be earned and verified consistently. In addition, laws and regulations are necessary to protect investors and ensure the integrity, transparency, and fairness of financial markets.

- The benefits for investors of a well-functioning investment industry include a broad range of investment products and services that meet their needs, competitive markets that provide liquidity and keep transaction costs low, timely and efficient disclosure of information, and the ability to modify their risk exposures.

Companies and governments use investment (merchant) banks to help them raise capital. Ensuring the investment industry performs at optimal levels involves a wide range of participants.

Exhibit 1 shows the key investment industry participants on the investing side.

Exhibit 1: Categories and Participants with Key Characteristics

Categories	Participants	Key Characteristics
Investors	Retail investors	Individual investors with the least amount of investible assets
Investors	High-net-worth investors	Individual investors with a higher amount of investible assets
Investors	Institutional investors	Organizations that invest to advance their mission or on behalf of their clients
Investment management service providers	Asset managers	Professionals who help their clients carry out investments to achieve their investment goals
Investment information service providers	Buy-side analysts	Professionals who collect and analyze information about companies and markets to inform the process of buying securities, typically working for institutional investors
Investment information service providers	Sell-side analysts	Professionals who collect and analyze information about companies and their competitors to report and share with potential investors
Trading service providers	Brokers	Professionals and their firms that facilitate trading between investors, acting as agents (they do not trade with their clients)
Custodial service providers	Custodians	Organizations that hold money and securities on behalf of their clients

Module 1 Glossary

In this module, you learned the following terms (in order of appearance):

Lesson 1: What Is the Financial Services Industry?

- **Financial services industry:** Industry that offers a range of products and services to those who have money to invest and those who need money and helps channel funds between them.
- **Financial capital:** Money provided to individuals, companies, and governments to finance their needs.
- **Assets**: Items that have value and include real assets and financial assets.
- **Real assets:** Physical assets, such as land, buildings, machinery, cattle, and gold.
- **Financial assets**: Claims on other assets; for example, a share of stock represents ownership in a company or a claim to some of the company's assets and earnings.
- **Physical capital:** The means of production; tangible goods such as equipment, tools, and buildings.
- **Portfolio:** The group of assets in which savings are invested.
- **Securities:** Financial assets that can be traded, such as shares and bonds.
- Financial markets: Places where buyers and sellers can trade securities; also called securities markets or capital markets.
- Direct finance: When providers and users of capital interact, the providers usually have a direct claim on the users (e.g., right to certain assets and earnings generated by the user). See also direct investments.
- Financial intermediaries: Organizations that go between those who have money and those who need money.
- Indirect finance: The use of financial intermediaries to find and channel funds between providers and users of capital. Because financial intermediaries sit between providers and users, the former do not have direct claims on the latter. See also indirect investments.

Lesson 2: How Economies Benefit from the Existence of the Investment Industry

- Investment industry: Subset of the financial services industry that comprises all the participants that are instrumental in helping savers invest their money and spenders raise capital in financial markets.
- Capitalism: An economic system that promotes private ownership as the means of production and markets as the means of allocating scarce resources.

Lesson 3: How Investors Benefit from the Existence of the Investment Industry

- No new terms were introduced in this lesson.

Lesson 4: Investment Industry Participants

- Investment banks: Financial intermediaries that assist companies and governments in raising capital and can provide other services, such as strategic advisory services, brokerage and dealing services, and research services; also known as merchant banks.
- Sell-side analysts: Analysts who work for the organizations, typically an investment bank, that sell securities. They collect and analyze information about a company and its competitors and then prepare a report that is shared with potential investors.
- Retail investors: Individual investors with the least amount of investible assets.
- High-net-worth investors: Individual investors with a higher amount of investible assets relative to retail investors.
- Pension plans: Institutional investors who hold investment portfolios for the benefit of future and current retirees.
- Endowment funds: Long-term funds owned by not-for-profit institutions.
- Foundations: Grant-making institutions funded by gifts and by the investment income that they produce.
- Sovereign wealth funds: Funds created by governments to invest surpluses for the benefit of current and future generations of their citizens.
- Buy-side analysts: Analysts who work for organizations, typically a type of institutional investor, that buy securities. They analyze data about companies and the markets in which they operate and review potential investments.

Up Next

Now that you have an overview of the investment industry, the next module will dive deeper into the structure of investment industry.

Module 2: Structure of the Investment Industry

Sell-side, buy-side, and front, middle, and back office are terms you may have come across without much clarity on what they mean. Although the distinctions between them are not absolute, these terms can be helpful to understand investment firms' relative positions, how their departments are categorized, and what roles are commonly held by their professionals. In Module 2, we will look at how banks, insurance companies, and the wider investment industry are structured.

By the end of this module, you will be able to:

- Distinguish between buy-side and sell-side firms and front-, middle-, and back-office functions in the investment industry
- Identify positions and responsibilities within firms in the investment industry
- Identify types of financial institutions, including banks and insurance companies

Let's get started!

Lesson 1: Structure of Investment Firms

Welcome to Lesson 1 of Module 2!

In this lesson, you will learn about how firms in the investment industry are classified according to the types of services that they provide. You will also learn about the responsibilities of portfolio managers.

By the end of this lesson, you will be able to distinguish between buy-side and sell-side firms and front-, middle-, and back-office functions in the investment industry.

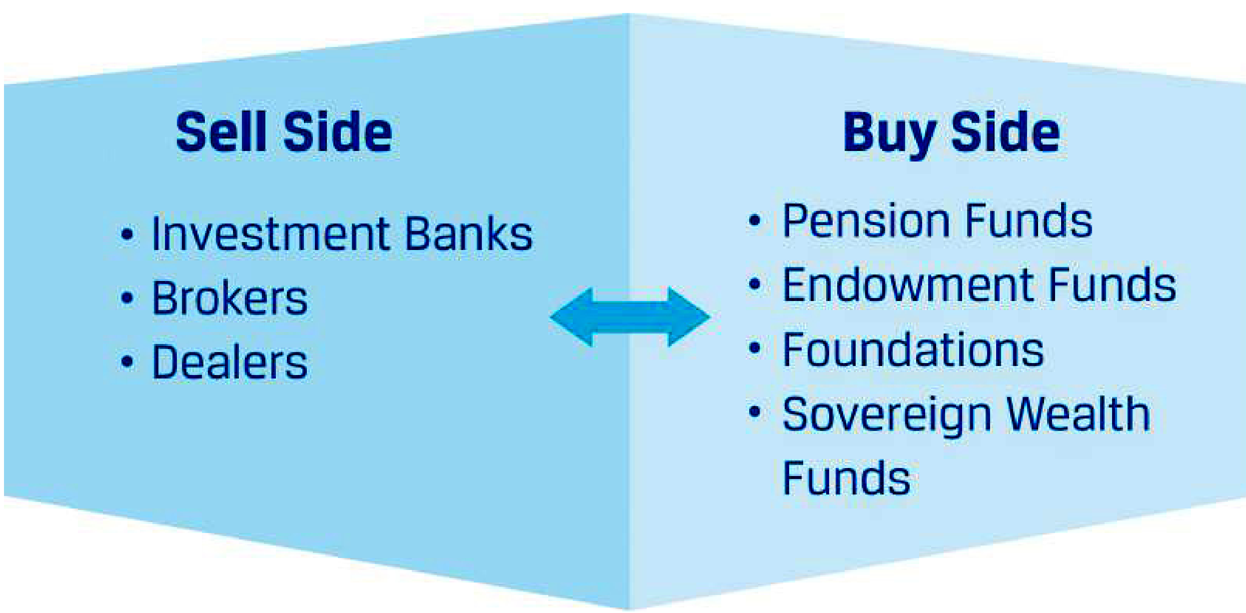
Let's get started!

Buy-Side and Sell-Side Firms

Practitioners classify many firms in the investment industry by whether they are on the sell side or the buy side.

Sell-side firms primarily provide investment products and services. They are typically investment banks, brokers, and dealers. **Buy-side firms** include firms that manage portfolios for clients, themselves, or both. This division of firms is shown in Exhibit 1.

Exhibit 1: Sell Side and Buy Side



Buy-side participants purchase investment products and services from sell-side firms. Institutional investors, such as pension plans, endowment funds, foundations, sovereign wealth funds, and insurance companies, are all considered to be on the buy side. Practitioners sometimes apply the buy-side label to consultants who provide services only to buy-side firms. For example, many buy-side consultants help buy-side institutional investors evaluate investment performance.

The buy-side and sell-side classifications do not apply to all firms in the investment industry. They are not relevant for firms providing investment information services, such as investment research providers, data vendors, or credit rating agencies.

The classifications are somewhat arbitrary and are not easily applied to many large, integrated firms. Many investment banks have divisions or wholly owned subsidiaries that provide investment management, which is a buy-side activity, although investment banks are sell-side firms.

Front, Middle, and Back Offices

Most sell-side firms organize their activities along similar lines. Activities are classified by whether they are in the **front office**, the **middle office**, or the **back office**.

Front Office

The front office consists of client-facing activities that provide direct revenue generation. Sales, marketing, and customer service are the most important front-office activities.

Some practitioners consider trading to be a front-office activity, especially if the traders regularly interact with clients. Some consider research to be a front-office activity because it generates revenue from clients.

Middle Office

The middle office includes the core activities of the firm. Risk management, information technology (IT), corporate finance, portfolio management, and research are generally considered to be middle-office activities, especially if these departments do not interact directly with clients.

IT is particularly important because most firms in the investment industry need to process and retrieve vast quantities of data efficiently and accurately. Risk management is also critical because it seeks to ensure that the firm and its clients are not intentionally, inadvertently, or fraudulently exposed to excessive risk.

Back Office

The back office houses the administrative and support functions necessary to run the firm. These functions include accounting, human resources, payroll, and operations.

For brokerage firms and banks that provide custodial services, the accounting department is especially important because it is responsible for clearing and settling trades and for keeping track of who owns what.

See Exhibit 2 for more information about the activities conducted in the front, middle, and back offices.

Exhibit 2: Front, Middle, and Back Office

Some activities are not easily classified as belonging to the front, middle, or back office. Compliance, for example, is relevant to the entire organization because this department ensures that the firm and its clients comply with the laws and regulations governing the investment industry.

The terms front office, middle office, and back office are not generally used when describing buy-side firms. But the main departments of buy-side investment management firms are similar to those of sell-side firms. These departments include sales and client relations, investment research and portfolio management, trading, compliance, accounting, and administration.

A Closer Look at Portfolio Management Activities

Jenna Egloff works as a portfolio manager at Primrose, a large bank. She manages the private wealth of 30 individual clients who have entrusted her and Primrose with full investment discretion over their assets.

As an investment professional directly responsible for managing client assets and for providing investment advice, Jenna must act in accordance with her client's best interests. The responsibilities of a portfolio manager can vary based on institutional objectives, client base, and geography. Professionals who perform these types of tasks can also be referred to as **investment managers**, wealth managers, or advisers.

Front-Office Activities

As a portfolio manager, Jenna's front-office responsibilities encompass a variety of activities:

- Jenna meets with her clients either quarterly or semiannually to:
    - check for any changes in their financial goals or circumstances,
    - review past performance, and
    - explain the bank's economic outlook and current investment strategies.
- Several of Jenna's clients have close relationships with their tax accountants. With her clients' permission, Jenna meets once or twice a year with the accountants to discuss appropriate tax strategies.
- Jenna participates in presentations made to prospective clients.
- Jenna is periodically asked by local media to speak on investment-related topics.

The following information illustrates Jenna's activities in the middle office and how she acts as an intermediary between her clients and the middle office.

> At client meetings, Jenna leads discussions on portfolio risk measurement, client risk tolerance, and expected returns. When appropriate, Jenna introduces software that simulates expected returns under a variety of economic and financial scenarios.
>
> Jenna discusses basic budgeting needs and financial planning strategies with clients. In complex situations, she may involve Primrose's full-time financial planner in the discussions.
>
> Taking into consideration her clients' risk tolerance, their financial needs, their goals, and any portfolio constraints (including taxes), Jenna constructs individual

portfolios that are consistent with each client's circumstances and Primrose's strategies for asset allocation and security selection.

Jenna is authorized to trade in her clients' accounts. Her orders go to Primrose's in-house trading desk where traders communicate with a wide range of institutional brokers. Primrose's traders ultimately select the broker they believe is best suited to match Jenna's trade with a counterparty.

Jenna sometimes interacts directly with brokers, Primrose's analysts, or other service providers to obtain investment reports, economic forecasts, access to useful databases, assistance with complex modelling, or performance measurement support.

Back-Office Activities

Jenna does not routinely interact with the back office, but her client-facing position means she gets involved when a client is dissatisfied with a back- or middle-office function. Problems that arise may be simple, such as a misspelled family name, but they can also be complex and time-consuming.

Jenna Egloff's day-to-day activities at Primrose are just one example of how a portfolio manager may interact with the front, middle, and back offices. Depending on the size of the institution, the role can have different responsibilities and interface with different departments.

Lesson 1 Knowledge Check

The following four knowledge check questions are designed to reinforce your understanding of the concepts taught in this lesson. They are ungraded, but you are strongly encouraged to complete them.

Question 1

Which of the following are considered sell-side firms? Select all that apply.

a. Front office brokers
b. Investment banks
c. Sovereign funds

Question 2

Categorize the following activity as taking place in the front office, middle office, or back office:

The account manager discusses a tax issue with a client's tax accountant (who works for another firm).

A. Front office
B. Middle office
C. Back office

Question 3

Categorize the following activity as taking place in the front office, middle office, or back office:

The firm's risk manager prepares a quarterly assessment of the client's exposure to risk factors.

A. Front office
B. Middle office
C. Back office

Question 4

Categorize the following activity as taking place in the front office, middle office, or back office:

Thanks to efficient administrative and operational controls, the firm experiences very few losses due to errors.

A. Front office
B. Middle office
C. Back office

Lesson 1 Knowledge Check Feedback

Question 1

a. & b. **Correct.** Sovereign wealth funds are institutional investors who typically purchase investment products and services from sell-side firms and are thus an example of buy-side firms. Brokers and investment banks primarily provide investment products and services and are examples of sell-side firms.

Question 2

a. **Correct.** Given that the tax accountant represents the client’s interests, this would be considered a client-facing activity. It is also a marketing opportunity to demonstrate the firm’s culture and capabilities to a potential source of business referrals.

Question 3

a. **Incorrect.** Internal risk management is usually invisible to the client, so it would not be in the front office, which deals with client-facing activities.

b. **Correct**. Like other middle- and back-office activities, internal risk management is usually invisible to the client but critical to the firm’s successful operation.

c. **Incorrect.** Internal risk management is usually invisible to the client but part of the firm’s core activities that are in the middle office.

Question 4

a. **Incorrect.** As with middle-office activities, back-office services are often invisible to clients and thus not considered front office.

b. **Incorrect.** Administration and operational controls are housed in the back office.

c. **Correct.** As with middle-office activities, back-office services are often invisible to clients but critical to the firm’s success.

Up Next

You have completed Lesson 1. We hope that this lesson has helped you understand how firms in the investment industry are classified according to the types of services that they provide.

You should now be able to distinguish between buy-side and sell-side firms and front-, middle-, and back-office functions in the investment industry.

In the next lesson, you will explore investment staff and leadership.

Lesson 2: Investment Firm Staff and Leadership

Welcome to Lesson 2 of Module 2!

In this lesson, you will explore the roles and responsibilities of staff and leaders in investment firms.

By the end of this lesson, you will be able to identify positions and responsibilities within firms in the investment industry.

Let's get started!

Leadership Titles and Responsibilities

Exhibit 1 lists the investment industry's major leadership titles and responsibilities. The actual titles used by different firms may vary.

Exhibit 1: Leadership Title and Its Responsibilities

Title	Responsibility
Chief Executive Officer (CEO)	Manages the firm
Chief Financial Officer (CFO)	Responsible for arranging financing for the firm and for financial reporting
Chief Operating Officer (COO)	Responsible for the day-to-day management of the firm
Chief Investment Officer (CIO)	Responsible for any investment advice that the firm provides to its clients and for the investment decisions that the firm makes for itself and on behalf of its clients
Head Trader	Responsible for all trading operations; at firms that engage in proprietary trading, the head trader is responsible for all positions, risks, and profits
Chief Accountant (also known as the Finance Controller)	Responsible for the accounting and financial systems

Title	Responsibility
Treasurer	Responsible for cash management, including the investment of receipts and payment of invoices
Chief Risk Officer (CRO)	Responsible for identifying and managing the risks to which the firm and its clients are exposed
Chief Compliance Officer (CCO)	Responsible for ensuring that the firm complies with all relevant laws and regulations and handles clients appropriately
Chief Audit Executive	Leads the internal audit department, which is responsible for providing independent assessments of the firm's operational systems as well as suggestions for improvement
General Counsel	Leads the legal department, which reviews and helps write contracts, responds to or initiates lawsuits, and interprets regulations

At many firms, especially smaller ones, some people hold multiple titles and responsibilities. For example, the chief investment officer of a smaller investment management firm may also be the chief executive officer.

Investment Staff

Firms in the investment industry employ many types of investment professionals, including the examples listed below:

- Portfolio managers at buy-side firms make investment decisions for one or more portfolios.
- Buy-side, sell-side, and independent research analysts produce the investment research that portfolio managers use to make decisions.
- Research assistants help research analysts with the collection and analysis of investment information.
- Buy-side traders interact with sell-side firms to trade orders created by their portfolio managers.
- Sales traders at sell-side firms help arrange trades for their buy-side clients.
- Sales managers handle sales for regions, products, or customer types.
- Salespeople identify potential clients and sell them the firm's products and services.
- Sales assistants provide administrative support to the salespeople.

- Client service agents and their assistants answer client questions and help clients open, close, and manage their accounts.

Investment professionals who interact with clients are known as account executives and account managers at many firms. The role of a research assistant is often the entry-level position for investment professionals interested in becoming portfolio managers. Assistants who acquire strong expertise in a particular area, have good quantitative skills, and write well may be promoted to research analysts. Those analysts who demonstrate excellent investment judgment often become portfolio managers. Likewise, sales assistants and account services assistants are entry-level positions for investment professionals interested in sales or account services.

Companies that provide investment management services also employ professionals with expertise in accounting, information services, marketing, and legal services.

Lesson 2 Knowledge Check

The following knowledge check is designed to reinforce your understanding of the concepts taught in this lesson. It is ungraded, but you are strongly encouraged to complete it.

Question 1

To fulfill orders given to them by portfolio managers, buy-side traders work with sell-side traders to arrange and execute trades.

a. True
b. False

Lesson 2 Knowledge Check Feedback

a. **Correct.** As an example, consider a portfolio manager in the investment division of Altgeld Bank, a buy-side institution, that is selling 100 shares of Faststart Inc. The order is given to Altgeld's trading desk, which selects a brokerage house, a sell-side institution, and contacts their trading desk. The sell-side traders find a buyer and execute the trade, often using an automated process.

b. **Incorrect.** This statement is true. As an example, consider a portfolio manager in the investment division of Altgeld Bank, a buy-side institution, that is selling 100 shares of Faststart Inc. The order is given to Altgeld's trading desk, which selects a brokerage house, a sell-side institution, and contacts their trading desk. The sell-side traders find a buyer and execute the trade, often using an automated process.

Up Next

You have completed Lesson 2. We hope that this lesson has helped you to expand your knowledge of roles and responsibilities of staff and leaders in investment firms.

You should now be able to identify positions and responsibilities within firms in the investment industry.

In the next lesson, you will explore financial institutions.

Lesson 3: Types of Financial Institutions

Welcome to Lesson 3 of Module 2!

In this lesson, you will explore the different types of financial institutions and how they serve their clients and customers.

By the end of this lesson, you will be able to identify types of financial institutions, including banks and insurance companies.

Let’s get started!

Financial Institutions

Financial institutions are types of financial intermediaries. Their role is to collect money from savers and invest it in financial assets. The two major types of financial institutions are **banks** and **insurance companies**.

Banks

Banks collect deposits from savers and transform them into loans to borrowers. In doing so, they indirectly connect savers with borrowers. The saver does not have a direct claim on the borrower but rather has a claim on the bank through its deposit, the bank’s liability, and the bank has a claim on the borrower through the loan, the bank’s asset.

Banks are also called **deposit-taking institutions** or depository institutions. In exchange for using the depositors’ money, banks offer transaction services, such as check writing and check cashing, and they may pay interest on deposits. Banks may also raise money to make loans by issuing and selling bonds or stocks.

Banks vary in whom they serve and how they are organized. They may have different names in different countries.

Savings and Loan Associations

Building societies, also called savings and loan associations in some countries, specialize in financing long-term residential mortgages.

Retail Banks

Retail banks provide banking products and services to individuals and small businesses. These products and services include current accounts (also known as checking accounts), savings accounts, debit and credit cards, mortgages, and personal loans.

An increasing number of retail banking transactions are now performed either electronically via automated teller machines (ATMs) or online. Commercial banks provide a wide range of products and services to companies and other financial institutions.

Cooperative and Mutual Banks

Cooperative and mutual banks are financial institutions that their members own and sometimes run. They may specialize in providing mortgages and loans to their members, and some offer a range of products and services similar to commercial banks. Depositors benefit because they earn a return in interest (from dividends), through transaction services, or from appreciation on their capital without having to locate the borrowers, check their credit, contract with them, or manage their loans.

If borrowers default, banks must still pay their depositors. If the banks cannot collect sufficient money from their borrowers, the banks will have to use their owners' capital to pay their debts.

In many countries, depositors benefit from government-guaranteed deposit insurance. This insurance gives depositors comfort that their savings are not at risk, although the amount that is guaranteed is usually capped. The risk of losing capital should focus the banks' attention so that they do not offer credit in a careless manner that increases the risk of not being paid back.

Notable lapses occasionally occur: In the run-up to the financial crisis of 2008, investors and regulators were frequently unaware of the risks that banks were taking, they ignored them, or the risks were beyond their control.

In 1982, Penn Square Bank, a small Oklahoma bank located in a shopping mall, was engaged in excessive risk-taking with borrowed funds.

Exhibit 1 demonstrates the consequences of excessive risk-taking with borrowed funds.

Exhibit 1: Circumstances

Circumstance	Description
Aggressive Growth	In the preceding years of high oil prices, Penn Square Bank had grown aggressively by making high-risk loans to companies in the oil and gas sector. It had leveraged its lending (i.e., used borrowed capital) by selling “participation” in these loans to other banks across the country.
Filing for Bankruptcy	Oil prices peaked in 1981 and then began to decline. By July 1982, Penn Square Bank declared bankruptcy, leaving uninsured depositors and banks that had participated in its loans with sometimes devastating losses.
Cascading Effect	The subsequent collapse in 1984 of Continental Illinois, once a top-tier US commercial bank, was attributed in part to losses stemming from Penn Square Bank’s failure.
Disregarding the Risks	Penn Square Bank’s aggressive lending program would not have been possible without funding from depositors and financial institutions who, in the quest for high returns, chose not to look closely at the associated risks.

Insurance Companies

Insurance companies offset risks. To protect themselves against a potential loss, individuals and companies buy insurance contracts, also known as policies, that provide payments in the event that losses occur. They typically pay insurance companies non-refundable payments, called premiums, when they purchase insurance contracts. If the insured risks materialize, the insured individual or company makes a claim to the insurance company and collects the insurance settlement.

Property and casualty insurers cover assets, such as homes, cars, and businesses, whereas legal liability and life insurers pay out sums of money upon the death or serious injury of the insured person. The business of insurance, also called insurance underwriting, is based on risk pooling and can be demonstrated with a simplified example of life insurance.

> Jon, who is 35 years old and married with two children, has determined that USD1 million would cover his family’s expenses and make up for his lost future income if he were to die unexpectedly.

Given that he does not have USD1 million, he cannot “self-insure,” so he buys life insurance. Insurance companies make it possible for Jon to essentially join a pool of 35-year-olds with similar health profiles to benefit from risk pooling.

With large pools of similar individuals, insurance companies can estimate the number of deaths that will occur in a given period with a high degree of confidence.

This ability allows the insurance company to insure the entire group while charging each insured person a reasonable rate, such that the premiums paid by all the members in the pool cover the payouts, also known as the settlements, for the small number of members who actually die. The premiums are also enough to cover the insurers’ administrative expenses and expected profits.

On top of the insurable risks they take on, insurance companies are exposed to price competition from other insurers, the effect of market volatility on their invested reserves, and other business risks, including the following examples

Fraud

Fraud occurs when people deliberately cause or falsely report losses in order to collect insurance settlements

Moral Hazard

Moral hazard occurs when people are less careful about avoiding losses once they have purchased insurance. Moral hazard leads to losses occurring more often with insurance than without.

Adverse Selection

Adverse selection occurs when only those who are most at risk buy insurance, causing insured losses to be greater than average losses.

Consider the following example:

Jon has always been a bit adventurous, but before buying the USD1 million life insurance policy, his hobbies were golf and tennis. Now that he is insured, Jon abandons those “safe” sports and indulges his long-standing interest in scuba diving and skydiving.

Jon’s wife expresses concern about this change in hobbies and attitude, to which he responds, “Don’t worry—we’re insured!” If other members of Jon’s insurance pool experienced similar changes in attitudes about engaging in risky hobbies, the insurance company would likely have to make higher policy payouts.

Based on the above list of insurable risks, which of these is Jon guilty of committing: fraud, moral hazard, or adverse selection?

Insurers are not only financial intermediaries but are also large institutional investors. They usually invest a significant portion of the premiums they receive, and they manage these investments to meet the cost of future claims.

Lesson 3 Knowledge Check

The following knowledge check is designed to reinforce your understanding of the concepts taught in this lesson. It is ungraded, but you are strongly encouraged to complete it.

Banks take deposits from their customers and then use the funds to make loans. From a bank's perspective, deposits represent:

a. debt obligations that the bank must repay.
b. a source of exposure to default risk.
c. proof of financial stability.

Lesson 3 Knowledge Check Feedback

a. **Correct.** The bank's customers are essentially making loans to the bank.

b. **Incorrect.** Customers' deposits are not a source of default risk for the bank. But depositors may be exposed to the risk of a bank default. (Deposit insurance often exists for depositors.)

c. **Incorrect.** Large deposits also imply large liabilities. Banks that lend imprudently may find that some loans will have to be written off and losses recognized.

Up Next

You have completed Lesson 3. We hope that this lesson has helped you to expand your knowledge of different types of financial institutions and how they serve their clients and customers.

You should now be able to identify types of financial institutions, including banks and insurance companies.

Module 2 Summary

To better understand the structure of the investment industry, in this module we explored where investment firms sit relative to each other, how their departments are categorized, and what roles are commonly held by their professionals. We first focused on buy-side versus sell-side firms and then looked at front-, middle-, and back-office functions.

- Sell-side firms, such as investment banks, brokers, and dealers, primarily provide investment products and services. Institutional investors, who are considered buy side, purchase these investment products and services from sell-side firms. Buy-side firms include firms that manage portfolios for clients, themselves, or both.
- Sell-side firms organize their activities into front-, middle-, and back-office functions. The front office consists of client-facing activities that provide direct revenue generation. The middle office includes the core activities of the firm, such as risk management, information technology (IT), corporate finance, portfolio management, and research. The back office houses the administrative and support functions necessary to run the firm, including accounting, human resources, payroll, and operations.
- The terms front, middle, and back office are not generally used when describing buy-side firms, but the main departments of buy-side investment management firms are similar to those of sell-side firms. These typically include such departments as sales and client relations, investment research, portfolio management, trading, compliance, accounting, and administration.

Next, we went a level deeper into the types of positions and associated responsibilities that exist in the investment industry, from leadership roles to investment staff.

- Typical titles that exist within a firm's leadership include, but are not limited to, Chief Executive Officer (CEO), Chief Financial Officer (CFO), Chief Investment Officer (CIO), Chief Operating Officer (COO), Chief Compliance Officer (CCO), Chief Risk Officer (CRO), Chief Audit Executive, Chief Accountant (also known as the Finance Controller), Head Trader, and General Counsel. Depending on the firm's size, some people may hold multiple titles and responsibilities.
- Firms in the investment industry employ many types of investment professionals, including portfolio managers, research analysts (buy-side, sell-side, and independent firms), research assistants, traders (buy side and sell side), sales managers, salespeople, sales assistants, client service agents, and account executives or account managers.

Lastly, we looked at two main financial institutions, banks and insurance companies, to understand the role they play in the investment industry.

- Banks collect deposits from savers and transform them into loans to borrowers, which is also why they are referred to as deposit-taking or depository institutions. They may also raise money to make loans by issuing and selling bonds or stocks. Banks vary in whom they serve and how they are organized, and they may have different names in different countries.
- Insurance companies are not only financial intermediaries that connect buyers of insurance contracts with providers of capital who are willing to bear the insured risks, but they are also among the largest investors.

Up Next

Now that you understand the structure of the investment industry, in the next module, you will explore the variety of services offered in the investment industry and how they fit within the larger financial services industry.

Module 2 Glossary

In this module, you learned the following terms (in order of appearance):

Lesson 1: Structure of Investment Firms

- Sell-side firms: Typically, investment banks, brokers, and dealers that provide investment products and services.
- Buy-side firms: Institutional investors and investment managers who purchase investment products and services from sell-side firms.
- Front office: Client-facing activities that provide direct revenue generation, such as sales, marketing, and customer service activities.
- Middle office: Core activities of a firm, such as risk management, information technology, corporate finance, portfolio management, and research.
- Back office: Administrative and support functions necessary to run the firm, including accounting, human resources, payroll, and operations.
- Investment managers: Investment professionals who receive authority from their clients to trade securities and assets on their behalf. Also called asset managers.

Lesson 2: Investment Firm Staff and Leadership

- No new terms were introduced in this lesson.

Lesson 3: Types of Financial Institutions

- Financial institutions: Financial intermediaries, such as banks and insurance companies, whose role is to collect money from savers and invest it in financial assets.
- Banks: Financial institutions that collect deposits from savers and transform them into loans to borrowers.
- Insurance companies: Financial institutions that help individuals and companies offset the risks they face; also are among the largest investors.
- Deposit-taking institutions: Financial institutions that take deposits, such as banks; also called depository institutions.
- Fraud: Intentional deception, such as deliberately causing or falsely reporting losses to collect insurance settlements.
- Moral hazard: A person's tendency to be less careful about avoiding losses after purchasing insurance, potentially leading to more losses when they are insured than when they are not.
- Adverse selection: The tendency of people who are most at risk to buy insurance, causing insured losses to be greater than average losses.

Module 3: Introduction to Investment Services

In this module, you will be introduced to types of investment management, investment information, and trading services as well as how the investment industry fits within the larger financial services industry.

By the end of this module, you will be able to:

- Describe investment management services
- Describe investment information services
- Describe trading services
- Compare the roles of brokers and dealers
- Compare the roles of clearinghouses and custodians

Let's get started!

Lesson 1: Investment Management Services

Welcome to Lesson 1 of Module 3!

In this lesson, you will learn about the different investment management services offered by investment managers.

By the end of this lesson, you will be able to describe investment management services.

Let's get started!

Investment Management Strategy

Once they have determined the relevant financial goals and constraints, individual and institutional investors must develop and implement a strategy for achieving their goals. Investment management services are frequently required, including asset allocation, investment analysis, and portfolio construction. The range of the services investors have access to depends largely on the amount of investable assets they have.

Services for High-Net-Worth and Institutional Clients

Some high-net-worth and institutional clients rely on investment professionals to take care of the entire investment process, from identifying potential investments to implementing the investments and evaluating their performance. Others use the services of investment professionals selectively. Many investment professionals receive authority from their clients to trade securities and assets on the client's behalf.

Those who have such discretionary authority are often called investment managers or asset managers. Depending on the context, these terms may refer to the individuals who make the investment decisions or to the investment firms for which they work.

Investment Management Activities

There are three major services that investment managers provide to clients.

- **Asset allocation:** Asset allocation indicates the proportion of a portfolio that should be invested in available asset classes and why. Asset classes include cash, equity and debt securities, and alternative investments, such as private equity, real estate, and commodities. The optimal allocation of equity and debt may include foreign and domestic securities. Investment managers assess the risk and return of various asset classes to determine the appropriate allocation.

- **Investment analysis:** Investment analysis involves determining the value of investments and identifying attractive securities and assets. An investment's fundamental value, also called its intrinsic value, can be the price that investors would pay for the investment if they had a complete understanding of its characteristics. A common approach to estimating the fundamental value is to calculate the value of all the cash flows—such as dividends—that the investment will generate in the future. (Present value and securities valuation will be covered in Course 3, Investment Instruments, and Course 4, Investment Inputs and Tools.) Provided that it fits the client's needs, an investment is financially appealing when its price is below its estimated fundamental value.
- **Portfolio construction:** Portfolio construction brings everything together. Investment managers invest in the attractive securities and assets that they identified through their investment analysis while taking into account the client's requirements and appropriate asset allocation. To do so, they must trade securities and assets; hold, manage, and account for these securities and assets during the periods of investment; and evaluate the performance of these investments.

The investment managers who assist high-net-worth and institutional investors typically work for investment firms. Clients pay management fees. The size of these fees typically depends on the total assets under management—bigger investments usually carry bigger fees. Clients may also pay performance fees that depend on how well the portfolio has done.

The Portfolio Construction Process

Please return to the online course to view a video on this process.

Passive and Active Investment Management

Investment managers typically take either a passive or an active approach to managing clients' investments. Select the following accordion tabs to learn more about these two approaches.

Passive Investment Managers

Passive investment managers seek to match the return and risk of an appropriate benchmark. Benchmarks include broad market indices that cover an entire asset class, indices for a specific industry, and benchmarks that are customized to the needs of a specific client.

Passive investing involves low-cost strategies because it entails buying and holding securities based only on their characteristics relative to a given index—such as the S&P 500 in the United States, FTSE 100 in Europe, or S&P Asia 50 in Asia—rather than on an analysis of their future return prospects. Index investing is a common passive investment strategy and is discussed further in Course 2, Types and Functioning of Markets.

Active Investment Managers

Active investment managers try to predict which securities and assets will outperform or underperform comparable securities and assets. The managers then act on their opinions by buying the securities and assets that they expect to outperform and selling (or simply not buying) the securities and assets that they expect to underperform.

Active investment strategies are more expensive than passive investment strategies because they require greater resources. So, clients hire active managers only when they believe that these managers have the skills needed to outperform the market after considering all the fees and commissions.

Active investment managers analyze as much relevant information as they can to predict which securities and assets will outperform or underperform their peers in the future. They often need the help of investment information service providers to gather the required data.

Services for Retail Clients

Retail investors do not typically require the full range of services used by high-net-worth and institutional investors. Retail clients frequently obtain assistance and advice on investment management—including asset allocation, investment analysis, and portfolio construction—from financial planners or brokers. Brokers are discussed in greater detail later in this module.

Unlike brokers and agents, who are paid commissions on the trades and contracts they recommend, fee-only professionals do not have incentives to generate commissions by recommending specific products or excessive trades. Alternatively, retail clients could implement their investment plans with a passive investment in pooled investment vehicles, such as mutual funds, that are professionally managed. Types and characteristics of pooled investment vehicles are discussed in Course 2, Types and Functioning of Markets.

Lesson 1 Knowledge Check

The following knowledge check is designed to reinforce your understanding of the concepts taught in this lesson. It is ungraded, but you are strongly encouraged to complete it.

Question 1

Match the active and passive investment managers with their investment objective.

Investment Managers

a. Passive managers
b. Active managers

Investment Objective

a. Seek to outperform a benchmark by analyzing the securities it tracks and investing in them selectively.
b. Seek simply to match the return of a benchmark.

Question 2

After analyzing the risk and return characteristics of appropriate asset classes, investment managers construct client portfolios that reflect the relative attractiveness of current pricing and:

a. provide maximum income.
b. correctly represent client goals and constraints.
c. offer downside protection.

Lesson 1 Knowledge Check Feedback

Question 1

Passive managers seek simply to match the return of a benchmark.

Active managers seek to outperform a benchmark by analyzing the securities it tracks and investing in them selectively.

Question 2

a. **Incorrect.** Income needs and goals are client specific. Generating maximum income will not be appropriate in every case. When investment goals focus on long-term objectives, maximizing income may come at the expense of reduced growth expectations.

b. **Correct.** Client goals and constraints determine how the analysis of asset classes and securities will be applied to construct the investment portfolio. Considerations include tax implications, risk appetite, income goals, funding for expenses, and planned gifting strategies.

c. **Incorrect.** Downside protection describes limiting the potential for losses and usually involves a trade-off or opportunity cost. The appropriateness of downside protection will vary with market conditions and client-specific circumstances.

Up Next

You have now completed Lesson 1! We hope that this lesson has helped you to expand your knowledge of the different investment management services offered by investment managers.

You should now be able to describe investment management services.

In the next lesson, you will explore investment information services.

Lesson 2: Investment Information Services

Welcome to Lesson 2 of Module 3!

In this lesson, you will learn about the types of information services available to investors and the types of providers that offer these services.

By the end of this lesson, you will be able to describe investment information services.

Investment Information Services

The investment research, financial data, and consultancy that investors need comes from investment research providers, credit rating agencies, financial news services, financial data vendors, and investment consultants. This section introduces these firms and discusses their business models.

Investment Research Providers

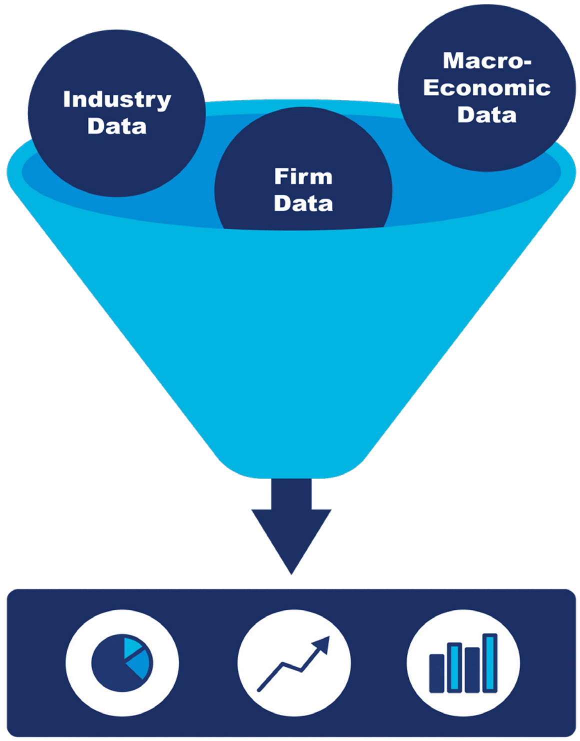
Many investors use research reports that dive deeper into the risk and return prospects of investments. Firms that provide research reports assemble information and opinions that most investors cannot easily produce themselves. To produce the reports, these firms employ data collectors, financial reporters, and expert analysts.

Research can be particularly valuable when it is written by industry experts who understand the financial implications of new technologies or industry regulation—for example, the regulatory process a biotech company navigates to get its new drug approved and what that means for when the drug will go to market.

Investment Research Data Inputs

Research reports are largely based on publicly available information, and they summarize text and data from lengthy disclosures, such as financial statements and regulatory filings, saving investors considerable time as show in Exhibit 1. Many reports estimate the fundamental value of securities.

Exhibit 1: Investment Data Research Inputs



Investors often get research reports from their brokers, who purchase them or use their own research departments to produce them. Brokers publish research to serve their clients, attract new clients, and encourage their clients to trade. Investors can also purchase reports from independent research firms, or the reports may be issued by research firms that were paid to produce reports by the companies.

Credit Rating Agencies

Credit rating agencies specialize in providing opinions about the credit quality of bonds and the companies or governments that issue them. A high bond credit rating indicates that the credit rating agency believes that the bond issuer has a high probability of making all future payments of principal and interest when they are due.

Most credit rating agencies do not charge investors for their ratings, although they may charge them for the detailed reports on which the ratings are based. Instead, companies pay credit rating agencies to rate their securities. They do so because having a rating generally makes a security more marketable.

Thus, an obvious conflict of interest arises because companies are likely to direct their business to those credit rating agencies that will provide higher ratings. Equally, the credit rating agencies may give companies high ratings to secure future business. But if they lose their independence, credit rating agencies run the risk that investors no longer respect their ratings. Such a situation would have a negative effect not only on credit rating agencies but also on the general economy—particularly the investment industry because flows of capital would be reduced.

Data Vendors

To invest and trade successfully, most investors need current, accurate data about companies and market conditions. Many data vendors provide such data, including historical and real-time data.

Following are some examples of historical data that may be of interest to investors and how investors may use the data to make decisions.

Examples and Potential Uses of Historical Data

Macroeconomic data: Investment professionals use macroeconomic data to better understand the environment in which companies operate and compete.
Example: Information about economic activity and international trade

Corporate accounting data: Investment professionals use corporate accounting data to assess a company's financial performance and to estimate the fundamental value of its securities, such as common shares.
Example: Information about a company's financial statements, including the balance sheet, income statement, and cash flow statement

Historical market data: Investment professionals use historical market data to evaluate the performance of their investments and to identify securities that may outperform in the future.
Example: Information about past market prices and trading volumes

News feeds and market data feeds are important real-time data resources used by investment professionals. News feeds provide news about companies and markets that investors need to know because such news may affect the value of the companies' securities. Market data feeds provide information on market quotes, the orders investors submit for securities, and the prices and volumes of recent trades—all of which are helpful for investors who want to trade.

Access to investment data was once very expensive and thus restricted to investment firms and institutional investors. The growth of information technologies, particularly those involving the internet, has substantially reduced the cost of accessing data, making more investment data now available to the general public. In many countries, some data, such as regulatory disclosures by issuers, can be freely accessed online. Other data are only available on a subscription basis from data vendors.

The widespread availability of investment data has greatly changed the investment industry landscape. Although access to data used to be a key driver of investment profits, investment profits now increasingly depend on the ability to analyze data.

Lesson 2 Knowledge Check

The following knowledge check is designed to reinforce your understanding of the concepts taught in this lesson. It is ungraded, but you are strongly encouraged to complete it.

Which of the following research reports have no obvious conflicts of interest?

a. Reports written by internal brokerage house analysts and provided at no charge to brokerage customers
b. Research reports sold directly to investors by independent research firms
c. Credit reports written by credit rating agencies that charge the company being rated for the research and rating

Lesson 2 Knowledge Check Feedback

b. **Correct**. Well done! There are no obvious conflicts of interest for independent research firms. They are economically motivated to deliver quality reports that clients are willing to pay for.

a. **Incorrect**. Brokers are exposed to multiple sources of potential conflicts of interest. A brokerage firm profits from trading activity, which may create subtle pressure on analysts to present their reports in ways that encourage trading. Brokerage firms may have divisions that generate significant fees by providing financial services to the same companies that the brokerage's research analysts are writing reports on. Senior management faces a conflict of interest if they know that a pending report from their research group will upset a company that generates fees for the firm, and they may be tempted to influence the tone of the report.

c. **Incorrect**. There is an inherent conflict of interest when the firm writing the credit report is hired and paid by the firm being rated. But credit rating firms have an economic interest in preserving their credibility and the perceived usefulness of their ratings.

Up Next

You have now completed Lesson 2. We hope that this lesson has helped you to expand your knowledge of investment information services.

You should now be able to describe investment information services.

In the next lesson, you will explore trading services and the types of professionals who provide those services.

Lesson 3: Trading Services

Welcome to Lesson 3 of Module 3!

In this lesson, you will learn about trading services, compare the roles of brokers and dealers, and compare the roles of clearinghouses and custodians.

By the end of this lesson, you will be able to:

- Describe trading services
- Compare the roles of brokers and dealers
- Compare the roles of clearinghouses and custodians

Let's get started!

Trading Services

There are various service providers that facilitate investment by holding assets for clients and by helping buyers and sellers of securities and investment assets arrange trades with each other, including the following:

- Brokers
- Dealers
- Clearinghouses
- Settlement agents
- Custodians
- Depositories

Exhibit 1: The Life of a Trade

Exhibit 1 illustrates the life of a trade, from transaction to custody services.

The following outlines the steps in further detail.

- Transaction

  Step 1: An investor places a sell order with Broker 1.

  Step 2: The sell order is matched with a buy order.

  Note: The broker may match the sell order with a buy order in-house. Small orders may go to an exchange for automated matching. For example, small trades in highly liquid securities are typically submitted to an exchange (such as the New York Stock Exchange) for automated matching with the best buy offer. Large trades in illiquid securities can affect the price. For example, large orders may have to be broken into smaller quantities and traded over time.

- Clearing and Settlement

  Step 3: The terms of the trade are confirmed and followed by a final exchange of cash and securities.

  Note: Time + 2 trade settlement takes place on the second day following a trade and is performed by a clearinghouse or a depository that also provides other services. In Europe, that could be Euroclear, and in the United States, it could be the Depository Trust Corp.

- Custody Services

  Step 4: Custodians interface between asset owners and their investments and provide safekeeping for securities, recordkeeping, and the distribution of dividends. Custody services may be provided by brokers or custodian banks. In the United States, Charles Schwab and Mellon Bank are providers.

Brokers

Brokerage services are provided to clients who want to buy and sell securities; they include not only execution services—that is, processing buy and sell orders on behalf of clients—but also investment advice and research.

Brokerage services are provided by brokerage firms or brokers. Brokers are agents who arrange trades for their clients. They do not trade with their clients but instead search for traders who are willing to take the other side of their clients' orders. Brokers help their clients by reducing the cost of finding counterparties for their clients' trades.

Brokers find sellers for their clients who want to buy and buyers for their clients who want to sell. For highly liquid securities, the search usually only involves directing a client's order to an exchange or to a dealer. Exchanges arrange trades by matching buy and sell orders and are discussed in Course 2, Types and Functioning of Markets. Dealers are discussed in the next section of this module. For less liquid securities and assets, brokers may spend substantial resources looking for suitable counterparties.

For complex trades, such as real estate transactions, for which effective negotiation is essential for a successful investment, brokers often serve as professional negotiators. In such transactions, skilled negotiators can increase the probability of arranging trades with favorable financial terms.

Clients pay commissions to their brokers for arranging their trades. The commissions vary widely but typically depend on the value or quantity traded. It is worth noting that commissions, particularly for stock trades, have decreased over the past 30 years, primarily because of deregulation, technological progress, and increased competition among brokers.

Brokers often ensure that their clients settle their trades. Such assurances are essential when exchanges arrange trades between strangers who do not have credit arrangements with each other. For such trades, brokers guarantee the settlement of their clients' trades.

Individual brokers may work, for example, at large brokerage firms or in the brokerage arms of investment banks or at exchanges. Some brokers match clients personally. Others use specialized computer systems to identify potential trades and help clients fill their orders. Many simply route their clients' orders to exchanges or to dealers.

Following is some insight into block brokers and prime brokerage.

Block brokers

Block brokers help investors who want to trade large blocks of securities. Large block trades are hard to arrange because finding a counterparty willing to buy or sell a large number of securities is often quite difficult. Investors frequently have to offer price concessions—an adjustment up or down to increase attractiveness—to encourage other investors to do a block trade with them. Buying a large quantity of securities often requires paying a premium on the market price, and selling a large quantity of shares often requires offering a discount on the current market price.

Prime brokerage

Prime brokerage refers to a bundle of services that brokers provide to some of their clients, usually investment professionals engaged in trading. In addition to the typical brokerage services mentioned, a prime broker helps these professionals finance their positions. Although the trades may be arranged by other brokers, prime brokers clear and settle them. Thus, prime brokerage allows for collateral requirements to be combined across all trades and thus lowers the costs of financing for the trader.

Dealers

Dealers make it possible for their clients to trade without having to wait to find a counterparty; they are ready to buy from clients who want to sell and ready to sell to clients who want to buy. In this way, they participate in their clients' trades, in contrast to brokers who do not trade with their clients but only arrange trades on their behalf.

Next is more information about the role and responsibilities of dealers.

Dealers profit when they can buy securities for less than they sell them—that is, when the price at which they buy securities (the bid price) is lower than the price at which they sell them (the ask price or offer price). If dealers can arrange trades simultaneously with buyers and sellers, they make risk-free profits. Dealers risk losses if prices fall after they purchase but before they can sell or if prices rise after they sell but before they can repurchase.

Dealers provide liquidity to their clients by allowing them to buy and sell when they want to trade. In effect, dealers match buyers and sellers who want to trade the same instrument at different times and are unable to trade directly with each other. In contrast, brokers must bring a buyer and a seller together to trade at the same time and place. Dealers are often called market makers because they are willing to make a market—that is, trade on demand—in specified securities at their bid and ask prices.

Dealers may organize their operations within investment banks, hedge funds, or sole proprietorships. Almost all investment banks have dealing operations ready to buy and sell currencies, bonds, stocks, and derivatives if no other counterparty can be found. Some dealers rely on people they hire to make trading decisions, whereas others primarily use computers.

Many dealers also broker orders, and many brokers also operate like dealers as they interact with their clients in a process called internalization. Internalization is when brokers fill their clients' orders by acting as proprietary traders rather than as agents, trading directly with their clients rather than arranging trades with others on behalf of their clients. Because the distinction between broker and dealer is not always clear, many practitioners often use the term "broker/dealer" to refer to them.

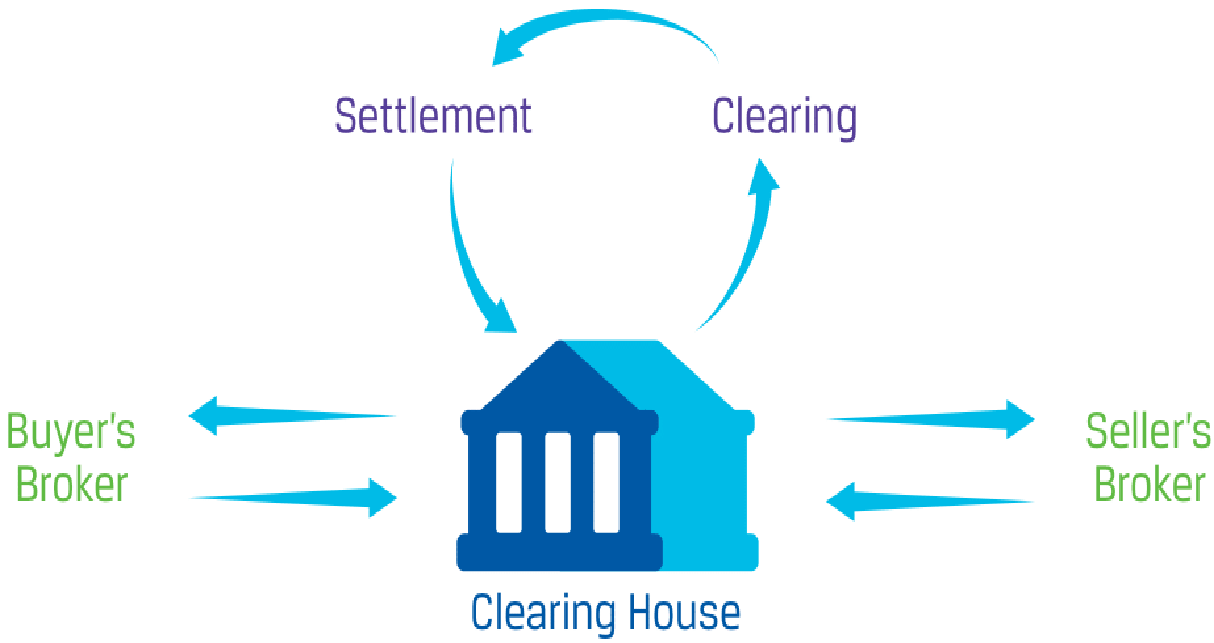
Broker/dealers face a conflict of interest with respect to how they fill their clients' orders. When acting as brokers, they must seek the best price for their clients' orders. When acting as dealers, however, they profit most when they sell to their clients at high prices or buy from their clients at low prices. This trading conflict of interest is most serious when clients allow their brokers to decide whether to trade their orders with other traders or to fill them internally. Consequently, when trading with broker/dealers, some clients may specify that they do not want their orders to be internalized. Or they may choose to trade only via brokers who do not also act as dealers.

Primary dealers are dealers that central banks trade with when conducting monetary policy. Monetary policy refers to central bank activities that aim to influence the money supply, interest rates, and credit availability in an economy. Central banks sell bonds to primary dealers to decrease the money supply. The primary dealers then sell the bonds to their clients. Central banks buy bonds from primary dealers to increase the money supply, with the primary dealers buying bonds from their clients and selling them back to the central banks. Monetary policy will be covered further in Course 4, Investment Inputs and Tools.

Clearinghouses

Clearinghouses and settlement agents settle trades after they have been arranged as show in Exhibit 2. Clearing refers to all activities that occur from the arrangement of the trade to its settlement. Settlement consists of the final exchange of cash for securities.

Exhibit 2: Clearinghouse



Clearinghouses arrange for the final settlement of trades. Members of a clearinghouse are the only traders for whom the clearinghouse will settle trades. Thus, brokers and dealers who are not members of the clearinghouse must arrange to have a clearing member settle their trades at the clearinghouse.

Trades fail when the exchange of cash and ownership does not take place on the scheduled settlement date. There can be understandable reasons for a failed trade, but corrections typically involve opportunity cost and lost time. Reliable trade settlement through a clearinghouse reduces settlement risk—the risk that counterparties will not settle their trades—for both buyers and sellers. A secure clearing system increases the number of counterparties available for a trader to safely arrange a trade.

Custodians and Depositories

The following provides definitions for custodians and depositories and outlines the roles and responsibilities of these investment participants.

> **Custodians:** Custodians are typically banks and brokerage firms that hold money and securities for safekeeping on behalf of their clients. They play an important role in reducing the risk of securities being lost or stolen. Security ownership records were once commonly held as actual paper certificates in secure vaults. Now, securities are almost exclusively held in electronic book-entry form as secure computer records, which greatly reduces the costs of clearing and settling

trades. Custodians may offer other services, including trade settlement and the collection of interest and dividends.

Depositories: Depositories act not only as custodians but also as monitors. They are often regulated, and their responsibilities are as follows:

- To prevent the loss of securities and payments through fraud, deficient oversight, or natural disaster
- To ensure securities cannot be pledged more than once by the same borrower as collateral for loans
- To ensure securities that are said to be purchased are actually purchased

Having reputable third-party custodians and depositories hold all assets managed by an investment manager helps prevent investment fraud—for example, Ponzi schemes, which use money contributed by new investors to pay purported returns to existing investors rather than to purchase additional securities.

Most individual investors and many smaller institutional investors hold securities in brokerage accounts that provide them with custodial services. Their brokers, in turn, hold the securities with custodians and depositories for safekeeping.

Comparison of Providers of Trading Services

The following information provides a recap of the roles and responsibilities of those that provide trading services:

Brokers

- Act as agents
- Find sellers for clients who want to buy and buyers for clients who want to sell
- Serve as professional negotiators
- Ensure clients will settle their trades

Dealers

- Participate in their clients' trades
- Allow clients to trade when they want by being ready to buy when their clients want to sell and to sell when their clients want to buy
- Provide liquidity because they are willing to trade on demand
- Are often proprietary traders

Clearinghouses

- Arrange for final settlement of trades
- Promote liquidity by reassuring investors that their trades will be settled

Settlement Agents

- Arrange final exchange of cash for securities

Custodians

- Hold money and securities for safekeeping on behalf of clients
- May offer other services for clients, such as trade settlement and collection of interest and dividends
- Depositories
- Act not only as custodians but also as monitors to prevent the loss of securities and fraud
- Often are regulated

Lesson 3 Knowledge Check

The following knowledge check is designed to reinforce your understanding of the concepts taught in this lesson. It is ungraded, but you are strongly encouraged to complete it.

Question 1

Which of the following services is not performed by a broker?

a. Finding counterparties for clients who want to buy or sell a security
b. Routing small trades in highly liquid securities to an exchange
c. Using a proprietary portfolio to transact directly with the client when it is in the client's best interest.

Question 2

Which of the following services are not provided by custodians and depositories?

a. Safekeeping and recordkeeping
b. Trade settlement
c. Dividend collection
d. Credit rating analysis

Lesson 3 Knowledge Check Feedback

Question 1

c. **Correct**. Well done! Dealers buy and sell from proprietary portfolios, providing market liquidity when counterparties may be otherwise hard to find. Dealers profit by buying at lower prices than they sell at. Dealers act as principals.

a. **Incorrect**. Brokers act as agents. They represent customers by identifying counterparties willing to take the other side of transactions. When a transaction is large or complex, the search may be time-consuming. Brokers profit by charging commissions.

b. **Incorrect**. Brokers often route smaller trades in highly liquid securities to an exchange in which they are automatically matched with a counterparty.

Question 2

d. **Correct**. Well done! Custodians and depositories do not provide credit ratings or investment analysis.

a. **Incorrect**. By providing reliable safekeeping and recordkeeping, custodians and depositories facilitate the efficient transfer of funds and records of ownership.

b. **Incorrect**. Trade settlement is a core function of centralized custodians and depositories.

c. **Incorrect**. Custodians and depositories process cash flows, such as quarterly dividends, for the securities they hold.

Module 3 Summary

You have now completed Lesson 3. We hope that this lesson has helped you understand trading services, the role of brokers and dealers, and the role of the clearinghouses and custodians.

You should now be able to describe trading services, compare the roles of brokers and dealers, and compare the roles of clearinghouses and custodians.

Once they have determined financial goals and constraints, individual and institutional investors must develop and implement a strategy for achieving their goals. Investment management services are frequently required. The range of these services that investors have access to depends largely on the amount of investable assets they have.

- Some high-net-worth and institutional investors rely on investment professionals to take care of the entire investment process, whereas other clients use the services of investment professionals selectively.
- Many investment professionals receive authority from their clients to trade securities and assets on their behalf. The three major services that investment managers provide to clients are asset allocation, investment analysis, and portfolio construction. Retail investors do not typically require the full range of services used by high-net-worth and institutional investors. They frequently obtain assistance and advice on investment management from financial planners or brokers.

The investment research, financial data, and consultancy that investors need comes from investment research providers, credit rating agencies, financial news services, financial data vendors, and investment consultants. We looked at each of these types of service providers.

- Many investors and their investment managers rely on investment information services to obtain investment research, financial data, and consultancy services that help them make decisions. The investors use research reports that dive deeper into the risk and return prospects of investments, which can be helpful when estimating the fundamental value of securities.
- Credit rating agencies specialize in providing opinions about the credit quality of bonds and of the companies or governments that issue the bonds. A high credit rating indicates that the credit rating agency believes that the bond issuer has a high probability of making all future payments of principal and interest when they are due. Most credit rating agencies do not charge investors for their ratings,

although they may charge them for the detailed reports on which the ratings are based.
- Data vendors provide investors with current, accurate data about companies and market conditions. Market data feeds provide information on market quotes, the orders investors submit for securities, and the prices and volumes of recent trades—all of which are helpful for investors who want to trade. Access to investment data was once very expensive and thus restricted to investment firms and institutional investors. The growth of information technologies, particularly those involving the internet, has substantially reduced the cost of accessing data and increased accessibility. The widespread availability of investment data has greatly changed the investment industry landscape. Although access to data used to be a key driver of investment profits, such profits are increasingly dependent on the ability to analyze data.

Finally, we examined how trading service providers—including brokers, dealers, clearinghouses, settlement agents, custodians, and depositories—facilitate investment by holding assets for clients and by helping buyers and sellers of securities and investment assets arrange trades with each other.

- Brokers act as agents, arrange trades for their clients, and ensure that clients settle their trades. For complex trades, they often serve as professional negotiators.
- Dealers participate on the opposite side of their clients' trades and are willing to trade on demand, thus providing liquidity.
- After a trade has been agreed on, clearinghouses arrange for final settlement of the trade, and then settlement agents organize the final exchange of cash for securities.
- Custodians and depositories hold money and securities for safekeeping on behalf of their clients and help prevent loss from securities investment fraud.

Up Next

In the next module, you will learn how investment professionals organize their efforts to help clients meet their financial goals and explore how fintech is transforming the world of finance.

Module 4: Financial Planning Services and Introduction to Fintech

The investment industry facilitates saving and investing. This module discusses how investment professionals organize their efforts to help clients meet their financial goals. It also describes how the rise of AI and fintech is revolutionizing the services that investment professionals and firms provide.

By the end of this module, you will be able to:

- **Describe** the needs served by the investment industry
- **Describe** financial planning services
- **Explain** fintech and the power of artificial intelligence (AI)
- **Describe** how fintech applications are currently transforming the world of finance

Let's get started!

Lesson 1: Needs Served by the Investment Industry

Welcome to Lesson 1 of Module 4!

In this lesson, you will be introduced to the reasons why investors seek out investment professionals to help them invest their money (capital).

By the end of this lesson, you will be able to describe the needs served by the investment industry.

Let's get started!

How the Investment Industry Promotes Successful Investing

The investment industry provides services to individual and institutional investors who have money (capital) to invest. As providers of capital, most investors seek professional assistance with activities that they cannot, or choose not to, do themselves, including the following:

1. Determine financial goals, particularly how much money needs to be invested to meet those goals
2. Identify potential investments
3. Evaluate the risk and return prospects of potential investments
4. Trade securities and assets
5. Hold, manage, and account for securities and assets during the periods of the investments
6. Evaluate the performance of the investments

These steps generally require information, expertise, and systems that few individual or institutional investors possess.

Investors are assisted by investment professionals to take these steps, either by hiring them or by investing in investment vehicles that the industry creates and oversees.

Some investment professionals specialize in a single service, whereas others provide a broad spectrum. Even though most investment firms and professionals provide multiple services, for the sake of clarity, this course considers each service separately.

Lesson 1 Knowledge Check

The following knowledge check is designed to reinforce your understanding of the concepts taught in this lesson. It is ungraded, but you are strongly encouraged to complete it.

True or False: Most investors seek professional assistance when they lack the necessary expertise, information, and systems support to do their own financial planning and asset management.

a. True
b. False

Lesson 1 Knowledge Check Feedback

Question 1

a. **Correct**. Financial skills and systems support can take considerable time and money to acquire. For individual investors, that can represent a poor allocation of personal time and capital. The financial services industry provides the following services:

- Budgeting and financial goal setting
- Market and security analysis
- Portfolio construction and rebalancing
- Risk management
- Performance measurement and interpretation
- Access to alternative investment vehicles

Up Next

You have now completed Lesson 1! We hope that this lesson has helped you to understand why investors seek out investment professionals to help them invest their money.

You should now be able to describe the needs served by the investment industry.

In the next lesson, you will learn about financial planning services.

Lesson 2: Financial Planning Services

Welcome to Lesson 2 of Module 4!

Financial planners assist their clients in understanding their financial needs, the risks involved, and the difference between preserving their capital and growing it at the same time.

By the end of this lesson, you will be able to describe financial planning services.

Let's get started!

Financial Planning Services

Investment clients often need advice to set their financial goals and determine how much money they should save. Some clients also need advice about how much money they can spend on expenses while still preserving their capital. **Financial planners** help their clients understand their current and future financial needs, the risks they will face when investing, their ability to tolerate risks, and their preferences for capital preservation versus capital growth. This process is described further in Course 5, Serving Client Needs.

Financial planners create savings and investment plans that are appropriate for their clients. The plans often require a complex analysis of many moving pieces:

1. The expected risk and rates of return for various securities and assets
2. The client's capacity and tolerance for bearing risk
3. Tax considerations
4. Forecasts of expenses

Expenses are often particularly difficult to forecast. They may depend on inflation and, in the case of retirement, uncertain life expectancy and health care costs. Analysis of pensions and health care is typically done by actuaries, who are professionals that specialize in assessing insurance risks using statistical models.

Many pension funds employ financial planners to help their beneficiaries make better savings decisions. Some employers contract with financial planners to make their services available to their employees. Increasingly, financial planners provide advice online to retail investors.

Organizations may require financial planning to meet their investment objectives. For example, foundations and endowment funds, which are not-for-profit institutions with long-term investment objectives, hire financial planners to create **payout policies**. Payout policies specify how much money can be taken out of long-term funds for current spending.

The payout policies depend on the assumptions the financial planners make about investment returns. Assuming high future returns allows for higher spending, but if these assumptions are too optimistic, payouts will exceed the returns, and spending will have to be curbed to preserve the fund's capital.

Lesson 2 Knowledge Check

The following knowledge check is designed to reinforce your understanding of the concepts taught in this lesson. It is ungraded, but you are strongly encouraged to complete it.

Which of the following inputs do financial planners use to help investors anticipate future expenditures and develop appropriate saving and investment strategies that reinforce financial security in retirement?

Select all that apply.

a. Life expectancy
b. Educational attainment
c. Inflation
d. Investment returns

Lesson 2 Knowledge Check Feedback

The correct answer is a. life expectancy, c. inflation, and d. investment returns. Financial planning often involves the complex modeling of inputs, such as life expectancy, inflation, and investment returns, among others. While b. "educational attainment" is helpful, it does not directly influence savings and investment decisions.

Up Next

You have now completed Lesson 2. We hope that this lesson has helped you to understand the different ways in which financial planners can help their clients.

You should now be able to describe financial planning services.

In the next lesson, you will learn about fintech and the power of AI.

Lesson 3: Introduction to Fintech

Welcome to Lesson 3 of Module 4!

Fintech refers to technology-driven innovation in the financial services industry. And as a part of fintech, artificial intelligence speeds up tasks, such as investment advice, financial planning, business lending, and payments.

By the end of this lesson, you will be able to explain fintech and the power of artificial intelligence (AI).

Let’s get started!

Fintech

In its broadest sense, **fintech** refers to technology-driven innovation in the financial services industry. Early forms of fintech included the automation of routine functions, such as data processing (as distinct from data analysis), but the focus now is on technological innovations that impact the design and delivery of financial services.

Fintech can also refer to companies—often new, startup companies—involved in developing these technologies and to the wider sector these companies are a part of. These innovations provide a real challenge to traditional business models and established financial services providers.

Fintech has advanced into decision-making applications based on complex machine learning algorithms, where computer programs are able to “learn” how to complete tasks over time. In some applications, advanced computer systems perform tasks at levels far surpassing human capability. Fintech has changed the financial services industry in many ways, giving rise to new systems for investment advice, financial planning, business lending, and payments.

There is often no clean separation of fintech components. AI is used to develop algorithmic trading software; robo-advisors, in turn, are supported by automated trading platforms that can seek out the best liquidity at the best price.

Artificial Intelligence (AI)

What do we mean by artificial intelligence? During its early development, AI typically referenced the ability of machines to solve problems and to display cognitive skills more closely associated with humans than machines. Today, it is generally taken to mean machines acting within the context of the intelligence they have iteratively improved as they collect information, which implies the ability to learn and adapt.

Learn more about how AI can be used to elevate the efficiency of various financial processes next.

Credit Assessment

AI assesses whether someone should be offered a loan and suggests the best type based on the client's circumstances. This is particularly important when it comes to supporting the historically unbanked and underbanked. The use of AI in this context can reduce losses from bad loans by approximately 20%.

Fraud Detection

The automated calls that cardholders receive from their credit card company about unusual activity are triggered by an AI analysis of their spending habits. Preventing cyber fraud and identifying money laundering are important AI applications.

Risk Management

AI is used to analyze unstructured data (i.e., information that either does not have a pre-defined data model or is not organized in a pre-defined manner) and to identify trends and risk signals.

Trading

Algorithmic trading programs often utilize AI. Some investment funds are totally automated, with AI making the trading decisions and executing the trades.

Personalized Banking

AI is seen in the form of virtual assistants and chatbots that use natural language processing (NLP) to suggest suitable financial recommendations and support 24-hour customer service.

Process Automation

AI can automate many repetitive, mundane, or time-consuming tasks, such as reviewing loan applications, in which human errors are common.

Ethics and AI

With AI-based process automation reducing human error, it can reduce human bias. But is it correct to say that AI has no bias? Moreover, what are the ethical questions that arise from the use of AI?

The ethical issues that arise from the use of AI include the following:

- **Bias:** AI programs are built on data, and it is possible, although likely not intentional, that the data itself reflects biases. An AI program developed to predict which patients require additional medical care was found to have a racial bias. The program itself did not consider race, but the underlying data that was used to build and train the AI program did have a racial bias.
- **Accountability:** Where does accountability lie if a decision made by AI results in damage or personal injury? This problem is very real in the context of self-driving cars. If a car being driven in a hands-off mode is involved in an accident, who is to blame—the driver, the manufacturer, or the software designer?
- **Transparency:** If an AI program turns down a loan application and the applicant wants to know why, the lender's response may be something along the lines of, "The computer said so." Finding out why an algorithm comes to a particular conclusion can be difficult.

Lesson 3 Knowledge Check

The following knowledge check is designed to reinforce your understanding of the concepts taught in this lesson. It is ungraded, but you are strongly encouraged to complete it.

Which of the following are served by fintech or AI applications?

Select all that apply.

a. Automation of routine functions
b. Portfolio construction
c. Loan credit analysis
d. Implementation of trading strategies

Lesson 3 Knowledge Check Feedback

All the options listed are correct. All the processes are served by fintech and AI applications.

- Automation of routine functions can reduce errors and costs.
- Many aspects of traditional portfolio construction can be replicated through fintech or AI applications.
- Fintech and AI applications allow for faster and more consistent loan credit analysis.
- Trading strategies driven by the analysis of rapidly changing data can be implemented more efficiently with fintech or AI applications.

Up Next

You have now completed Lesson 3. We hope that this lesson has helped you to understand what fintech is and how AI is shaping the future of financial services.

You should now be able to explain fintech and the power of AI.

In the next lesson, you will learn about how fintech applications are currently transforming the world of finance.

Lesson 4: Fintech Applications

Welcome to Lesson 4 of Module 4!

Large amounts of personal data are analyzed daily by analytical tools to establish client behavior, which is integrated into the decision-making process. There are different types of intelligent automation applications, such as fintech applications, automated/algorithmic trading, automated advice and robo-advisors, mobile banking, decentralized finance (DeFi), and crypto assets. We will look at each of these to get a better understanding of how they are transforming the world of finance.

By the end of this lesson, you will be able to describe how fintech applications are currently transforming the world of finance.

Let's get started!

Analysis of Large Datasets

Why do social media companies provide “free” services? The answer is that they are not really free. We pay for them by providing large amounts of personal data, which includes data about our shopping habits, our likes and dislikes, where we go, and how we get there.

Financial analysts no longer limit themselves to studying financial statements, economic reports, and market indicators. They also dive deeply into the stories told by non-traditional data from non-traditional sources. These sources produce a huge amount of data that must be integrated into the decision-making process, a task that fintech supports through the development of analytical tools.

Big Data

Big data (or alternative data) encompasses data generated by financial markets, businesses, governments, individuals, sensors, and the “Internet of Things.”

Big data has exploded over the past decade, especially in the form of unstructured data generated from social media, email and texts, web traffic, online news sites, and other electronic information. Investment practitioners are increasingly using big data in their decision making as they augment structured data with unstructured data to improve their financial models and forecasts.

The characteristics of big data differ from those of traditional data in several ways. Four core areas of difference—the four Vs—are volume, variety, velocity, and veracity.

- Volume: The quantity of data available
- Variety: The wide array of available data sources
- Velocity: The speed at which data is created
- Veracity: The credibility and reliability of different data sources

One of the first problems to consider when using big data is deciding what data is actually relevant and where it comes from. For example, if an analyst is trying to determine the likely profitability of a chain of coffee shops, what data could be useful?

If a financial analyst is trying to determine the likely profitability of a chain of coffee shops, what data could be useful? Review each form of data and select "Yes," "No," or "Unlikely."

1. Structured data from the firm's financial statements
2. Management expertise
3. The volume of pedestrian traffic past the shops
4. The demographics of those pedestrians
5. The height of the pedestrians
6. Voting habits of the pedestrians

Take a moment to consider it:

Structured data from financial statements (Yes)
Management expertise (Yes)
The volume of pedestrian footfall past the shops (Yes)
Demographics of those pedestrians (Yes)
Height of the pedestrians (Unlikely) (tall and less-tall people are equally likely to drink coffee)
Voting habits of the pedestrians (Unlikely) (voters and non-voters are equally likely to drink coffee)

Analytical models must be able to distinguish between data that adds value to the analysis and data that detracts from it. To test the explanatory power of the data, analysts will combine traditional techniques, such as regression and time-series analysis, with AI and machine learning–based techniques.

> Is AI valuable when dealing with complex and nonlinear relationships that are beyond the scope of traditional quantitative methods and statistical analysis?
>
> Take a moment to consider this question, and then compare your response to this response:
>
> Yes. Advances in AI-based techniques are enabling new data analysis approaches. For example, analysts are turning to AI to sort through the enormous amount of data from company filings, annual reports, and earnings calls. AI helps determine which data is the most important, uncovers trends, and generates insights on human sentiment and behavior. Natural language processing also comes into play here.

Natural Language Processing (NLP)

Stated concisely, NLP is a set of techniques that begins with transforming unstructured text data into structured (i.e., machine-readable) data and then classifies or clusters it to extract useful information. Consider the following two investment examples.

Stock Sentiment Analysis

> NLP is applied to the classification of company news into “positive” or “negative” sentiment to quickly and robustly answer the question, Does the stock market perceive this news as good or bad for the company’s stock price? Sentiment analysis can augment analysts’ forecasting and valuation models to improve the timing of stock purchases and sales.
>
> **Theme Extraction and Identification**
>
> NLP can be used to quickly analyze quarterly reports and regulatory filings for numerous companies, such as S&P 500 Index companies, to extract and cluster information that reveals underlying trends or themes. These include supply chain issues, profit margins becoming thinner, or weakening consumer demand. These themes may then become the basis for trading or investment strategies.

Automated/Algorithmic Trading

Algorithmic trading is the computerized buying and selling of financial instruments, in accordance with pre-specified rules and guidelines. It is widely used in both institutional and retail environments to implement trading strategies.

Institutions may use algorithmic trading to execute large orders, slicing them into smaller pieces to avoid market impact and executing them across a number of different trading venues to get the best price. During the day, these algorithms may continuously update and revise their execution strategy on the basis of changing prices, volumes, and market volatility.

Algorithmic trading is not restricted to institutions. Many investment firms offer retail funds in which all trading decisions and trade executions are automated. This type of fund could use a simple strategy based on technical analysis, with trades being based on moving averages and trading volumes. These rules can be easily programmed and automated.

Algorithmic trading provides speed, anonymity, and lower transaction costs.

High-Frequency Trading

High-frequency trading (HFT) is a particular form of algorithmic trading that pulls in large amounts of data from traditional and non-traditional sources to automatically place trades when certain conditions, such as mispricing, are met. The algorithm directs trades to ultra-high-speed, low-latency networks in fractions of a second. The fragmentation of traditional trading venues has facilitated the development of HFT algorithms that can exploit small, transitory pricing differences across exchanges.

It is estimated that in the United States, for example, 50% of trading volume is driven by HFT. The ultra-high speed of these systems allows a computer to analyze, process, and execute a trade in approximately 10 milliseconds. As a consequence, some companies locate their servers as close as possible to key exchanges to reduce the time needed to execute a trade by a fraction of a millisecond.

Automated Advice and Robo-Advisors

Robo-advisors make algorithmically driven investment suggestions based on client data. They have a role to play in the automation of personal wealth management services and in the provision of investment services to a larger number of retail investors at a lower cost than is possible with traditional adviser models.

Engaging a robo-advice service typically begins with an investor questionnaire that may cover income, time left until retirement, anticipated spending, risk tolerance, financial knowledge, and investment experience. Once assets, liabilities, risk preferences, and target investment returns have been digitally entered by a client, the robo-adviser produces recommendations based on algorithmic rules, historical market data, and the client's responses to the questionnaire.

Although robo-advice covers both active and passive management styles, most robo-advisers follow a passive investment approach. These robo-advisers typically charge low fees and recommend low-cost, diversified, index-based mutual funds or exchange-traded funds (ETFs).

Two services dominate robo-advice: fully automated digital wealth managers and adviser-assisted digital wealth managers.

Learn more about these services:

Fully Automated Digital Wealth Managers

The fully automated model does not rely on assistance from a human financial adviser, usually presents a low-cost solution to investing, and often recommends a portfolio composed of ETFs. The service package may include direct deposits, periodic rebalancing, and dividend reinvestment options.

Adviser-Assisted Digital Wealth Managers

Adviser-assisted digital wealth managers provide automated investment services along with a financial adviser, who is available virtually (by phone or online) to offer basic financial planning advice and periodic reviews. Adviser-assisted digital wealth managers' services may involve a more holistic analysis of a client's assets and liabilities.

As robo-advisors have been incorporated into the investment landscape, they have drawn the attention of regulatory authorities.

In the United States, robo-advisors must be established as registered investment advisers, and they are regulated by the Securities and Exchange Commission.

In the United Kingdom, they are regulated by the Financial Conduct Authority.

In Australia, all financial advisers must obtain an Australian Financial Services license, with guidance on digital advisers coming from the Australian Securities and Investments Commission.

Although regulatory conditions vary, there are indications that robo-advisors will be held to a similar level of scrutiny and code of conduct as other investment professionals.

Mobile Banking

For many people, the impact of fintech is most obvious when it comes to mobile banking. The ability to transfer money from one account to another, view bank statements, make investments, and pay bills with a banking app on a phone is a benefit of fintech for most retail banking customers. But behind this improved digital experience, financial transactions still settle in traditional, centralized depositories and clearinghouses out of view.

Fintech banking via mobile phones has generally made bank services more convenient and, in communities not well served by physical banks, more accessible. As with AI, however, an increasingly cashless society carries societal impacts. Cashless transactions may result in certain jobs becoming irrelevant, such as cashiers in shops and ticket clerks at railway stations. And because cashless transactions make spending easy, critics worry it encourages reckless spending.

Decentralized Finance (DeFi)

Decentralized finance (DeFi) is an umbrella term for financial services and financial instruments that do not rely on intermediaries, such as brokerages, exchanges, or banks. By using smart contracts on public blockchains, DeFi platforms allow people to lend or borrow funds from others, insure against risks, earn interest in savings-like accounts, and trade crypto currencies.

As of early 2022, the value of assets used in decentralized finance amounted to USD200 billion. **Blockchain** technology provides the foundation for DeFi and crypto currencies. Broadly speaking, blockchains are created by networks of users (decentralized) who use digital ledgers to securely engage in transactions and maintain records of ownership. The costs and inefficiencies of intermediaries are reduced or eliminated as transactions are decentralized; transactions may settle in hours rather than days.

Because the digital ledgers are duplicated and distributed across the entire peer-to-peer network, making the information about the transaction immutable, blockchain technology promises to create the same level of investor trust that historically required reliable third parties, such as clearinghouses.

Blockchains store data in “blocks” that are “chained” together. In the case of public blockchains, such as Bitcoin, this structure creates an irreversible chronological record. Different types of information can be stored on blockchains, but financial transactions predominate.

Crypto currencies are typically accounted for using blockchain technology. Advantages include the efficiency of transacting and the relative independence from government fiscal and monetary policy. Although initially thought to be suited for commerce and retail trade, the volatility of crypto currencies has drawn comparisons with commodities.

DeFi, blockchain, and the decentralized protocols they support are examples of technology’s impact on finance. They will all be discussed in more detail in Course 2, Types and Functioning of Markets.

To be successful, DeFi must provide solutions to some of the problems identified with centralized and traditional finance, such as the following:

- Centralized control
- Limited access
- Inefficiency
- Lack of interoperability
- Opacity

We briefly consider each of the problems and how DeFi addresses them.

Learn more about the problems of traditional finance and how DeFi addresses them in Exhibit 1.

Exhibit 1: Problems in Traditional Finance Addressed by DeFi

Problem	Traditional	DeFi
Centralized control	Retail banking can be highly concentrated, with many markets dominated by a small number of banks. Although there is competition, these banks tend to offer similar products at similar rates, and if one bank turns down a request for a business loan, then it is quite likely that most of the others will. Bank transactions, even the simple task of transferring money from one account to another, are not instantaneous and can be costly because of their reliance on a centralized clearing system. The complexity of clearing large daily volumes of transactions requires a costly infrastructure typically built around a national central bank, such as the US Federal Reserve (the Fed), the Bank of England, or the People's Bank of China. It is worth noting that every US dollar transferred across the national banking system is ultimately recorded at the Fed as an increase in dollars held by one bank and a reduction in dollars held by another bank.	Blockchain offers peer-to-peer transactions without the reliance on any traditional financial intermediaries.
Limited access	It is estimated that around 1.7 billion people around the world are unbanked and have no access to the banking system. It can also be very difficult for entrepreneurs and startups to obtain financing at an affordable rate as banks have become collectively more risk averse.	In a public blockchain without permissions, like Bitcoin or Ethereum, opening an account to trade assets or instruments on the blockchain is easily done and available to everyone.

Problem	Traditional	DeFi
Inefficiency	Examples of inefficiency include the time and cost involved in moving money internationally. For instance, it usually takes two days to settle a simple stock transaction.	Blockchain transactions are usually executed in a matter of minutes, although speed can depend on how busy the blockchain is at any given moment.
Lack of interoperability	The banking system is siloed and isolated from other systems, which makes it vulnerable to inefficiencies. The silo effect can be observed when buying real estate because there is no link between the banking system, which is required to pay for the property, and the registration system, which records transfers of ownership.	Using a special standard for creating and issuing "smart" contracts on a network, such as the Ethereum blockchain, fungible tokens that represent an asset, a right, an ownership, access, or a **crypto currency** can be used on any protocol across that network, thus ensuring a high level of transactional interoperability.
Opacity	There is little transparency into the financial health and security of financial institutions. A bank that is offering a high interest rate on its short-term debt may have ample funds but may want to attract new customers. Alternatively, the bank could be desperate for funds and near collapse. The consequence of this lack of transparency is excessive and costly regulation.	Everything is done on a transparent public blockchain, from the smart contract code to the details of every transaction.

Crypto Assets

Bitcoin is probably the most well-known crypto asset that trades on a blockchain. Supporters of Bitcoin have claimed it is the future of money.

Satoshi Nakamoto's 2008 paper, "Bitcoin: A Peer-to-Peer Electronic Cash System," proposed an asset that could be used as an alternative to traditional fiat currency to settle transactions. Although Bitcoin has not fulfilled this purpose, it has found a purpose as a store of value, although a very volatile one.

An arguably bigger impact of the blockchain is the Ethereum network and the development of the smart contracts that are deployed on it. These smart contracts, a couple of which are discussed in the next course, are at the heart of decentralized finance.

The ability technology has brought to execute swaps, loans, and derivative transactions with a program, the smart contract, and the removal of intermediaries and counterparties has a real impact on finance. Throw in open access to all users, and blockchain technology appears truly revolutionary.

Whether this technology will truly revolutionize the world of finance remains to be seen. The potential is real, but so are the risks and pitfalls.

Lesson 4 Knowledge Check

The following knowledge check is designed to reinforce your understanding of the concepts taught in this lesson. It is ungraded, but you are strongly encouraged to complete it.

Decentralized finance refers to financial products and services that promise to eliminate what are seen as inefficiencies in centralized financial systems. Which of the following criticisms is not associated with traditional, centralized financial systems?

a. Centralized controls can lead to complex and costly operational infrastructure.
b. In some markets, the high costs associated with building and staffing physical banks result in limited access to banking services.
c. Money laundering regulations impede small cash transactions.
d. Interoperability—the ability of service providers to interact smoothly across their various financial platforms—is constrained when institutions have different software systems and accounting protocols.

Lesson 4 Knowledge Check Feedback

c. **Correct**. Money laundering regulations impede small cash transactions. Money laundering is illegal. Regulatory tools for monitoring cash transactions are better developed in centralized financial systems. Because of their access to information and many years of experience, financial institutions may be required to report, not impede, large cash transactions or unusual activity. In the course of normal business, small cash transactions are not considered unusual activities.

a., b., and d. are **incorrect.** Each are associated with centralized financial systems

Up Next

You have now reached the end of Lesson 4! We hope that this lesson has helped you to understand the different types of automation in the finance world.

You should now be able to describe how fintech applications are currently transforming the world of finance.

Module 4 Summary

In this module, we focused on financial planning services. Often investors seek professional assistance with activities that they either cannot or choose not to do themselves. Some of these activities include determining financial goals; identifying potential investments; evaluating the risk and return prospects of potential investments; trading securities and assets; holding, managing, and accounting for securities and assets during the periods of the investments; and evaluating the performance of those investments.

- Financial planners help investors set their financial goals and determine how much money they should save for future expenses and/or how much money they can spend on current expenses while still preserving their capital.

Next, we looked at how fintech is changing the financial services industry and, in particular, how it is changing investment advice, financial planning, business lending, and payments.

- AI generally refers to machines acting within the context of the intelligence they have iteratively improved as they collect information, which implies the ability to learn and adapt. The use and application of AI has led to rapid innovations within financial services.
- AI has been instrumental in advancements in credit assessment, fraud detection, risk management, trading, personalized banking, and process automation.
- The prevalence of the use of AI in financial services has led to the rise of many ethical implications around the issues of bias, accountability, and transparency.

Lastly, we went over the specific ways in which fintech applications are transforming the world of finance.

- Financial analysts no longer limit themselves to studying financial statements, economic reports, and market indicators but also dive deeply into the stories told by non-traditional data from non-traditional sources, which must be integrated into the investment decision-making process. Fintech supports the development of analytical tools that aid in this process.
- Big data (or alternative data) encompasses unstructured data generated by financial markets, businesses, governments, individuals, sensors, and the "Internet of Things." The characteristics of big data differ from those of traditional data in four core ways—volume, variety, velocity, and veracity.
- AI is especially valuable when dealing with complex and nonlinear relationships that are beyond the scope of traditional quantitative methods and statistical

analysis, such as insights on human sentiment and behavior. Natural language processing is a set of techniques that begins with transforming unstructured text data into structured (i.e., machine-readable) data and then classifies or clusters it to extract useful information. Stock sentiment analysis and theme extraction and identification are two examples of such applications.

- Algorithmic trading is the computerized buying and selling of financial instruments, in accordance with pre-specified rules and guidelines. It provides speed, anonymity, and lower transaction costs. High-frequency trading (HFT) is a particular form of algorithmic trading that pulls in large amounts of data from traditional and non-traditional sources to automatically place trades when certain conditions, such as mispricing, are met.
- Robo-advisors make algorithmically driven investment suggestions based on client data, and they have a role to play in the automation of personal wealth management services and in the provision of investment services to a larger number of retail investors at a lower cost than is possible with traditional adviser models. Two services dominate robo-advice: fully automated digital wealth managers and adviser-assisted digital wealth managers.
- Fintech banking via mobile phones has generally made bank services more convenient; however, an increasingly cashless society carries societal impacts, including people losing their jobs and the potential for reckless spending.
- DeFi is an umbrella term for financial services and financial instruments that do not rely on intermediaries, such as brokerages, exchanges, or banks. Blockchain technology provides the foundation for DeFi and crypto currencies. DeFi platforms, via smart contracts on public blockchains, allow people to lend or borrow funds from others, insure against risk, earn interest in savings-like accounts, and trade crypto currencies. Broadly speaking, blockchains are created by (decentralized) networks of users who use digital ledgers to securely engage in transactions and maintain records of ownership. Because the digital ledgers are duplicated and distributed across the entire peer-to-peer network, making the information about the transaction immutable, blockchain technology promises to create the same level of investor trust that historically required third parties, such as clearinghouses.
- DeFi provides solutions to some of the problems identified with centralized and traditional finance, such as centralized control, limited access, inefficiency, lack of interoperability, and opacity.

Module 4 Glossary

In this module, you learned the following terms (in order of appearance):

Lesson 2: Financial Planning Services

- **Financial planners:** Investment professionals who help their clients set their financial goals and determine how much money they should save for future expenses and/or how much money they can spend on current expenses while still preserving their capital.
- **Payout policies:** Guiding principles that specify how much money an institution, such as a foundation or an endowment fund, can take from long-term funds to use for current spending.

Lesson 3: Introduction to Fintech

- **Fintech:** Technology-driven innovation in the financial services industry. The term can also refer to companies—often new, startup companies—involved in developing these technologies and to the wider sector they are a part of.

Lesson 4: Fintech Applications

- **Blockchain:** A system of recordkeeping that is maintained by independent computers in a peer-to-peer (decentralized) network.
- **Crypto currency:** A digital currency composed of encrypted data that acts as a medium of exchange.

Up Next

In the next module, we will look at how ethics and regulation are key forces driving the investment industry, how critical trust is, and the benefits that ethical decision making brings to the investment industry and society at large.

Module 5: Investment Professionalism and Regulations

Trust is essential to the functioning of the investment industry as well as the broader financial services industry.

Investment professionals are responsible for providing access to information about investment opportunities and financial markets. Regulations govern who is allowed to operate as an investment professional and how financial products are marketed. Failure to comply with regulations, policies, and procedures can have significant consequences for employees, managers, customers, the company, the investment industry, and the economy.

By the end of this module, you will be able to:

- **Describe** how ethics and regulation are key forces driving the investment industry
- **Define** why regulations for the financial industry are so important
- **Describe** the objectives of regulation
- **Identify** specific types of regulation and describe the reasons for each
- **Describe** the potential consequences of compliance failure
- **Describe** why trust and ethics in the investment industry are important
- **Identify** common obligations that individuals as professionals in the investment industry have toward various stakeholders
- **Identify** elements of the CFA Institute Code of Ethics and explain how standards of professional conduct that are based on the CFA Institute Code of Ethics can be applied in practice
- **Explore** the benefits of ethical conduct as well as the consequences of unethical conduct for various parties in the investment industry and the economy as a whole
- **Describe** a framework for making ethical decisions

Let's get started!

Lesson 1: Key Forces Driving the Investment Industry

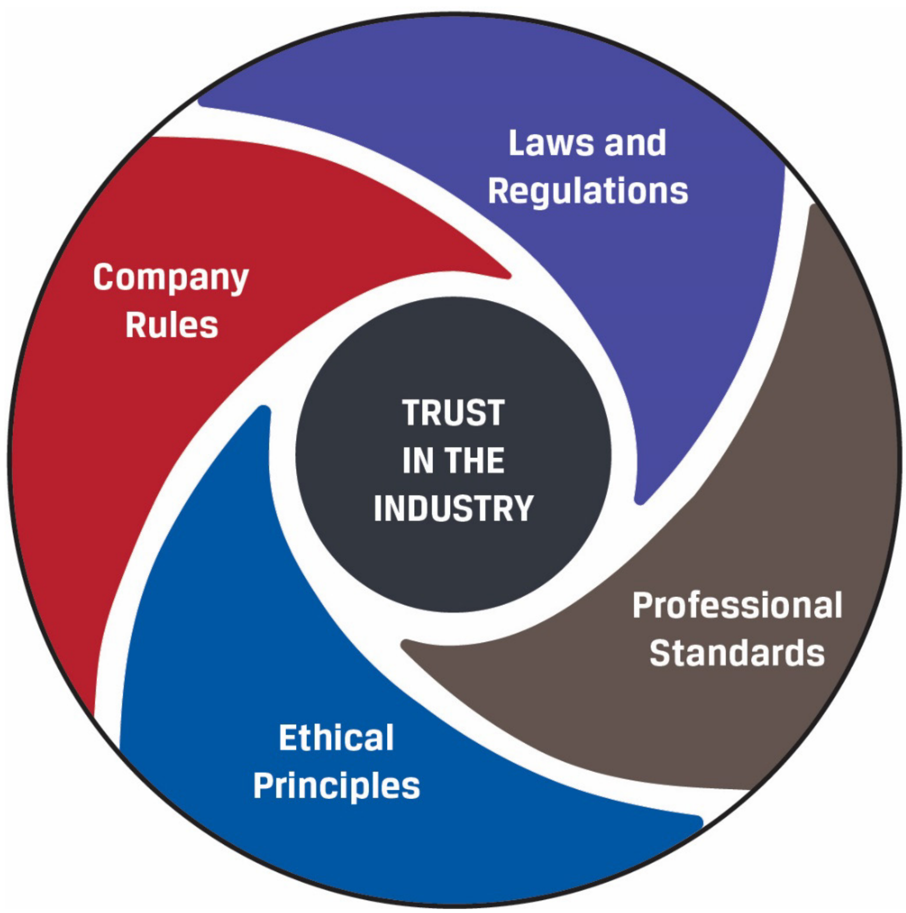
Welcome to Lesson 1 of Module 5!

In this lesson, you will learn about ethics and regulations in the investment industry. You will dive deeper into such concepts as investor trust and ethical guidelines as well as the need for rules, regulations, and laws that promote ethical behavior.

By the end of this lesson, you will be able to describe how ethics and regulations are key forces driving the investment industry.

Let's get started!

Exhibit 1: The Need for Trust, Laws, and Regulations



At its best, the investment industry efficiently matches those who need money with those who have savings to invest, allowing money to support the most productive businesses and projects. Like many industries, the investment industry evolves in response to internal and external drivers of change as shown in Exhibit 1. Competition and technology are internal drivers; regulation (laws) and globalization are external drivers. When the investment industry is efficient and trustworthy, economies and people benefit.

Trust is essential to the proper functioning of the financial system in general and the investment industry in particular. Investors should be confident they will be treated fairly by the people and institutions they transact with as well as by those who advise them, sell them investment products, and manage their investments. If trust is lacking, investors will be reluctant to commit their capital to viable investment opportunities and the economy and society will suffer.

Laws and **regulations** are necessary to ensure that investors are treated fairly and honestly. Usually, laws are passed by a legislative body, such as Congress in the United States, Parliament in the United Kingdom, and the Diet in Japan. Regulations are created by agencies, such as the Canadian Securities Administrators in Canada, the Autorité des marchés financiers in France, and the Securities and Futures Commission of Hong Kong.

Note: Laws and regulations must be enforceable to be effective.

The form and extent of laws and regulations vary among countries and change over time, but a few general principles are widely applied. Laws and regulations are designed to do the following:

1. Prevent fraud
2. Protect investment industry participants—in particular, investors
3. Promote and maintain the integrity, transparency, and fairness of financial markets

For example, trading based on non-public information that could affect a security's price, known as **insider trading**, is forbidden in most jurisdictions.

This means that an analyst who learns during a private meeting with a company's management that the company is about to acquire a competitor is not allowed to buy or sell shares in the company or its competitor until the company has officially announced the acquisition. If the analyst trades before this information is available to all market participants, he could gain from this inside information, and the integrity and fairness of the financial market would be compromised. In many jurisdictions, the analyst could also face legal or regulatory punishment.

The Need for Ethical Behavior

The adoption and successful application of ethical and professional standards, regulations, and corporate policies create trust.

Although the investment industry is subject to them, laws and regulations cannot cover every eventuality, nor can they completely prevent fraud and market abuse. For this reason, it is important that investment industry professionals behave professionally and ethically, in accordance with a set of moral principles, and that financial institutions proactively promote cultures of integrity.

The story of Wells Fargo Bank shows why individual behavior at all levels of management is so important in the financial industry. It is up to the individuals working in financial institutions to inspire public trust by always behaving professionally and ethically.

In 2016, Wells Fargo Bank was found to have fraudulently created millions of new savings and checking accounts in the names of their customers over a period of years. The accounts had been opened without customers' knowledge or permission. Initial reports indicated that rogue employees or offices had responded inappropriately to sales incentives. As the scale of the fraud became apparent, however, it was determined that much of the blame lay with upper management's pressure on employees to show successful cross-selling of Wells Fargo's services, such as creating credit card accounts for clients who were taking out home loans or opening savings accounts. Wells Fargo's CEO was subsequently barred from the banking industry. In 2020, Wells Fargo agreed to pay USD3 billion to settle criminal and civil investigations. Because of the scandal, Wells Fargo suffered reputational damage, and public trust in the financial industry suffered.

Lesson 1 Knowledge Check

The following knowledge check is designed to reinforce your understanding of the concepts taught in this lesson. It is ungraded, but you are strongly encouraged to complete it.

Question

Laws and regulations that promote integrity, transparency, and fairness in financial markets are necessary for which of the following reasons?

Select all the correct choices that apply.

a. Regulations are needed to assure investors that they will be treated fairly.
b. Investors who lack confidence in the financial markets will be reluctant to invest, raising the costs for seekers of capital to find providers.
c. Regulations create global standards for determining fair prices.

Lesson 1 Knowledge Check Answer Feedback

Correct answers

a. **Correct**. Regulations are needed to assure investors that they will be treated fairly.
b. **Correct**. Investors who lack confidence in the financial markets will be reluctant to invest, raising the costs for seekers of capital to find providers.

Incorrect answer

c. **Incorrect.** Fair pricing is determined by willing buyers and sellers in liquid, well-functioning financial markets, not by regulations. However, regulations are necessary to assure investors that markets will remain fair and efficient. And sellers lacking confidence in the financial markets will demand higher prices to compensate them for the perceived risks; buyers will offer lower prices to buy.

Up Next

You have now completed Lesson 1. We hope that this lesson has helped you understand the key forces that drive the investment industry.

You should now be able to identify the key forces driving the investment industry.

In the next lesson, you will learn about financial industry regulation.

Lesson 2: Financial Industry Regulation

Welcome to Lesson 2 of Module 5!

In the financial industry, rules and regulations are in place to prevent not only consumers from losing their life savings but also large financial services companies from failing, which would cause serious damage to the economy. Regulations frequently drive industry change and underpin attempts to prevent, identify, and punish undesirable behavior. Financial services and products are highly regulated because industry failure or disruption can have devastating consequences.

By the end of this lesson, you will be able to define why regulations for the financial industry are so important.

Let's get started!

What Exactly Are Regulations?

Rules are important to the investment industry. Without them, customers could be sold unsuitable products and lose their life savings. Customers can also be harmed by the investment industry misusing their assets. Furthermore, the failure of a large financial services company can lead to a systemic chain reaction that results in the failure of many other companies, causing considerable damage to the economy.

Regulations are rules that set standards for conduct and that carry the force of law. They frequently drive industry change and underpin attempts to prevent, identify, and punish undesirable behavior.

Financial services and products are highly regulated because industry failure or disruption can have devastating consequences.

Set and enforced by government bodies and by entities authorized by government bodies, regulations differ from ethical principles or professional standards. Violations of ethical principles and professional standards have consequences, but those consequences may be less severe than what arises from violations of laws or regulations. Laws and regulations can be used to reinforce ethical principles and professional standards.

Companies and employees that fail to comply with regulations face sanctions that can be severe on top of the harm that the noncompliance can have on companies, employees, and trust in the industry. When companies enforce rules for their employees to ensure compliance, and as they guide employees with matters outside the scope of regulation, their corporate policies serve to establish good behavior and business practices. An understanding of the regulatory environment and the corporate policies of one's employer is essential for success in the investment industry.

In this module, many of the examples are drawn from developed economies, primarily because the regulatory systems in these economies have had longer to evolve. Many of these systems have been adjusted over the years, so they not only protect investors and the financial system but also allow the investment industry to innovate and prosper.

Lesson 2 Knowledge Check

The following knowledge check is designed to reinforce your understanding of the concepts taught in this lesson. It is ungraded, but you are strongly encouraged to complete it.

Question

True or False: Regulations are rules that set standards for conduct and carry the force of law. They can be used as effective tools to reinforce ethical principles and professional standards.

Question choices

a. True
b. False

Lesson 2 Knowledge Check Feedback

Correct answer feedback

a. **Correct**. Financial regulations reinforce ethical behavior and attach a cost to unethical practices.

Incorrect answer feedback

b. **Incorrect**. Financial regulations reinforce ethical behavior and attach a cost to unethical practices.

Up Next

You have now completed Lesson 2. We hope that this lesson has helped you understand the consequences of unethical behavior and why regulation is necessary.

You should now be able to define why regulations for the financial industry are important. In the next lesson, you will learn about the objectives of regulation.

Lesson 3: Objectives of Regulation

Welcome to Lesson 3 of Module 5!

By the end of this lesson, you will be able to describe the objectives of regulation.

Objectives of Regulation

Regulators act in response to a perceived need for rules. Regulation is needed when market solutions are insufficient. Understanding the objectives of regulation makes it easier for industry participants to anticipate and comply with regulation. The following information describes these objectives.

Protect Consumers

Consumers in this space include borrowers, depositors, and investors, and although they may be able to quickly determine the quality of goods or services, many of them may not have the skills or information needed to determine the quality of financial offerings. Regulators seek to protect consumers from abusive and manipulative practices, including fraud. Regulators may, for instance, prevent investment firms from selling complex or high-risk investments to individuals lacking sufficient knowledge or assets.

Foster Capital Investment and Economic Growth

Financial markets match suppliers of capital with users of capital, such as companies and governments. The allocation of capital to productive uses is essential for economic growth. Regulators seek to ensure fair and efficient financial markets to foster economic development.

Support Economic Stability

The increasingly complex interconnections among global financial institutions may increase the consequences, if not the risk, of a **systemic failure**, which is a failure of the entire financial system. Regulators thus seek to ensure that financial firms, both individually and as an industry, do not engage in practices that could put the whole industry at risk.

Ensure Fairness

Not all market participants have the same information. Sellers of financial products might choose not to communicate negative information about the products they are selling. Insiders who know more than the rest of the market might trade on their inside information. These information asymmetries, or differences in available information, can deter investors from investing, thus harming economic growth.

Regulators attempt to deal with these asymmetries by requiring fair and full disclosure of relevant information on a timely basis and by enforcing prohibitions on insider trading. Regulators seek to maintain fair and orderly markets in which no participant has an unfair advantage.

Enhance Efficiency

Regulations that standardize processes can enhance economic efficiency by reducing duplication and confusion. Regulations that ensure an efficient dispute resolution process can also reduce costs and increase economic efficiency.

Improve Society

Governments may use regulations to achieve social objectives. These objectives can include increasing the availability of credit financing to a specific group, encouraging homeownership, or increasing national savings rates.

Another social objective is preventing criminals from using financial firms to transfer money from illegal operations to other, legal activities, a process known as **money laundering**. As a consequence of the transfer, the money becomes “clean.” Regulations help prevent money laundering and detect and prosecute criminal activity.

Specific regulations are developed in response to the broad objectives of regulation. Regulation can help to achieve multiple objectives. For example, rules about insider trading protect consumers (investors) and promote fairness in financial markets. Specific types of regulation are discussed in the following lessons.

Lesson 3 Knowledge Check

The following knowledge check is designed to reinforce your understanding of the concepts taught in this lesson. It is ungraded, but you are strongly encouraged to complete it.

Question

In addition to reinforcing ethical principles and standards, regulations are needed when market solutions are not sufficient to meet economic and societal goals. Which of the following is not a broad objective of regulation?

Question choices

a. Consumer protection
b. Capital formation and economic growth
c. Population growth
d. Economic stability

c. **Correct**. Population growth is not a broad objective of financial regulation.

a. **Incorrect**. Consumer protection against unethical products and sales tactics increases public trust in financial institutions.

b. **Incorrect**. Efficient access to fairly priced markets and services facilitates capital formation, which, in turn, drives economic growth.

d. **Incorrect**. Economic stability contributes to the confidence necessary for providers and users of capital to commit to long-term growth opportunities.

Up Next

You have now completed Lesson 3. We hope that this lesson has helped you understand the need for regulation in the financial industry.

You should now be able to describe the objectives of regulation. In the next lesson, you will learn about the types of regulation.

Lesson 4: Types of Regulation

Welcome to Lesson 4 of Module 5!

If regulators do not enforce rules of conduct, it can cause irreparable damage to the economy through a chain reaction from customer to company to industry failure. To help prevent this failure, sets of rules that focus on specific types of activity in the industry have evolved to ensure the smooth operation of the financial system.

By the end of this lesson, you will be able to identify specific types of regulation and describe the reasons for each.

Let's get started!

Types of Financial Market Regulation

When regulations fail to achieve their intended purpose, the consequences for individuals, institutions, and the economy can be severe. Trust in the financial services industry, which includes the investment industry, can also be damaged.

Customers may lose their life savings when they are sold unsuitable products, or customers could be harmed if an investment firm misuses customer assets. Furthermore, the failure of one large company in the financial services industry can lead to a catastrophic chain reaction (contagion) that results in the failure of many other companies, causing serious damage to the economy.

In response to this risk, multiple groups of related regulations have evolved to ensure the smooth operation of financial systems.

The broad objectives of regulation are attained by creating sets of rules that focus on specific types of investment industry activity. These rules include the following:

1. Gatekeeping rules
2. Operations rules
3. Disclosure rules
4. Sales practice rules
5. Trading rules
6. Anti-money-laundering rules
7. Business continuity rules

1. Gatekeeping Rules

Gatekeeping rules govern who is allowed to operate as an investment professional and how financial products are marketed. A primary activity of regulators is screening investment industry personnel to ensure they meet established standards for integrity and competence. Regulators in most financial markets require industry personnel to pass licensing exams to ensure a sufficient understanding of financial products and relevant financial law.

Financial products typically must comply with numerous regulations before they can be sold to the public. Gatekeeping rules are necessary because some financial products are difficult to understand and should not be sold to clients who are unable to fully appreciate the associated risks.

2. Operations Rules

Regulations may dictate certain aspects of how financial firms operate. One example is the case of net capital. It is important that financial firms have sufficient resources to honor their obligations.

History shows that highly leveraged companies, which are companies with a high amount of debt relative to equity, pose a risk not only to their own shareholders but also to their customers and the economy. Regulators seek to maintain financial industry stability by imposing capital requirements that limit the amount of leverage and risk that companies assume. Minimum equity capital ratios are an example.

3. Disclosure Rules

For markets to function properly, market participants require information, including the following:

- Information about companies and governments that are raising funds
- Information about financial instruments that are being sold and traded
- Information about the markets the financial instruments trade in

Note: Disclosure rules specify what information is to be passed on to participants and how.

Here are some examples of the types of disclosure rules.

- **Corporate issuers:** Regulators typically require corporate issuers of securities to disclose detailed information to potential buyers before the securities are offered for sale. The disclosures generally include audited financial statements, information about the general business of the company, the intended use of the buyers' proceeds, information about management, and important risk factors.

- **Market transparency:** Information about what other investors are willing to pay for a security, or the price they just paid, is valuable, but investors often do not want to reveal private information. Regulators generally require the dissemination of at least some information regarding the trading environment for securities.
- **Disclosure triggers:** Stock exchanges and market regulators may require public disclosure of trading activity when a trigger event occurs or when a threshold is reached. Disclosures about shares may describe potential takeover activity, directors' dealings in those shares, or short positions. In the United States, for example, investors are required to file a public disclosure when their ownership in a public company exceeds 5% of the outstanding shares. A real-world example of this occurred in April 2022, entrepreneur Elon Musk filed a Schedule 13D with the US Securities and Exchange Commission that reported his beneficial ownership of 73,115,038 shares, or 9%, of Twitter, Inc (NYSE: TWTR).

4. Sales Practice Rules

Some consumers seeking financial advice may not have sufficient knowledge to assess the quality of the advice they are receiving, and as a result, they may be vulnerable to abusive sales practices. For instance, some providers may be incentivized to recommend products that pay them high commissions rather than products that best suit the consumer.

The following are examples of different sales practice rules:

Advertising

Regulators may control the form and content of advertising to ensure it is not misleading. Regulators often disapprove of advertised promises, such as "guaranteed" returns and "sure win" situations, and they seek to ensure fair representation of past and anticipated future returns by creating industry standards for presenting performance.

Fees

Regulators may impose price controls to limit the commissions that can be charged on the sale of financial products as well as to limit the markups and markdowns that occur when investment firms trade securities directly with their customers.

Information Barriers

Many large investment firms offer investment banking services to corporate issuers while publishing investment research and providing financial advice. This creates potential conflicts of interest. Biased investment advice could help them win more lucrative investment banking business from any corporate client they publicly praise. Similarly, research analysts may be under pressure to publish favorable reports on securities in which the firm has large positions in its inventory. Regulators attempt to resolve these conflicts of interest by requiring firms to create both virtual and physical barriers between investment banking and research teams.

Suitability Standards

Regulation seeks to hold those in the investment industry accountable for the advice that they give to their clients. Any advice or recommendation should be suitable for the client and consistent with the client's interests. Regulators may set an even higher standard and require that the financial service provider act in the client's best interest. In most common law countries, including the United Kingdom and the United States, a well-established body of law has developed around the role of fiduciaries and their fiduciary duty to act in the client's best interest ahead of their own. You will learn more about fiduciary duty in Course 6, Serving the Greater Good.

Restrictions on Self-Dealing

Self-dealing is the practice of a fiduciary trading directly with the investor, rather than matching the trade with a third-party buyer or seller. Many firms in the investment industry sell financial products, such as securities, directly out of their own inventories, which allows the firm to provide faster service and better liquidity to their customers. However, self-dealing can create conflicts of interest because the firm is incentivized to charge the highest price to the customer, who wants to pay the lowest price. There can also be confusion among some customers as to whether the firm is acting as a principal (i.e., buying or selling for the firm's inventory) or as an agent (i.e., working for the client but not trading with the client).

Regulators may impose the best execution requirements, they may require that the firm disclose the conflict, or they may ban self-dealing altogether.

5. Trading Rules

Regulations are often designed to set investment industry standards and to prevent abusive trading practices.
The following are some trading rules.

Market Standards

Government regulation can be used to set the standard length of time between a trade and the settlement of the trade, which is three business days for equities in most global markets.

Market Manipulation

Regulators attempt to prevent and prosecute **market manipulation**. Market manipulation involves actions intended to move the price of a stock to generate a short-term profit.

Insider Trading

A market in which some participants have an unfair advantage over other participants lacks legitimacy and thus deters investors. For this reason, most jurisdictions have rules designed to prevent insider trading. Regulators often expect companies to have policies and procedures in place to restrict access to such information and to deter parties with access from trading on this information.

Front-Running

Front-running is the act of placing a personal order ahead of a customer's order to take advantage of the price impact that the customer's order will have. For example, if you know a customer is ordering a large quantity of stock that is likely to drive up the price, you could take advantage of this information by buying in advance of that customer's order. As with insider trading, regulations may help ensure that companies have procedures in place to deter front-running and to monitor employees' personal trading.

6. Anti-Money-Laundering Rules

Companies in the financial services industry can be used by criminals to launder money or otherwise facilitate illegal activities. Governments naturally want to deter such activities, and they may use their regulatory power over companies in the financial services industry to do so. Regulations may require companies to confirm and record the identity of their clients; to report payments, such as dividends, to tax authorities; and to report other pertinent activity, such as large cash transactions.

7. Business Continuity Rules

Given the essential nature of financial services to the economy, regulators may be concerned about business continuity in the event of disasters, such as fires, floods, earthquakes, and epidemics. Regulators want to be assured that customer records are adequately backed up and that companies have plans in place to recover from a disaster.

Lesson 4 Knowledge Check

The following knowledge check is designed to reinforce your understanding of the concepts taught in this lesson. It is ungraded, but you are strongly encouraged to complete it.

Question 1

Regulatory failures undermine investor confidence in the financial system and the willingness to engage in transactions. Which of the following is not an example of regulatory failure?

Question choices

a. For an extended period of time, margin deposits at a futures exchange remain below the minimum levels required by regulators. Failure to identify and correct the inadequate capital levels results in a major loss to the exchange.
b. A bank whose net capital has fallen below regulatory minimums continues to make high-risk loans. When the economy slows and borrowers' default, the bank announces that it will be unable to fully repay its depositors.
c. A brokerage firm is confronted by clients who believe that the firm should have advised them to sell their holdings of Deepdive Corporation before the stock market dropped.

Question 2

Regulations that attempt to prevent market manipulation are examples of:

Question choices

a. trading rules.
b. operational rules.
c. sales practice rules.

Lesson 4 Knowledge Check Feedback

Question 1

Correct answer feedback

c. **Correct**. Brokers act as agents. They are not responsible for subsequent portfolio performance or the timing of purchases and sales.

Incorrect answer feedback

a. **Incorrect**. The exchange failed to maintain the required capital reserves (margin balances). Regulatory monitoring and enforcement were not sufficient to prevent the loss from occurring.

b. **Incorrect**. The bank engaged in an inappropriate lending strategy that ignored regulatory standards. Regulatory monitoring and enforcement were not sufficient to prevent the loss from occurring.

Question 2

Correct answer feedback

a. **Correct**. Regulations that attempt to prevent market manipulation are examples of trading rules. Trading rules focus on trading practices in the market and trading activity of financial participants to ensure fair, organized, and efficient markets. Market manipulation involves investors taking actions intended to move the price of a stock to generate a short-term profit.

Incorrect answer feedback

b. **Incorrect**. Operational rules are related to how a company operates; operation rules include rules with respect to financial leverage and how customer accounts are managed.

c. **Incorrect**. Sales practice rules are intended to ensure that industry professionals treat clients appropriately with regard to the products recommended and the fees charged. Furthermore, such rules are intended to reduce potential conflicts of interest an industry professional might face in the sale and recommendation of a financial product.

Up Next

You have now completed Lesson 4. We hope that this lesson has helped you understand the types of financial market regulation.

You should now be able to identify a variety of rules and regulations that impact the financial markets. In the next lesson, you will learn about how to identify potential consequences of compliance failure.

Lesson 5: Consequences of Compliance Failure

Welcome to Lesson 5 of Module 5!

Regulation helps maintain market stability and enables participants to thrive. Regulatory rules together with company policies and procedures help ensure compliance and that all parties can benefit.

By the end of this lesson, you will be able to describe the potential consequences of compliance failure.

Let's get started!

Consequences of Compliance Failure

Failure to comply with regulations and policies and procedures can have significant consequences for employees, managers, customers, the company, the investment industry, and the economy.

Companies may fire employees and managers who fail to comply. When a regulatory action occurs, even if no formal charges are brought, the legal costs to the individuals and firms who deal with it can be high.

Regulators have many ways of disciplining firms and individuals who violate informal rules. Sanctions for individuals include fines, imprisonment, the loss of their financial licence, and lifetime bans from working in the investment industry. When subordinates violate rules, managers may also face consequences for failure to adequately supervise them. Even long after the issue is resolved, the regulatory sanctions remain a matter of public record that can tarnish the reputations of the individuals involved for the rest of their lives. The economic impact of a damaged reputation can be huge.

Companies also face sanctions, including fines, loss of licences, and forced closure. A company may be forced to spend significant resources on corrective actions, such as hiring outside consultants, to demonstrate compliance. Companies interact with regulators on an ongoing basis, so running afoul of a regulator's opinion in one area can lead to problems in other areas.

Compliance failures affect more than just the company and its employees. Customers may lose all their money, counterparties can suffer losses, and trust in the industry and markets can be seriously damaged.

Lesson 5 Knowledge Check

The following knowledge check is designed to reinforce your understanding of the concepts taught in this lesson. It is ungraded, but you are strongly encouraged to complete it.

Question
The consequences of failure to comply with regulations and corporate policies and procedures:

Question choices

a) include costs to only the firm and employees.
b) range from individual costs to damage to the global economy.
c) are borne by the employee who failed to comply and not by the employee's supervisor or employer.

Lesson 5 Knowledge Check Feedback

b) Correct. The consequences of failure to comply with regulations and corporate policies and procedures can have far-reaching consequences.

Incorrect answer feedback

a) Incorrect. These failures affect more than just the company and its employees.

c) Supervisors and the employer may be assigned some responsibility for the failure.

Up Next

You have now completed Lesson 5! We hope that this lesson has helped you understand more about the consequences of compliance failure.

You should now be able to describe the ethics and investment industry.

Lesson 6: Ethics and the Investment Industry

Welcome to Lesson 6 of Module 5!

Because of the growth of the financial markets, there is an increase in the complexity of ethical dilemmas, especially in business conducted across borders under different regulatory and cultural frameworks. Therefore, guidance is needed to inform the acceptable behavior of market participants. While the ideas of “do the right thing” and “always act ethically” seem simple, in the world of investments, the right path isn’t always as clear. For this reason, and given the importance of maintaining ethical behavior in the investment profession, Lesson 6 is the longest lesson of this module.

By the end of this lesson, you will be able to describe why trust and ethics in the investment industry are important.

A Deeper Dive on the Importance of Ethics in the Investment Profession

Unethical behavior, whether legal or not, can have significant consequences for the reputation of professional bodies, the financial system, and the economy at large. When faced with ethical dilemmas and decisions that can affect multiple stakeholders, professionals working within the investment industry must have a well-developed set of principles.

Establishing and applying an ethical decision-making framework in practice is crucial to engaging in ethical conduct and can result in engendering stakeholder trust and positive sentiment for the industry at large.

By the end of this lesson, you will be able to:

- Identify common obligations that individuals as professionals in the investment industry have toward various stakeholders (clients, prospective clients, employers, and coworkers)
- Identify elements of the CFA Institute Code of Ethics
- Explain how the Standards of Professional Conduct that are based on the CFA Institute Code of Ethics can be applied in practice
- Describe benefits of ethical conduct
- Describe consequences of conduct that is unethical or unprofessional
- Describe a framework for making ethical decisions
    - Describe benefits of ethical conduct
    - Describe consequences of conduct that is unethical or unprofessional
    - Describe a framework for making ethical decisions

Let's get started!

Ethics and Trust

Ethics form a personal moral philosophy that is related to a person’s upbringing and to the cultural, social, economic, and legal environments the person moves through. Because there are no universally agreed-on standards of ethical behavior, we need additional guidance.

We can find additional guidance for ethical behavior in the following places.

- **Professional standards: Professional standards**, including codes of ethics, are established by professional associations to guide the behavior of individuals within the profession.
- **Regulations:** Regulations carry the force of law and set standards for conduct. Regulations can be used to reinforce ethical and professional standards.
- **Policies and procedures:** Corporate policies and procedures are put in place by companies to promote desired behaviors and to guide employees.

Guidance may include rules that cover many situations, but they cannot reasonably be expected to cover every contingency. It is important to recognize that ethics and rules are interlinked: **Ethical standards** help guide the development of rules, and equally, rules help individuals think about and form ethical standards. A two-way feedback process between individual behavior and rules is critical to creating an ethical investment industry that inspires trust among investors.

As mentioned, ethics is a set of moral principles or the principles of conduct governing an individual or a group. Ethics plays an essential role in protecting financial market integrity and the functioning of the investment industry. Financial market integrity refers to financial markets that are fair and transparent and that provide investor protection.

> *Note:* Trust in the investment industry is enhanced when workers in the industry demonstrate ethical decision-making.

Case Study

In the 2022 CFA Institute Investor Trust Study “Enhancing Investors’ Trust” (CFA Institute 2022), researchers examined trust in the financial services industry.

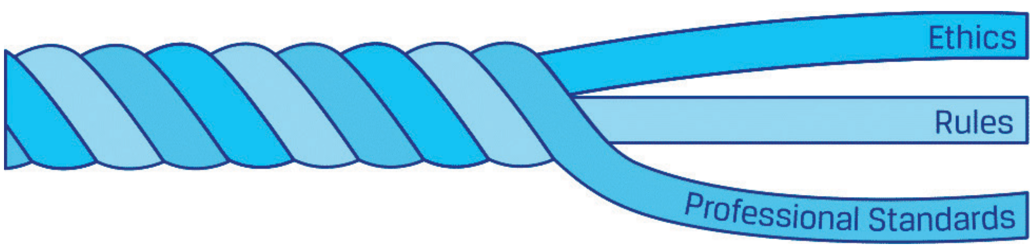
They determined that trust levels among institutional and retail investors were relatively high, but they also found that only 33% of those surveyed believed that their advisers would “always” put the client’s interests first.

A more positive finding was that trust in the financial services sector was significantly higher among retail investors who used advisers than among those who did not.

The study further noted a “trust multiplier” effect from technology. Investors indicated that the increased use of financial tools and access to financial data had increased their trust in the industry. The study summarized its findings with the observation that “trust, in some form, is at the center of all financial transactions.” As show in Exhibit 1:

- Ethical standards and some professional standards are based on principles that support and promote desired values or behaviors.
- Professional organizations, such as CFA Institute, establish codes of ethics and professional standards based on ethical principles to guide practice.

Exhibit 1: Ethics, Rules, and Standards



Ethics and rules (i.e., regulations) are intertwined; ethical standards help guide the development of rules, and rules help individuals and groups, such as professional associations, think about, develop, and apply ethical standards.

A culture of integrity based on ethical standards can be built at a personal and business level by applying four basic steps:

1. Set ambitious standards, and put them in writing.
2. Get adequate and ongoing training on professional and ethical standards.
3. Assess the integrity of individuals and groups you encounter.
4. Take action when integrity and ethical standards are breached.

These steps help people assess and deal with **ethical dilemmas**. Ethical dilemmas are situations in which values, interests, or rules conflict. Sometimes the dilemma and an appropriate response are obvious; sometimes they are not. The challenge of resolving ethical dilemmas is addressed further in Course 6, Serving the Greater Good.

Why Is Ethical Behavior Important?

Global financial markets have grown in size and complexity over the last few decades. Investment products and financial services have also increased in breadth and complexity. Such growth and complexity increase the likelihood of ethical dilemmas, especially in business conducted across borders under different regulatory and cultural frameworks.

Investment professionals provide access to and information about investment opportunities and financial markets. They are involved in making and helping clients make investment decisions and in creating products that help with and add value to the investment decision-making process.

Individuals who work in the investment industry, but outside the investment management functions, are also critical to the functioning of the industry. These individuals include employees working in fund administration, securities trading and account services, and other support activities, including legal, human resources, marketing, sales, and information technology.

The decisions and actions of all investment industry professionals may directly or indirectly affect clients, prospective clients, employers, and coworkers. These professionals have a responsibility to make ethical decisions and to act appropriately. In other words, they must be trustworthy.

For example, clients seeking wealth management advice trust that the investment professionals they consult will provide suitable recommendations. Typically, these professionals rely on the support of others to provide investment services to clients. This support may be provided from within the investment firm or, in some cases, by third parties, such as legal or tax consulting firms. Such support also extends to the use of third-party information, such as credit ratings and investment research.

When relying on these sources, investment industry professionals must be careful and conduct due diligence to ensure the reliability of the information. If all parties are committed to acting in the best interests of the client, the client’s trust in the professional relationship is more likely to be sustained.

Lesson 6 Knowledge Check

The following knowledge check is designed to reinforce your understanding of the concepts taught in this lesson. It is ungraded, but you are strongly encouraged to complete it.

Question

Regulations cannot anticipate every situation in which questions of integrity or fairness will arise. For this reason, individuals and institutions should be guided by ethical principles and a culture of ethical behavior. Which of the following should not be considered a reliable source of guidance for the application of ethical principles?

Question choices

a. Professional standards addressing activities specific to the financial industry
b. Financial regulations carrying the force of law
c. Peer group behavior
d. Corporate policies and procedures designed to promote ethical behavior

Lesson 6 Knowledge Check Answer Feedback

c. **Correct**. Peer group behavior can have a strong effect on individual decision-making but may not always be aligned with ethical conduct or professional standards.

Incorrect answer feedback

a. **Incorrect**. Professional standards reflect ethical principles applied to specific industry practices.

b. **Incorrect**. Financial regulations reinforce ethical behavior and attach a cost to unethical practices.

d. **Incorrect**. Corporate policies, procedures, and culture can reinforce ethical behavior and attach a cost to unethical practices**.**

Up Next

You have now completed Lesson 6! We hope that this lesson has helped you understand more about the consequences of compliance failure.

You should now be able to describe the potential consequences of compliance failure.

Lesson 7: Trust and Stakeholder Obligations

Welcome to Lesson 7 of Module 5!

Establishing trust among stakeholders is paramount for the credibility of both professions and professionals. Upholding high ethical standards can both engender trust and enable effective collaboration in the pursuit of serving clients.

By the end of this lesson, you will be able to identify common obligations that individuals as professionals in the investment industry have toward various stakeholders (clients, prospective clients, employers, and coworkers).

Let's get started!

How Professions Establish Trust

For a profession to be credible, its primary goal must be to establish trust among clients and society in general. In doing so, professions have to fulfill some common obligations that, when combined, greatly increase confidence and credibility in professionals and their organizations. The following sections go into more detail about those obligations.

Professions Normalize Practitioner Behavior

Professionalism is underpinned by a code of ethics and standards of conduct developed by professional bodies. Regulators typically support professional ethics and recognize the framework for ethics that professions can provide. Many regulators around the world have engaged closely with professional bodies to understand their codes of ethics (codes) and standards of conduct (standards) as well as how they are enforced.

Codes and standards developed by practitioners can be complementary to regulations, codifying many more individual practices than the high-level principles set by regulation. Many governments have recognized that a profession can develop a more sophisticated system of standards than a regulator can via continuous practitioner input and a strong mutual interest within the profession to maintain good standards and adopt best practices. Government support of professions is attributable to the role of professions in helping the public and ensuring expert and principled performance of complex services.

Professions Provide a Service to Society

There is an obligation for professionals to go beyond codes and standards. Professionals should advocate for higher educational and ethical standards in their industry—individually and through their organizations. Professions can widen access to services and support economic activity by encouraging trust in the industries they serve.

Professions have realized that earning community trust not only creates professional pride and acceptance but also delivers commercial benefits.

A profession that earns trust may ultimately have greater flexibility and independence from government and regulators to manage its own affairs, which allows members of the profession to develop service models that are both useful to clients and beneficial to members.

Professions Are Client Focused

An integral part of a profession's mission is to develop and administer codes, best practice guidelines, and standards that guide an industry. These codes, standards, and guidelines help ensure that all professionals place the integrity of their profession and the interests of clients above their own personal interests.

At a minimum, professionals must act in the best interests of the client, exercising a reasonable level of care, skill, and diligence. The obligation to deliver a high standard of care when acting for the benefit of another party is called fiduciary duty, as we learned in Module 1, Industry Overview and Structure. Other entities, including employers, regulators, trade associations, and not-for-profit organizations, may also support an industry but are not the same as professional bodies. Unlike professions, these other entities generally do not exist to set and maintain professional standards. Most employers encourage employees to be members of relevant professional bodies, and many financially support this membership to ideally improve the quality of client service and reinforce ethical awareness.

Professions Have High Entry Standards

Membership in a profession is a signal to the market that the professional will deliver high-quality service of a promised standard, going beyond academic and technical credentials. Professions develop curricula that equip future professionals to be competent, including in their knowledge, technical skills, and ethics.

Professions Possess a Body of Expert Knowledge

A repository of knowledge, developed by experienced and skilled practitioners, is made available to all members of a profession. This knowledge helps members work effectively and ethically and is based on best practice.

Professions Encourage and Facilitate Professional Development

Entry into a profession does not, on its own, guarantee that an individual will maintain competency and continue to uphold professional standards. After qualification and throughout the working life of a professional, there will be changes in knowledge and technical skills to perform certain jobs, in technology, in standards of ethical behavior, in services that can be offered, and in the legal and business environment in which professional services are delivered.

These changing aspects of a profession all require the development of competence and ethical awareness. Most professional bodies make it a condition of membership that a specific amount of new learning is undertaken each year.

Typically, such conditions specify a time commitment, which may be separated into different competencies and types of learning activity. This is often referred to as continuing professional development (CPD) and is seen as an important part of maintaining professional standards. The training and education that professionals undertake increase the value of human capital, which can contribute to economic growth and social mobility.

Professions Monitor Professional Conduct

Members of a profession must be held accountable for their conduct to maintain the integrity and reputation of an industry. Doing so often involves self-regulation by professional bodies through monitoring and imposition of sanctions on members.

Professions Are Collegial

Professionals should be respectful to each other, even when they are competing. At the very least, they must respect the rights, dignity, and autonomy of others, especially coworkers.

Lesson 7 Knowledge Check

The following knowledge check is designed to reinforce your understanding of the concepts taught in this lesson. It is ungraded, but you are strongly encouraged to complete it.

Question 1

According to fundamental ethical and professional principles applicable to the investment industry, which group should have their interests ranked first?

a. Clients
b. Employers
c. Coworkers

Lesson 7 Knowledge Check Feedback

a. **Correct**. Individuals working in the investment industry have the obligation to place client interests ahead of personal interests or the interests of employers.
b. and c. are incorrect as these individuals work in the investment industry and have the obligation to place client interests first.

Up Next

You have now completed Lesson 7. We hope that this lesson has helped you to understand the importance of establishing trust among stakeholders for both professions and professionals.

You should now be able to identify common obligations that individuals as professionals in the investment industry have toward various stakeholders (clients, prospective clients, employers, and coworkers).

In the next lesson, you will learn about ethics and standards of professional conduct.

Lesson 8: Ethics and Standards of Professional Conduct

Welcome to Lesson 8 of Module 5!

Laws and regulations alone cannot fully protect the integrity of the financial system. Professional codes of ethics complement laws and regulations by establishing universal standards to which professionals are held accountable by the professional body.

By the end of this lesson, you will be able to identify elements of the CFA Institute Code of Ethics and explain how standards of professional conduct that are based on the CFA Institute Code of Ethics can be applied in practice.

Let’s get started!

The Ideal Asset Management Professional

Having obtained a high-level understanding of how a profession establishes trust within society, in this lesson we will take a closer look at the key attributes of a successful investment industry professional and provide an overview of how industry employees and employers can maintain high ethical standards in practice.

Laws and regulations help to ensure that those working in the investment industry fulfill their obligations. They also help protect the integrity of the financial system and promote fairness and efficiency of financial markets. But laws and regulations alone are not sufficient to protect the financial system. Laws and regulations may not extend to all areas of finance and can be vague or ambiguous, making their interpretation a challenge. Laws and regulations are often slow to catch up with market innovations. Activities that occur in different jurisdictions can be complicated by inconsistencies in legal obligations in different countries. Situations may arise in which no applicable law exists or the existing law is inconclusive. The effectiveness of laws and regulations depends on how market participants interpret and comply with them.

The need for ethical standards is particularly apparent in situations in which vague or ambiguous legal rules provide room for unethical behavior that could affect the integrity of the investment industry and result in a loss of clients’ and/or investors’ trust. To protect the financial system in these cases, ethical standards should guide the behavior of market participants. The principles embedded in codes of ethics and professional standards should help guide the behavior of industry participants and allow them to adapt to a continuously changing investment industry.

Professionalism at the Individual Level

We are all familiar with job advertisements. Often, some of the required attributes are described as “preferred but not essential” because employers are looking for the right balance of qualifications, experience, and personal qualities. In the investment industry, the need to maintain high ethical standards means that specific characteristics are definitely “essential.”

Learn more about the specific non-exhaustive characteristics outlined in the CFA Institute Code of Ethics and Standards of Professional Conduct:

- **Knowledge of the law:** Related to Standard I(A): Professionalism – Knowledge of the Law, professionals should have an understanding of the various laws and rules issued by governments, regulatory bodies, licensing authorities, and professional associations that govern the industry.
- **Independence and Objectivity:** Related to Standard I(B): Professionalism – Independence and Objectivity, professionals must use sound judgement to maintain independence and objectivity at all times. This behavior means not offering, soliciting, or accepting any gift, benefit, inducement, or compensation that could be interpreted as compromising their objective evaluation.
- **Misrepresentation:** Related to Standard I(C): Professionalism – Misrepresentation, professionals must never knowingly make any misrepresentations relating to investment analysis, recommendations, or other activities.
- **Misconduct:** Related to Standard I(D): Professionalism – Misconduct, professionals must not engage in any dishonest, fraudulent, or deceitful activity that reflects adversely on their reputation, integrity, or competence.
- **Remain informed with regular reviews:** Related to Standard V(A): Investment Analysis, Recommendations, and Actions – Diligence and Reasonable Basis, investment professionals usually deal with a substantial information load, so it is often challenging for them to keep pace with industry developments. Employers can help by making sure their employees are regularly informed about changes in laws, rules, regulations, and case law. In most firms, compliance departments or legal advisers can provide this information in a concise, easily digestible form. Regular workshops or seminars are also an effective way of keeping staff up to date.
  But in a busy work environment, some information can easily be missed or overlooked. Professionals and their employers should, therefore, carry out regular reviews of the company’s compliance procedures to ensure they reflect current law and provide guidance on permissible conduct.

- **Unconflicted opinions matter:** Related to Standard VI(A): Conflicts of Interest – Disclosure of Conflicts, professionals need to be aware that when producing reports, investment analysis, or client notes, it is easy for personal opinions to filter through and potentially compromise objectivity. To prevent this from happening, employers should establish a clear policy to ensure that every research report, especially those covering the securities of a corporate client, reflects the unbiased opinion of the analyst. Firms should also design compensation systems that encourage integrity in the investment decision-making process.
- **Beware of contacts bringing gifts:** Also related to Standard VI(A), gifts are nice, but they often come with strings attached. As such, investment professionals must restrict their acceptance of gratuities and gifts to only token items. This issue can be a sensitive topic because local customs in many countries perceive the refusal of a gift as impolite; refusing a gift could run the risk of damaging business relationships. Therefore, companies should set unambiguous guidelines on gifts for employees in each jurisdiction, perhaps by setting a strict value limit or imposing an aggregate annual value ceiling. Broadly speaking, customary business-related entertainment is acceptable, provided its purpose is not to influence or reward, and individual professionals should be able to use their experience and judgement to gauge what is acceptable. Accepting a cup of coffee or a glass of wine is acceptable; taking a bottle of vintage champagne is not.

Now that we have described several key principles of professional conduct contained in the CFA Institute Code of Ethics and Standards of Professional Conduct, we will explore a case study to help bring these principles to life.

Case Study 1: Cross-Referral of Clients

Raphael, an investment adviser for the fictional firm Enright Financial Solutions (EFS), enters into an understanding regarding the referral of clients with a friend, who is a lawyer.

Raphael will refer EFS clients needing legal services to the lawyer in return for the lawyer recommending clients needing financial advisory services to Raphael and EFS.

What do you think of this arrangement?

a. It is acceptable because there are no payments involved.
b. It is acceptable as long as the lawyer discloses the arrangement to the clients he refers to Raphael.
c. It is acceptable as long as EFS is aware of Raphael's agreement with the lawyer.
d. It is unacceptable.

Let's take a look at which option is correct and which options are incorrect.

Choice A

Option A is incorrect. Although this choice addresses the need for disclosure, it places the onus of disclosure on the lawyer and not on Raphael. Standard VI(C) requires Raphael to disclose the referral arrangement to any clients he refers to the lawyer and any potential clients referred to him by his friend.

Choice B

Option B is incorrect. Although this choice also addresses the disclosure issue by correctly stating that Raphael must disclose the arrangement to his employer, it does not go far enough because Standard VI(C) requires disclosure to be made to clients, prospective clients, and the employer.

Choice C

Option C is incorrect. The facts of the case do not mention whether Raphael has made all of the appropriate disclosures required by Standard VI(C). Assuming Raphael has not made the required disclosures to all applicable parties, this arrangement would be unacceptable.

Choice D

Option D is correct. Although there is no money changing hands between Raphael and his friend in this arrangement, there is mutual consideration and benefit. The fact that no money is exchanged does not preclude the obligation for disclosure.

Analysis

This case concerns a mutually beneficial referral arrangement, whereby service professionals refer clients to one another. Although such an agreement is not necessarily unethical and may ultimately be beneficial for the clients, there is a potential for a conflict of interest that must be disclosed. CFA Institute Standard VI(C): Conflicts of Interest – Referral Fees requires members to disclose to their employer, clients, and prospective clients "any compensation, consideration, or benefit received from or paid to others for the recommendation for products of services." This disclosure allows both clients and the employer to evaluate any partiality shown in the recommendation of services and the full cost of those services.

Professionalism at the Firm Level

Professionalism goes beyond the remit of the individual. There is also the need to uphold professionalism at the firm level.

Learn more about the essential, but non-exhaustive, characteristics of professionalism at investment firms outlined in the CFA Institute Code of Ethics and Standards of Professional Conduct:

- **Restrict investments:** Related to Standard VI(B): Conflicts of Interest – Priority of Transactions, all firms need clear, formal policies covering the timing of investment transactions by their employees. In particular, firms should require prior approval for employee participation in initial public offerings (IPOs), prompt disclosure of investment action taken following any equity offering, and limits on any involvement in private placements. There should also be strict supervisory procedures to ensure employees comply with these policies.
- **Diligent and reasonable effort to safeguard independence:** Related to Standard V(A): Investment Analysis, Recommendations, and Actions – Diligence and Reasonable Basis, firms that have developed a strong reputation for financial research are those that guard their independence. Individual professionals must take the same approach and fiercely defend their integrity. All firms should, therefore, adopt formal written policies that protect the objectivity of their researchers.
- **Responsibilities of supervisors:** Related to Standard IV(C): Duties to Employers – Responsibilities of Supervisors, once compliance standards have been adopted, it can be tempting to sit back and assume employees will remain alert to them, which is not always the case. So, firms should appoint a senior officer with compliance oversight responsibilities to enforce their code of ethics and all regulations concerning the business. Firms should provide every employee with procedures and policies to report potentially unethical or illegal behavior.
  In addition, all businesses should operate a no-tolerance policy toward violations. Companies should ensure that all employees are aware of the consequences of violations by distributing a list of potential infringements and associated sanctions, up to and including dismissal.

- **Safeguarding the integrity of capital markets:** Related to Standard II: Integrity of the Capital Markets, firms should thoroughly check the references of all potential employees to ensure they are not ineligible to work in the investment industry because of past infractions. Protecting the integrity of the industry and its professionals is a two-pronged task. First, firms and employees need to ensure they are armed with the right knowledge. Second, that knowledge needs to be put into practice, every day.

Following these guidelines can help ensure the industry, and the professionals working within it, protect their most valuable asset—their reputation.

Now that we have described several key principles of professionalism at the firm level contained in the CFA Institute Code of Ethics and Standards of Professional Conduct, we will explore a case study to help bring these principles to life.

Case Study 2: Misrepresented Return Performance

Prosper Funding (Prosper) is a (fictional) privately owned marketplace lender that enables borrowers to obtain unsecured consumer loans and allows investors to purchase securities linked to the performance of those loans. Prosper creates individual account pages for each investor on its website, which provides information on the consumer loans, gives the annualized net return (ANR) performance of each investor's account, and includes a link through which they can invest in Prosper securities.

Prosper calculates the ANR for each investor using an automated process, which was developed some years ago by personnel no longer with the firm. ANR results are reported prominently on each investor's online account page.

After several years, Prosper implements a "debt sale program" through which eligible non-performing, charged-off consumer loans linked to Prosper securities are sold. Because of the unforeseen effect of a previous, unrelated change to the original coding in the program, Prosper reports an ANR to its clients that excludes the impact of the worst-performing securities that they previously held but that have since been sold to third-party debt purchasers. Periodically, Prosper reviews all the computer processes used by the company, and through that review, it becomes clear that current employees lack an understanding of the ANR calculation process. The review is always general and at a high level, and thus Prosper does not identify the error in the ANR calculation. Because of the error, Prosper reports to a majority of its investors that their investments are earning up to double the returns they had actually earned.

When certain affected investors raise questions about the accuracy of the ANR calculations they received, Prosper's customer service department handles the complaints and confirms the reported figures were calculated erroneously because of a flawed computer program. The customer service employees do not elevate any of these complaints to Prosper's product, engineering, or compliance departments. Prosper has also solicited new investors for its securities based on the miscalculated ANR by sending hundreds of emails that highlight their (erroneous) ANR performance and has recommended that investors "add funds and build on their solid returns."

Prosper does not identify the error for several years and discovers it only after receiving a complaint from a large institutional investor. Prosper notifies investors that it miscalculated and misstated their ANR and provides a current, correct calculation of the ANR to investors at that time.

Analysis

Learn more about the different Prosper personnel, how certain employees may have violated the CFA Institute Code of Ethics and Standards of Professional Conduct (Code and Standards), and why:

- **Employee responsible for the computer program that calculates the ANR:** Prosper's issues begin with ANR performance that is calculated incorrectly and posted to client accounts. The CFA Institute Standard of Professional Conduct III(D): Performance Presentation requires CFA Institute members, and those abiding by the Code and Standards, to make reasonable efforts to ensure that investment performance is presented in a fair, accurate, and complete manner. The ANR is calculated using a computer program that was created many years earlier, and over the years, it has developed a defect that leads to dropping underperforming accounts from the calculation. The person currently responsible for calculating the ANR likely did not create the computer program, and they do not understand the program's process and have not adequately reviewed the calculation method to ensure that it is accurate, thus allowing the error to continue.
- **Customer service employee responding to client complaints about the ANR:** Although the customer service employee responding to client complaints is not responsible for the ANR calculation error, that employee is responsible for acting in accordance with Standard III(A): Duties to Client – Loyalty, Prudence, and Care to protect client interests once the issue has been raised. That employee failed to elevate any of the investor complaints to Prosper's personnel responsible for the calculation and apparently simply recalculated the ANR using

the same flawed code. The customer service employee did not engage with or investigate the client complaints with sufficient reasonable care, diligence, competence, or professionalism, as required by Standard V(A): Investment Analysis, Recommendations, and Actions – Diligence and Reasonable Basis and the Code of Ethics.

- **Marketing employee using the ANR performance calculations to solicit investments:** The marketing employee using erroneous ANR performance is not responsible for the calculation error. Team members should be able to rely on the work of others in their firm (in this case, those calculating the ANR) to fulfill their responsibilities in an appropriate manner to produce accurate work. But in this case, the marketing employee is ultimately responsible for distributing misleading information that misrepresents the true performance of the securities, which violates Standard I(C): Professionalism – Misrepresentation. Once the marketing employee becomes aware that the information is inaccurate, that employee must take steps to remediate the error and distribute accurate and complete information.
- **Compliance employee responsible for firm policies and procedures:** The compliance employee has a responsibility to review the firm's policies and procedures to ensure that they meet relevant regulatory and ethical requirements. Prosper only sporadically reviews all the computer programs and processes used by the company, uses employees who lack an understanding of the ANR calculation process to calculate the ANR performance, and conducts only a high-level review that would not catch the error in the ANR calculation. Similar to the customer service employee, the compliance employee did not exercise sufficient care, diligence, competence, or professionalism as required by Standard V(A): Investment Analysis, Recommendations, and Actions – Diligence and Reasonable Basis and the Code of Ethics to ensure that an adequate compliance program was in place to address errors or properly investigate problems once discovered.
- **CEO of Prosper:** The Prosper CEO likely has no direct involvement with the calculation and presentation of performance, addressing client complaints, or creating and implementing compliance procedures. But the CEO does have overall responsibility to ensure that the firm complies with the law (Standard I(A): Professionalism – Knowledge of the Law) and protects client interests (Standard III(A): Loyalty, Prudence, and Care) as well as overall supervisory responsibility for firm employees (Standard IV(C): Duties to Client – Responsibilities of Supervisors). The CEO is allowed to delegate this responsibility. But the Code and Standards require those with supervisory responsibility to make reasonable efforts to ensure that those under their supervision comply with applicable laws and the Code and

Standards. The breakdown of the policies and procedures of the firm designed to accurately calculate performance, adequately address complaints, and implement an appropriate compliance program are all red flags indicating a potential failure of the CEO to properly supervise employees of the firm.

Lesson 8 Knowledge Check

The following knowledge check is designed to reinforce your understanding of the concepts taught in this lesson. It is ungraded, but you are strongly encouraged to complete it.

Question 1

The fundamental ethical principle to act with independence and objectivity is *best* upheld when:

a. client confidentiality is maintained.
b. personal education and professional development are undertaken.
c. work and opinions are unaffected by any potential conflict of interest.

Lesson 8 Knowledge Check Feedback

c. **Correct**. Conflicts of interest can adversely affect objectivity and independence by creating a bias that may affect judgement.

a. and b. are not incorrect in this case, but are less critical than c. therefore ‘c’ is the *best* upheld.

Up Next

You have now completed Lesson 8. We hope that this lesson has helped you to understand the critical importance of professional codes of ethics in safeguarding the integrity of the financial system.

You should now be able to identify elements of the CFA Institute Code of Ethics and explain how standards of professional conduct that are based on the CFA Institute Code of Ethics can be applied in practice.

In Lesson 9, we will explore how the investment industry and wider economy are impacted by both ethical and unethical conduct.

Lesson 9: Benefits of Ethical Conduct and Consequences of Unethical Conduct

Welcome to Lesson 9 of Module 5!

In this lesson, you will explore the benefits of ethical conduct as well as the consequences of unethical conduct for various parties in the investment industry and the economy as a whole.

By the end of this lesson, you will be able to describe the benefits of ethical conduct and the consequences of conduct that is unethical or unprofessional.

Let's get started!

The Benefits of Ethical Conduct

Unethical behavior, whether legal or not, can have significant consequences for the financial system and the economy. Compliance with ethical standards is important to prevent financial crises that can affect economic development and society's welfare. Accordingly, one of the benefits of complying with ethical standards is the increased stability of the financial system and thus of the entire economy. Importantly, ethical standards complement existing legal obligations, professional standards of conduct, and organizational policies and procedures to prevent behavior that can affect other industries and even the global economy.

Learn more about the beneficial effects of ethical conduct in the investment industry on the market and greater economy:

- **Trust:** The liquidity, profitability, and efficiency of markets and economies are rooted in trust. Compliance with ethical standards increases trust in investment professionals and in the entire investment industry.
- **More trust equates to lower transactional friction:** Increased investor trust in the investment industry is an outcome of ethical behavior by the industry's professionals. It also strengthens the fairness of financial markets. This increased trust increases market participation and market efficiency (by which prices adjust quickly to reflect new information about the value of assets in the marketplace), which, in turn, help investors achieve their investment goals.
- **Greater transactional volume equates to increased liquidity:** Increased market participation benefits all market participants and stakeholders. These benefits include increased liquidity (investors are able to trade assets without affecting prices significantly) and increased market efficiency.

Increased market participation also promotes public awareness and understanding of the financial system. This understanding, in turn, leads to additional participation in and efficiency of financial markets.

- **Increased market efficiency improves access to both equity and debt financing:** Increased market efficiency and trust can increase access to equity and debt funding and decrease the cost of capital for companies and governments requiring capital. Increasing the availability of capital and decreasing the cost of capital may positively influence the profitability and growth of companies as well as the development of the investment industry and the overall economy.
- **An ethical environment better aligns the goals of companies and employees:** Increased market efficiency and participation can directly support the goals of companies in the investment industry. These goals include economic objectives, such as profitability and share value, and non-economic objectives, such as reputation and customer satisfaction.
  In addition, an employee following ethical standards is less likely to take excessive or unauthorized risk or to misappropriate company assets. Such misappropriation can be of tangible assets (for example, using the company car for personal trips without authorization) or intangible assets (for example, sharing information about customers or company research).

  If employees behave ethically, their actions are less likely to have far-reaching (including legal) consequences for their employers.

Without trust, the public is unlikely to use investment professionals to manage investments, to obtain advice on issues related to financial goals, or to provide services to support the overall investment process. Therefore, compliance may enhance employment security and increase certainty in career development.

Complying with ethical standards may directly and positively affect a professional’s position, compensation, and reputation. It may also provide indirect benefits through the increased reputational and business success of a professional team and entire firm, thereby providing long-term career and skill set development opportunities.

Consequences of Unethical Conduct

When ethical and professional standards are violated, there can be significant consequences. These consequences include failure to achieve goals on behalf of a client or an employer, negative effects on the entire investment industry through reduced market participation, and a loss of trust in the integrity of financial markets and the investment profession. Such consequences can have long-lasting effects on the investment industry and the economy.

Consequences for Industry and Economy

When people in the investment industry act unethically, it can lead to changes in the behavior of market participants. If investors lose trust in the investment industry, they may cease to invest. Potential consequences include companies being unable to raise capital in financial markets, investment firms losing business or even going out of business, and the economy slowing down.

Unethical and/or illegal behavior can be responsible for these kinds of negative consequences.

> **Example: Madoff Ponzi Scheme**
>
> Former US businessman Bernard L. Madoff ignored his duty to put his clients' interests first when he turned an allegedly legitimate wealth management firm into a Ponzi scheme, in which investors were paid "investment returns" with money from other investors. Madoff acted deceitfully and defrauded clients. The unveiling of this fraud in 2008 resulted in Madoff being sentenced to 150 years in prison and caused a widespread loss of investor trust in investment industry professionals.
>
> In addition to the immediate consequences of unethical behavior on companies in the investment industry and their clients, the transmission of financial shocks from one company to another can be dangerous for interconnected companies and, potentially, for the economy and the financial system.
>
> Therefore, the consequences of unethical behavior at one company can affect other companies (and their clients) despite them not being involved in the unethical behavior.

For example, if a company goes bankrupt as a result of unethical behavior, the company's unfulfilled liabilities could result in a widespread crisis for a larger group of related companies, thereby potentially damaging the economy. Such spillovers or financial contagion may result in decreased economic output, increased unemployment, and reduced long-term economic growth expectations.

Example: Financial Crisis of 2008

As a consequence of unethical behavior—in the form of aggressive mortgage lending by some market participants coupled with other financial events—the US housing and stock markets declined in 2008. As a result, the US-based global investment firm Lehman Brothers went bankrupt. These events led to a liquidity crisis (a shortage of available funds) in the financial markets.

The ensuing global financial crisis almost resulted in the bankruptcy of American International Group (AIG), a large multinational insurance company, which required a regulatory bailout to survive. The crisis negatively also affected many more companies and reduced the output and growth expectations of several economies around the world. This extreme case demonstrates how unethical behavior, such as aggressive mortgage lending, by some market participants can lead to the bankruptcy of a company and a negative impact on other interlinked companies, their clients, and the entire financial system.

Consequences for Clients

When people in the investment industry act unethically, clients may suffer both financially and emotionally. They may receive inappropriate investment advice or services, lose confidence in the investment profession and in financial markets, lose personal wealth and current or future income, and experience personal distress.

Unethical behavior resulting in inappropriate investment advice or services may expose clients to the following:

- Excessive risks
- Ownership of unsuitable assets
- Lack of diversification
- Inflated costs of investment management services
- Unjustified transaction and management costs

For example, excessive trading of client assets to maximize trading commissions will result in clients incurring excessive transaction costs and may result in clients owning assets that are not consistent with their objectives and needs. Because of the central role of trust in a client relationship, the violation of legal and ethical standards generally decreases a client's trust in investment professionals and possibly in the investment industry as a whole. Violations related to financial market transactions (for example, insider trading) further decrease a client's trust in the integrity of financial markets.

In all of these cases of unethical behavior, clients can lose wealth and income.

For example, if a client owns shares in an investment bank and there is a trading scandal, the value of the shares may fall, and the client may lose money. The client may also suffer a decrease in current income because of lower dividend payments. In addition, the client's future income can be reduced if retirement funds have been affected.

Faced with financial damage to wealth and income, clients can experience a great level of personal distress along with severe mistrust in the investment industry, regardless of whether that mistrust is justified or misplaced.

Consequences for Employers

From an employer's point of view, the consequences of ethical standard violations include negative effects on current and future client relationships, loss of reputation and company value, and legal liabilities and increased scrutiny by regulators, which creates additional administrative and analysis costs. The worst-case scenario includes going out of business.

Often, ethical violations become apparent externally only when the conduct results in legal scrutiny, including threat of lawsuits, legal charges, and prosecution. Because clients associate a company's employees with the company's brand, illegal or unethical conduct can result in a loss of company reputation that can damage or destroy current and future client relationships.

> **Example: Enron and Barings Bank**
>
> In some cases, an employer can face closure because of the unethical and/or illegal behavior of one or more individuals within the company. US energy and commodity trading company Enron, for instance, collapsed in 2001 as a result of unethical, aggressive accounting practices, which were later identified to be illegal. As a result, Enron CEO Jeffrey K. Skilling received a 24-year prison sentence. In addition, Arthur Andersen, Enron's audit firm, was negatively affected and later dissolved because of questions of integrity with respect to some of its partners involved in the Enron audit.
>
> In a different case, unethical and ultimately illegal behavior by former British derivatives trader Nicholas Leeson caused losses exceeding GBP800 million, resulting in the bankruptcy of the United Kingdom's oldest investment bank, Barings Bank, in 1995. Leeson, who had executed unauthorized trades, was sent to prison for six and a half years.

Interestingly, if the back-office accounting person had refused to comply with Leeson's orders on accounting matters, the losses might have been identified earlier when they were significantly lower.

These extreme examples show how the unethical behavior of a few individuals can have detrimental effects on coworkers and employers. A loss of reputation can result in a loss of profits and a loss of shareholder value. In cases of illegal conduct, a company in the investment industry may be held liable for financial losses sustained by customers or other market participants. If prosecuted, the employer may also be subject to fines, lose their operating licenses, and may be obliged to compensate clients or other market participants for financial losses.

These consequences can result in additional loss of company value, which is amplified if the company loses the right to provide some investment services.

The employer's profits may further decrease as a result of expenditures required to assess, manage, and prevent future occurrences of compliance failures. Lastly, a company may become subject to increased regulatory scrutiny and required to use company resources to administer and provide additional costly analysis and information.

Consequences for Individuals

For the individual, the legal, professional, personal, and economic consequences for violating ethical standards can be significant. In circumstances where there has been a breach of ethical standards that results in legal scrutiny, the individual may be subject to civil or criminal prosecution. The consequences can include expenses to defend a prosecution, fines, imprisonment, and loss of current and future employment in the investment industry. Unethical behavior resulting in legal offences includes investment fraud, insider trading, accounting fraud, and unauthorized decision making—all of which can result in significant prison sentences for the perpetrator.

In addition to the previously mentioned cases involving Jeffrey Skilling of Enron, Nicholas Leeson of Barings Bank, and Bernard Madoff, each of whom received long prison sentences, other examples illustrate the consequences of unethical and/or illegal behavior for individuals.

Example: Martha Stewart

US businesswoman Martha Stewart was accused of making false statements to federal investigators about a suspiciously timed share sale. Although she was not charged with insider trading, she was charged with conspiracy, obstruction of justice, and perjury. In 2004, she was found guilty of some of these charges and sentenced to a five-month prison term and fined USD30,000. In 2007, she settled a civil action suit for insider trading brought by the US Securities and Exchange Commission (SEC) by paying a fine of USD195,000 and accepting a five-year ban on serving as an officer or director of a public company.

Consequences for breaches of ethical and professional standards also include disciplinary action by the employer and/or a professional or regulatory body and disapproval from clients, colleagues, and peers within the industry. This discipline can result in the decreased ability to advance a career because of a loss of reputation, which has personal and economic consequences. In particular, the individual can suffer a loss of income as a result of a restricted ability to provide current or future services. The individual could also face the loss of job and career, and even alienation of family and friends.

Lesson 9 Knowledge Check

The following knowledge check is designed to reinforce your understanding of the concepts taught in this lesson. It is ungraded, but you are strongly encouraged to complete it.

Question 1

Which of the following outcomes of acting with high ethical standards will *most likely* directly benefit both clients and investment professionals?

a. Enhanced employment security
b. Reduced risk of adverse legal consequences
c. Increased trust in the fairness of financial markets

Question 2

Which of the following is *most likely* a potential effect of unethical behavior by people in the investment industry?

a. Increased employment
b. Decreased economic output
c. Decreased regulatory scrutiny

Lesson 9 Knowledge Check Feedback

Question 1

c. Correct. Acting with high ethical standards increases clients’ trust in the fairness of financial markets, thus promoting market efficiency, which, in turn, helps clients achieve their investment goals. This outcome also benefits investment professionals because increased trust from clients increases the likelihood that clients will seek their advice, thus enhancing the security of their employment.

a. Incorrect. Enhanced employment security. An advisor's employment security is less critical than their duty to clients.
b. Incorrect. Reduced risk of adverse legal consequences. Advisors' focus should be on the client's well being, and not putting their own risk ahead of the client.

Question 2

b. Correct. Unethical behavior in the investment industry may lead to a loss of investor confidence, which may reduce the availability of capital to companies and lead to decreased economic output.

a. Incorrect. Unethical behavior is more likely to discourage investment. Lower investment would likely reduce employment.

c. Incorrect. Unethical behavior is more likely to increase regulatory oversight.

Up Next

You have now reached the end of Lesson 9. We hope that this lesson has helped you understand the benefits of ethical conduct as well as the consequences of unethical conduct for various parties in the investment industry and the economy as a whole.

You should now be able to describe the benefits of ethical conduct and the consequences of conduct that is unethical or unprofessional.

In the next lesson, you will learn about using a framework for ethical decision-making.

Lesson 10: Ethical Decision-Making Framework

Welcome to Lesson 10 of Module 5!

By the end of this lesson, you will be able to describe a framework for making ethical decisions.

Let's get started!

Introduction

Laws, regulations, codes of ethics, and professional standards of conduct can guide ethical behavior, but individual judgement is a critical ingredient in making principled choices and engaging in appropriate conduct. One strategy to increase trust in the investment profession is to increase the ability and motivation of market participants to act ethically and help them minimize the likelihood of unethical actions.

Firms can enhance the ability and the motivation of their employees to act ethically by integrating ethics into all decision-making activities. An investment professional's natural desire to "do the right thing" can be reinforced by building a culture of integrity and accountability in the workplace.

> *Note:* Development, maintenance, and demonstration of a strong culture of integrity within the firm by senior management may be the single most important factor in promoting ethical behavior among the firm's employees.

Adopting a code that clearly lays out the ethical principles that guide the thought processes and conduct the firm expects from its employees is a critical first step. But a code of ethics, although necessary, is insufficient. We need to exercise ethical decision-making skills to develop the ability necessary for fundamentally ethical people to make good decisions despite the reality of conflicts.

A strong ethical culture built on a defined set of ethical principles that helps honest people engage in ethical behavior will foster the trust of investors, lead to robust global financial markets, and ultimately benefit society. That is why an ethical decision-making framework is important.

The CFA Institute Ethical Decision-Making Framework

When faced with decisions that can affect multiple stakeholders, investment professionals must have a well-developed set of principles. Otherwise, their thought processes can lead to, at best, indecision and, at worst, fraudulent conduct and destruction of the public trust. Establishing an ethical decision-making framework to guide an investment professional’s internal thought process regarding how to act is a crucial step to engaging in ethical conduct.

Investment professionals are generally comfortable analyzing and making decisions from an economic (profit/loss) perspective. Given the importance of ethical behavior in carrying out professional responsibilities, it is also important to analyze decisions and their potential consequences from an ethical perspective.

Learn more about the benefits of ethical decision-making frameworks:

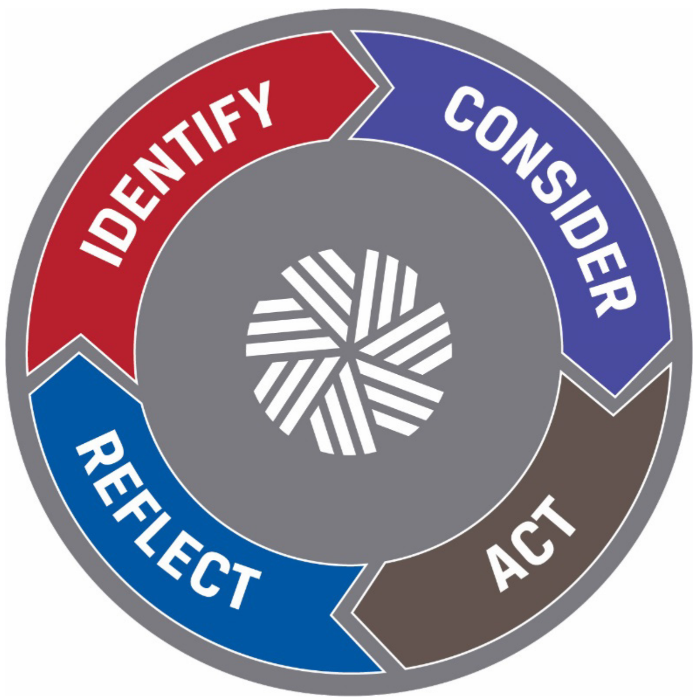
1. Using a framework for ethical decision-making will help investment professionals to examine their choices effectively in the context of conflicting interests common to their professional obligations (e.g., researching and gathering information, developing investment recommendations, and managing money for others). Such a framework provides investment professionals, as well as those that support them, with a tool to help them adhere to a code of ethics and allows them to analyze and choose options to meet high standards of ethical behavior.
2. By applying the framework and analyzing the particular circumstances of each available alternative, professionals working in the investment industry are able to determine the best course of action to fulfill their responsibilities in an ethical manner.
3. An ethical decision-making framework helps a decision maker see the situation from multiple perspectives and pay attention to aspects of the situation that may be less evident with a short-term, self-focused perspective. The goal of getting a broader picture of a situation is to be able to create a plan of action that is less likely to harm stakeholders and more likely to benefit them.
4. If a decision maker does not know or understand the effects of her or his actions on stakeholders, the likelihood of making a decision and taking action that harms stakeholders is more likely to occur, even if unintentionally.
5. Finally, an ethical decision-making framework helps decision makers explain and justify their actions to a broader audience of stakeholders.

An ethical decision-making framework is designed to facilitate the decision-making process for all decisions. It helps people look at and evaluate a decision from multiple perspectives to identify important issues they might not otherwise consider. Using an ethical decision-making framework consistently helps professionals develop sound judgement and decision-making skills and avoid making decisions that have unanticipated ethical consequences. Ethical decision-making frameworks come in many forms, with varying degrees of detail.

Examining an Ethical Decision-Making Framework

An ethical decision-making process includes multiple phases, each of which has multiple components. The process is often iterative, and the decision maker may move between phases in an order different from that presented. For simplicity, we will discuss the phases sequentially. A general ethical decision-making framework developed by CFA Institute[1] is shown in Exhibit 1.

Exhibit 1: Phases of the CFA Institute Decision-Making Framework



1. **Identify:** In the first phase, you will want to identify the important facts that you have available to you as well as information that you may not have, but would like to have, to give yourself a more complete understanding of the situation.
    - It is helpful to distinguish between facts and personal opinions, judgements, and biases.
    - You will also want to identify the stakeholders and the duties you have to each of them.

      You will then identify relevant ethical principles and/or legal requirements that might apply to the situation.

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[1] The CFA Institute ethical decision-making framework is based on "A Framework for Thinking Ethically" from the Markkula Center for Applied Ethics at Santa Clara University (www.scu.edu/ethics/practicing/decision/framework.html).

    - You should also identify any potential conflicts of interest inherent in the situation or conflicts in the duties you hold to others. For example, your duty to your client may conflict with your duty to your employer.
2. **Consider:** In the second phase of the framework, you will take time to consider the situational influences as well as behavioral biases that could affect your thinking and thus decision-making. During this phase, you may seek additional guidance from trusted sources, turn to your compliance department for assistance, or possibly consult outside legal counsel.

   Seeking additional guidance is a critical step in viewing the situation from different perspectives. You should seek guidance from someone, possibly external to the firm, who is not affected by the same situational influences and behavioral biases as you are and can, therefore, provide a fresh perspective. You should also seek guidance from your firm's policies and procedures and the CFA Institute *Code of Ethics and Standards of Professional Conduct*.[2]

   A helpful technique can be to imagine how an ethical peer or role model might act in the situation.

3. **Act:** The third phase of the framework is to decide and act. After all the thinking and information collection, it is still "theory" at best. If no action is initiated, nothing will change. At this stage, one should leverage the holistic and thoughtful preparation of Phases 1 and 2. Doing so should prevent unnecessary hesitation or deliberation.
4. Finally, subject to the unique circumstances surrounding different situations, the timing of execution can be further fine-tuned to achieve optimal impact.
5. **Reflect:** After you have acted on your decision, the fourth phase is to reflect on and assess your decision and its outcome:
    - Was the outcome what you anticipated? Why or why not?
    - Had you properly identified all the relevant facts, stakeholders and duties owed, relevant ethical principles and/or legal requirements, and conflicts of interest?
    - Had you considered the situational influences?
    - Did you identify behavioral biases that might affect your thinking?
    - Had you sought additional guidance?
    - Had you considered and properly evaluated a variety of alternative actions?

[2] Please visit cfainstitute.org/ethics for full text of the CFA Institute Code and Standards.

You may want to reflect on the decision multiple times as the immediate and longer-term consequences of your decision and actions become apparent over time.

The ethical decision-making process is often iterative. After identifying the relevant facts and considering situational influences, you may, for example, decide that you cannot decide without more information. When faced with such a situation, you may want to do the following:

- Seek additional guidance on how to obtain the information you need
- Begin considering alternative actions regarding how to proceed based on expectations of what the additional information will reveal
- Wait until you have more information, reflect on what you have done and learned so far, and start the process over

Application of an Ethical Decision-Making Framework

Sometimes situations can be complicated, and multiple iterations may reveal that no totally acceptable solution can be created. Applying an ethical decision-making framework can help you evaluate the situation so you can make the best possible decision.

Using an ethical decision-making framework will help you evaluate situations from multiple perspectives, avoid poor decision making, and avoid the consequences that can result from taking an ill-conceived course of action. Using an ethical decision-making framework is no guarantee of a positive outcome but may help you avoid making unethical decisions.

After all, there is nothing perfect in life. As long as you make a best effort attempt to minimize blind spots and act in full alignment with your holistically thorough consideration, you are not likely to be too wrong.

Case Study: Applying an Ethical Decision-Making Framework

Carlos experienced the following dilemma:

1. Several colleagues and their friends from outside of the office go out after work with Carlos, a newly hired employee.
2. The more experienced employees tell the “new guy” to charge the meal and drinks to the company credit card. Carlos’s colleagues tell him the friends are “prospective clients” and that they have charged similar outings on the card before, and the boss always has always approved the charges.
3. Carlos is confused. During orientation, the presenter from human resources made it clear that it is against company policy to charge personal expenses on the company’s credit card.

4. Based on the orientation, he wonders if using the company credit card would be wrong. But if it is common practice at his firm, maybe it is okay?

Using an ethical decision-making framework may help him make an ethically sound decision.

Learn more about using the ethical decision-making framework to work through Carlos's dilemma below.

1. **Identify the ethical issue(s) and relevant duties/obligations**
   What is the ethical dilemma?

   Is it appropriate for Carlos to use the company credit card to pay for the dinner with colleagues and their friends?

   To whom is a duty owed, or who might be affected by the decision?

   Often a decision is easier if you can put a face to the party that might be affected. Carlos owes a duty to his employer, which includes displaying loyalty, following its policies, and acting with integrity. His decision will potentially affect his relationships with his employer, his boss, and his colleagues.

   Are any duties in conflict, and which duty takes precedence?

   In some situations, duties to multiple parties may exist for the individual facing the dilemma. In this case, Carlos's sole duty is to his employer.

2. **Identify conflicts of interest**
   Are personal interests affecting the decision/action?

   In this case, Carlos may struggle with wanting to fit in with his new colleagues versus following his understanding of the policies of the employer.

3. **Get the relevant facts**
   What are the relevant facts in the situation?

   During Carlos's orientation, limitations on the use of the company credit card were identified. During the dinner, more experienced colleagues told Carlos it is okay to use the card for business expenses of this nature. The friends were identified by them as prospective clients.

   Are there any facts not known that should be known?

Carlos would like to know the company policies on allowable client entertainment expenses and for classifying and documenting someone as a potential client. He would like to know if the friends are, in fact, prospective clients.

The following are other questions that might be asked in fact gathering:

- What resources are available to learn more about the situation?
- Is there enough available information to make a decision?

In this situation, Carlos does not have the luxury of gathering more information about the policy. Carlos may wonder if the friends actually represent prospective clients for the company but may find it difficult to question his colleagues further. As a result of the orientation, Carlos is concerned that charging the meal to the card may be in violation of the company policy and be unethical.

4. **Identify applicable ethical principles**
   What are the fundamental ethical principles involved in the situation?

   Carlos is expected to act with integrity in using the resources of his employer. He should use reasonable care in determining whether the friends actually represent prospective clients. He should use independent judgement. He should treat his colleagues with respect, but this does not mean he cannot question their guidance.

5. **Consider factors that could be affecting judgement**
   What factors are affecting judgement?

   Outside factors, such as authority figures, a vocal group, and incentives, can affect judgement. Internal factors, such as overconfidence and rationalization, can also affect judgement. Carlos may be influenced by his more experienced colleagues. He may rationalize that it is just one dinner and thus unimportant. He should pause and think through his decision before acting.

6. **Consider and evaluate alternative actions**
   What are the options? Have creative options been identified?

   Carlos's options include the following:

   a. Complying with his colleagues' request and charging the dinner to the company card.
   b. Complying with his colleagues' request and charging the dinner to the company card but indicating that he will speak to the boss about it the next day.

c. Charging the dinner to his personal card.
d. Charging the dinner to his personal card and indicating that he will speak to the boss about it the next day.
e. Declining and suggesting that it is better for a more senior colleague to charge the dinner.
f. Declining and suggesting that those in attendance split the bill.

What criteria affect choosing among the options? How should conflicts of interest be managed? How will an alternative affect the relevant parties identified in Step 1?

Carlos wants to behave ethically and abide by company policy. He also wants to be on good terms with his colleagues. If Carlos chooses to pay with either the company card or his personal credit card and talk to the boss, he may end up paying the entire bill personally. He may not be willing or able to do this.

Furthermore, if he ends up paying the entire bill personally, he may distrust his colleagues and their advice in future. If he indicates that he is going to speak with the boss or declines to pay, it has consequences for his relationships with his colleagues. They may not view him as a trusted member of the team. Speaking to the boss may affect his colleagues.

Paying also has consequences for his relationships. The colleagues may lose respect for him, and his boss may disapprove. This consequence may result in censure or even dismissal.

If a particular alternative were shared with a respected individual or reported to a wider audience, what might be the expected response?

The orientation should lead Carlos to understand that using the company card is unacceptable and would reflect poorly on him. Carlos might be uncomfortable sharing the decision to charge the company card.

7. **Consider seeking additional guidance**
Would discussing the issues with a supervisor, colleagues, or legal (compliance) personnel be appropriate or helpful?

Seeking guidance to gather more facts about company policy or discussing the issue with a supervisor, colleague, or compliance officer would be helpful but is not possible in this case. Carlos has to make a decision and act now based on the information available.

8. **Act and review the outcome**
   After making his decision, Carlos should take the time to reflect on how his decision turned out and what he has learned from this specific situation. He should also apply what he learned to future dilemmas he will experience in the course of his work.

Lesson 10 Knowledge Check

The following knowledge check is designed to reinforce your understanding of the concepts taught in this lesson. It is ungraded, but you are strongly encouraged to complete it.

The last step in an ethical decision-making process should be to:

a. assess how a decision turned out and learn from it.
b. assess how others might view a possible course of action.
c. determine whether the decision is affected by outside factors.

Lesson 10 Knowledge Check Feedback

a. **Correct**. The last step in an ethical decision-making process should be to assess how a decision turned out and to learn from it.
b. **Incorrect**: assessing how others might view a course of action could be an early stage consideration.
c. **Incorrect**: assessing whether a decision is affected by outside factors should be an early stage consideration.

Up Next

You have now reached the end of Lesson 10. We hope that this lesson has helped you to understand frameworks for ethical decision-making as well as how to apply them in real-life dilemmas.

You should now be able to describe a framework for making ethical decisions.

In the Module 5 Summary, we will review all the important concepts that were covered in this module.

Module 5 Summary

In this module, we discussed professionalism and its impact on the industry and investing.

Professions are occupational groups that have specific education, expert knowledge, and a framework of practice and behavior that underpins community trust, respect, and recognition. Most professions emphasize an ethical approach, the importance of good service, and empathy with the client

- For a profession to be deemed credible, it must be viewed as trustworthy by clients, participants, and society in general. Professions can establish trust through a variety of activities, including the following:
    - Developing codes of ethics and standards of conduct designed to normalize practitioner behavior and encourage high educational and ethical standards. Regulators typically support professional ethics and recognize the framework for ethics that professions can provide.
    - Providing a service to society by widening access to services and supporting economic activity by encouraging trust in both the industry at large and the professionals within it.
    - Encouraging professionals to place the integrity of their profession and the interests of their clients above their own personal interests.
    - Ensuring high entry standards for professionals and that future professionals are equipped to be competent in their knowledge, technical skills, and ethics.
    - Encouraging the development and dissemination of a professional body of expert knowledge.
    - Encouraging and facilitating the continuing professional development of its professionals.
    - Holding professionals accountable for their conduct through self-regulatory actions, including monitoring and imposing sanctions in instances of misconduct.
    - Encouraging a collegial and respectful environment among professionals within the workplace.

We then explored some specific elements in the CFA Institute *Code of Ethics and Standards of Professional Conduct*, to which its membership is held accountable, including the following:

- Understanding the laws and rules that govern the industry.
- Consistently maintaining independence and objectivity in work practices.
- Never knowingly making misrepresentations in analysis, recommendations, or other activities.
- Avoiding the perception of misconduct by not engaging in any dishonest, fraudulent, or deceitful activity that can reflect adversely on the individual's reputation, integrity, or competence.
- Staying up-to-date by completing compliance-related training provided by or required in organizations and conducting regular reviews of corporate compliance procedures to ensure they reflect current law and provide guidance on permissible conduct.
- Ensuring that objectivity is not compromised by inclusion of personal opinions in investment analysis, reports, client notes, and other activities, such as acceptance of gratuities and gifts that could create or be perceived as a conflict of interest.

We explored the benefits of ethical conduct, consequences of unethical or unprofessional conduct, and a framework that can be used for making ethical decisions.

- Unethical behavior, whether legal or not, can have significant consequences for the financial system and the economy. Compliance with ethical standards is important to prevent financial crises that can affect economic development and society's welfare.
- The liquidity, profitability, and efficiency of markets and economies are rooted in trust. Trustworthy behavior increases investor trust, strengthens the fairness of the markets, and encourages and supports market participation and efficiency.
- The establishment of ethical cultures within organizations can help to better align the goals of companies and employees and decrease misconduct and potential for conflicts of interest.
- When ethical and professional standards are violated, there can be significant consequences, including failure to achieve goals on behalf of a client or an employer, negative effects on the entire investment industry through reduced market participation, and a loss of trust in the integrity of financial markets and the investment profession.

- Laws, regulations, codes of ethics, and professional standards of conduct can guide ethical behavior, but individual judgement is a critical ingredient in making principled choices and engaging in appropriate conduct.
- When faced with ethical dilemmas and decisions that can affect multiple stakeholders, investment professionals must have a well-developed set of principles. Establishing an ethical decision-making framework is a crucial step in engaging in ethical conduct.
- Using an ethical decision-making framework consistently can help investment professionals develop sound judgement and avoid making decisions that have unanticipated ethical consequences.
- The CFA Institute Ethical Decision-Making Framework presents four key phases of ethical decision making, which may be done in an iterative process:
    - Identify the important facts or additional information needed to make an informed decision, the fundamental ethical principle potentially at issue, to whom a duty is owed, and any potential conflicts of interest at play.
    - Consider situational influences or biases that may affect decision-making, brainstorm alternative actions and solutions at an individual level, and seek objective feedback and perspective from others.
    - Determine the most appropriate course of action (which may also include not taking an action or elevating concerns to a higher authority), and act.
    - Reflect on what was learned in the process, regardless of outcome, and use the lessons learned to improve future ethical decision making.

Finally, we looked at how ethics and regulation are key forces driving the investment industry. Trust, the importance of rules and regulations, and the benefits that ethical decision-making bring to the investment industry and society at large were significant themes we covered.

Trust is essential to the functioning of the investment industry as well as to the broader financial services industry. Trust depends on participants complying with rules and acting ethically. Increased trust in the fairness of financial markets and the ethical conduct of market participants leads to increased market participation. Increased market participation leads to increased liquidity, market efficiency, and availability of capital at a reduced cost. As a result, the overall economy thrives.

- Regulations are rules carrying the force of law that are set and enforced by legislative bodies and other entities authorized by legislative bodies. It is important that all investment industry participants comply with relevant regulations. Those that fail to do so face sanctions that can be severe.

- Financial services and products are highly regulated because a failure or disruption in the financial services industry, which includes the investment industry, can have catastrophic consequences for individuals, companies, and the economy.
- Regulation is necessary for many reasons, including to protect consumers; to foster capital formation and economic growth; to support economic stability; to promote fair, efficient, and transparent financial markets; and to improve society.
- The types of regulations that have been developed in response to perceived needs include rules on gatekeeping, operations, disclosure, sales practice, trading, anti-money-laundering, and business continuity planning.

Module 5 Glossary

Ethical standards: Principles that support and promote desired values or behaviors.

Front-running: The act of placing an order ahead of a customer's order to take advantage of the price impact that the customer's order will have.

Insider trading: Trading while in possession of material non-public information.

Market manipulation: Abusive trading practice that involves taking actions intended to move the price of a stock to make a short-term profit.

Money laundering: A process in which criminals use financial services to transfer money from illegal operations to other, legal activities; the money becomes "clean" in the process.

Professional standards: Guidelines set by professional associations based on fundamental ethical principles to guide the behavior of individuals within the profession.

Regulations: Rules that set standards for conduct and carry the force of law. They are set and enforced by legislative bodies and by other entities authorized by legislative bodies.

Systemic failure: Failure of the financial system as a whole, including loss of access to credit and collapse of capital markets.